ROYAL COLLEGE OF SURGEONS OF EDINBURGH

Membership in Implant Dentistry

M Imp Dent RCSEd

A Candidate's Revision Notebook

Blueprint-mapped core revision notes

800 single best answer questions with explanations

58 OSCE stations with model answers and checklists

51 structured oral cases with examiner question flows

Written throughout to the standard of a general dentist with an interest in implant dentistry

For candidates sitting the examination from September 2026 onwards

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Mapped throughout to Appendix A — Learning Outcomes and Blueprint (April 2026), the Regulations and the Guidance for Candidates

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Membership in Implant Dentistry (M Imp Dent RCSEd)

A Candidate's Revision Notebook

Blueprint-mapped core notes · 300 single best answers · 40 OSCE & SOE stations with model answers

Prepared for candidates sitting the examination from September 2026 onwards

How to use this notebook

This notebook is written to one specification only: **Appendix A — Learning Outcomes and Blueprint (April 2026)**. Every section carries the blueprint domain it serves, and every question is tagged to the component that tests it (W = Written, OS-P = OSCE Practical, SCE = Structured Oral).

Three points about *level* govern everything that follows, and they are the single most common reason candidates fail:

> **Regulation 18.2:** *"It is important to note that the examination is not a specialist examination and that the level of knowledge expected in any area will not exceed that which would be expected of a general dental practitioner who has three years' experience of clinical dental practice together with a particular interest in implant dentistry."*

> **Regulation 2.1:** The examination *"is not intended to test the candidate's competency at the execution of these procedures or the detailed knowledge of the more advanced procedures such as treatment planning with computerised scanning software, major bone grafting (block onlay grafting), external sinus grafting or full arch prosthodontic rehabilitation."*

> **Regulation 2.1 (aims):** *"…harmless, independent overall patient care."*

Read those three quotations again before every practice session. The examiners are not looking for the most sophisticated answer; they are looking for the **safest defensible answer, delivered with insight into your own limits**. A candidate who says "I would place an immediate implant with simultaneous horizontal GBR and an immediate provisional" in a high-aesthetic-risk case will score *lower* than one who says "this is a Complex case on the SAC classification; the aesthetic risk is high because of the thin phenotype and high smile line, so I would either refer, or treat with a delayed protocol after ridge preservation, and I would explain both to the patient." **Recognising complexity and knowing when to refer is an examinable learning outcome in its own right** (Appendix A: *"Recognise the complexity of the case and the need for referral"*).

Start here: calibrate before you revise

Three short sections are written to set your expectations before you start working through the material, and they are the most useful pages in this notebook if your time is short:

- **"Where the marks are in the OSCE"**, at the start of Part 6 — how a practical circuit is constructed, and the six habits that separate a clear pass from a bare one.
- **"Where the marks are in the SOE"**, at the start of Part 7 — the case archetypes that recur, and what each is really testing.
- **The three set-pieces**, also at the start of Part 7 — placement timing and loading protocol; bone grafting materials with their advantages, disadvantages and complications; and incision and flap design. Each is written as a spoken 60–90 second answer with the follow-up questions you should expect.

Read all three before you plan your revision. They will change your priorities — in particular, the procedural outcomes in the blueprint are weighted far more towards **restorative and prosthodontic procedure, temporisation, communication with the technician and problem-solving around existing restorations** than most candidates assume. Candidates who prepare only for surgery prepare for the wrong examination.

A note on scope. Nothing in this notebook is a record of any past examination. Every station, case and question is written by the author from the published learning outcomes and blueprint, the Regulations, the Guidance for Candidates and the standard texts, as teaching material. Candidates are reminded that

the College's regulations prohibit the disclosure of examination content, and that this obligation binds you both before and after you sit.

Suggested six-week revision plan

Week	Focus	This notebook
0	**Calibration**	**"Where the marks are" at the start of Parts 6 and 7, and the three set-pieces — read first**
1	Assessment, risk, investigations, anatomy	Part 2 §1–§4; SBAs 1–86, 301–440
2	Biology, materials, implant design	Part 2 §5–§6; SBAs 87–132, 441–515
3	Planning, consent, SAC, sequencing	Part 2 §7; SBAs 133–180, 516–590; SOE cases 1–10
4	Surgery and minor augmentation	Part 2 §8–§9; SBAs 181–246, 591–700; OSCE 1–25
5	**Prosthodontics, occlusion and temporisation**	Part 2 §10; SBAs 247–280, 701–760; **OSCE 52, 53, 54, 56**; OSCE 26–50
6	Complications, peri-implant disease, governance	Part 2 §11–§12; SBAs 281–300, 761–800; SOE cases 11–51

Then: Part 3 (numbers and classifications) daily for the last ten days, and one full timed SOE circuit out loud with a colleague.

1. The examination at a glance

(*Source: Regulations and Guidance for Candidates, April 2026, and the Important Announcement of Changes, April 2026.*)

Three components. Each must be passed independently. No compensation between components.

Component 1 — Written

Format	2 papers × 2 hours (4 hours total)
Questions	Single Best Answer (SBA), **maximum 90 per paper** (≈180 total)
Marking	Machine marked; standard set by panel of examiners
Compensation	**Allowed between the two papers** (but not with other components)
Result	Pass / fail against the standard-set cut score

Implication: at ~180 questions in 240 minutes you have **~80 seconds per question**. That is enough time to read a stem once, eliminate two options, and commit — but not enough to agonise. There is no negative marking in a standard-set SBA paper; **answer every question**.

Component 2 — OSCE (Practical & Procedural Knowledge and Skills)

Format	Circuit of **10 stations**, ~70 minutes total
Timing	**2 minutes reading** outside the station + **5 minutes** inside
Content	Either *undertake a practical task* on a simulated model, or *describe the steps* of a procedure to the examiner
Marking	**10 marks per station**, checklist-based, equal weighting; a global mark is recorded for quality assurance
Extras	There may be a rest station and/or a **pilot station** (pilot marks do not count)
Examiner	Present at every station; **will not interact unless the task requires it**

Implication: the checklist rewards **completeness and sequence**, not elegance. Five minutes is short. Narrate continuously, in order, and cover the whole pathway even if you cannot finish the physical task. Silence scores zero. If you finish early, **stay in the station** and add detail (post-operative instructions, what you would record, what could go wrong) — those are almost always checklist items.

Component 3 — Structured Oral Examination (SOE) on Clinical Scenarios

Format	**8 stations**, each **25 minutes = 10 minutes preparation + 15 minutes oral**
Break	5 minutes between cases
Materials	Patient history, clinical examination, investigations, radiographs, scans, photographs, study casts, wax-ups
Notes	You may make notes in the prep room and **take them in**; they are surrendered at the end
Examiners	**Two per case**, both will question you, both mark
Marking	**5-point scale $\times$ 4 domains = 20 marks per examiner per case**
Pass	**Mean of 3 across all domains and all cases**

The four SOE domains — memorise these, they are the marking sheet:

1. **Clinical Interpretation and Reasoning**
2. **Clinical Decision Making and Treatment Planning (including consent)**
3. **Decision Making in Relation to Treatment Procedures** (surgical placement & prosthetic reconstruction)
4. **Interpretation & Problem Solving of Clinical Outcomes**

> **The single most valuable habit in this notebook:** in your 10 minutes of preparation, write four headings on your paper — *Interpret / Plan & Consent / Procedures / Outcomes* — and fill each with three or four bullets. You will then have an answer ready for whichever domain the examiners open with, and you will have structured your own 15 minutes.

Change from September 2026

The **Communications OSCE has been discontinued**. The examination is now the three components above only. Communication is *not* untested, however — it has migrated into the SOE, where consent, explaining risk, managing expectation and handling dissatisfaction are all explicitly blueprinted to the SCE column.

Eligibility (Regulation 4 / Appendix B self-declaration)

- Primary dental qualification acceptable to Council
- **3 years / 36 months** post-qualification full-time experience (part-time equivalent within 6 years)
- **15 cases** treated with surgery personally performed or under supervision, **2 with $\geq$1 year follow-up**, comprising:
  - 4 single-unit replacements
  - 5 multiple implants in different quadrants **or** 2 implants supporting a 3-unit bridge
  - $\geq$1 edentulous case
  - $\geq$5 cases with **some evidence of minor augmentation**
- **$\geq$30 hours implant CPD with reflection** in the past 18 months
- **A completed quality improvement project** related to implant dentistry in the past 18 months

Failure to submit the signed declaration = ineligible. **Maximum four attempts** (Regulation 9.1). Failed components may be re-sat alone **for one diet only**; after that all three must be retaken (Regulation 14.2). Feedback must be requested within **28 days** of results.

> **SOE tip:** your quality improvement project is fair game as a discussion opener under domain 4 ("*Demonstrate an understanding of the importance of quality assurance measures*"). Have a 60-second account ready: the problem, the standard you audited against, your sample, your finding, your change, and your re-audit.

2. What the blueprint actually asks of you

Appendix A groups content under **five domains**. This notebook maps to them as follows.

Blueprint domain	Blueprint sub-headings	Notebook sections	Tested in
Clinical Interpretation & Reasoning	History taking; Patient examination	§1 Assessment & risk; §2 Examination, occlusion & space; §3 Investigations; §4 Anatomy; §5 Biology; §6 Materials & design	W, SCE (some OS-P)
Clinical decision making & treatment planning incl. consent	Diagnosis & clinical reasoning; Drug therapy & safe prescribing; Record keeping; Consent; Treatment planning	§7 Planning, SAC & consent; §7.6 Prescribing; §12.5 Records & governance	W, SCE, some OS-P
Decision making in treatment procedures & basic procedural knowledge/skills	Surgical placement, handling of tissues & complications; Restoring implants & managing complications	§8 Surgery; §9 Minor augmentation; §10 Prosthodontics	**W, OS-P, SCE**
Interpretation & problem solving of clinical outcomes	Patient safety; Quality & maintenance; Complications & clinical outcomes	§11 Complications; §12 Peri-implant disease, maintenance & outcomes	W, SCE
General professional skills	(threaded throughout)	§12.5–§12.7	SCE

Learning outcomes that appear in all three columns (W + OS-P + SCE)

These are the highest-yield topics in the whole syllabus, because they can be examined three different ways. Know them cold:

1. **Diagnostic wax-ups** — role in planning, discussion and execution (*also the published sample OSCE station*)
2. **Selection of guides** (radiographic and surgical) and **transfer of planning information into surgery**
3. **Bone quality and quantity** and their influence on the plan
4. **Need for soft and hard tissue augmentation**, and differentiating augmentation **materials**
5. **Types of bone defect** and how to manage them
6. **Minor augmentation** — appropriateness and execution
7. **Flap design** and **atraumatic flap elevation**
8. **Choice of suture material and technique**
9. **One-stage vs two-stage** decision making
10. **Principles of site preparation**
11. **Infection control and aseptic technique**
12. **Abutment selection; impression techniques; screw- vs cement-retained; laboratory prescription; digital workflows**
13. **Local anaesthetics** — range, side effects, methods; and **management of drug interactions**

Every one of these has a dedicated OSCE station in Part 6 of this notebook.

What is explicitly out of scope

You are **not** examined on detailed knowledge or planning of:

- **Block onlay grafting** (major autogenous augmentation)
- **External (lateral window) sinus floor elevation** — *you must still know what it is, when it is indicated, and that it warrants referral*
- **Full-arch prosthodontic rehabilitation and reconstruction**
- **Treatment planning with computerised scanning/planning software** at an advanced level
- Execution competency of any procedure (this is a knowledge, judgement and procedural-knowledge examination, not a clinical competency test — *no patient work is undertaken*, Guidance 4.2)

But — and candidates get caught here — you *are* expected to *recognise* when these are indicated and to say that the case falls outside the scope of a GDP with a special interest and requires referral. "I would refer" is a scoring answer when it is justified; it is a losing answer when used to dodge something a GDP should manage.

3. The FGDP training standards context

The examination is calibrated against the UK training standards for implant dentistry (*Training Standards in Implant Dentistry*, FGDP(UK), the successor to the GDC-convened 2005/2008/2012 standards). Two things from that document recur in the SOE:

Implant dentistry is divided into two levels: 1. **Straightforward** placement and/or restoration 2. **Complex** placement and/or restoration

The GDP with a special interest — i.e. you, for the purposes of this examination — works in level 1 and refers or team-manages level 2.

The thirteen skills a dentist must have (paraphrased from the standards) map almost one-to-one onto the blueprint: clinical assessment and risk–benefit analysis; judging case complexity and knowing when to refer; communication and valid consent at every stage; imaging in line with **IR(ME)R**; aseptic surgical technique; raising mucoperiosteal flaps and suturing; **minor alveolar augmentation using autogenous bone or substitutes**; appropriate pharmaceutical agents; conventional restorative procedures; **straightforward** implant-supported restorative procedures; diagnosing and dealing with complications; monitoring and maintaining implants over time; record-keeping, documentation and audit.

GDC Standards Principle 7 is quoted in the training standards and is quotable in your SOE: - **7.1** provide good quality care based on current evidence and authoritative guidance - **7.2** work within your knowledge, skills, professional competence and abilities - **7.3** update and develop your professional knowledge and skills throughout your working life

> **SOE gold:** when an examiner asks "would you treat this patient?", a strong answer names the complexity level, applies SAC, references GDC Principle 7.2, and states what would have to change for you to take it on (e.g. "if I had a mentor available, or if the patient accepted a delayed protocol").

4. Reference key

Throughout this notebook, references are abbreviated. **Where a topic is best learned from a single source, that source is named after the section heading.** Use this key to know which book to open.

Abbrev.	Full reference	Best used for
Misch	*Misch's Contemporary Implant Dentistry*, Resnik RR (ed.), 4th edn, Elsevier	Bone density classification, biomechanics, treatment planning, anatomy, complications, medical assessment. **The single best "one book" for the written paper.**
Lindhe	*Clinical Periodontology and Implant Dentistry*, Lang NP & Lindhe J (eds), 6th edn, Wiley-Blackwell	Biology of osseointegration, peri-implant tissues, peri-implant disease, ridge alterations after extraction, evidence base. **The authority for anything biological.**
SAC	*The SAC Classification in Implant Dentistry*, 2nd edn, Dawson A, Chen S, Buser D, Cordaro L, Martin W, Belser U (eds), ITI/Quintessence	Case complexity, normative and modifying factors, aesthetic risk. **Learn this book — it is the language of the SOE.**
ITI TG	*ITI Treatment Guide*, vols 1–13, Quintessence	Vol 1 (aesthetic zone), Vol 2 (loading protocols), Vol 3 (posterior), Vol 5 (sinus floor elevation), Vol 7 (ridge augmentation), Vol 8 (biological/hardware complications), Vol 10 (implants in extraction sockets), Vol 11 (digital), Vol 13 (peri-implantitis)
ITI DT	ITI Decision Trees (immediate placement; atrophic posterior maxilla; sinus floor elevation; peri-implant mucositis & peri-implantitis)	Single-page algorithms — ideal for last-week revision and for structuring SOE answers
OIR	*Oral Implantology Review: A Study Guide*	Rapid factual recall, definitions, numbers
CCID	*Clinical Cases in Implant Dentistry*, Karimbux N & Weber H-P (eds), Wiley 2017	Case-based rehearsal for the SOE. Chapters: 1 Examination & Diagnosis; 2 Implant Design; 3 Prosthetic Design; 4 Soft Tissue Management; 5 Ridge Site Preparation; then implant placement, complications, maintenance

Abbrev.	Full reference	Best used for
NID	*Notes in Implant Dentistry*, Chu AKY et al., 1st edn 2021	Condensed evidence summaries; excellent for revision-density
Termeie	*Periodontal Review Q&A*, Termeie DA, Quintessence	Periodontal underpinning, classification, wound healing; strong Q&A format
PCS	*Periodontics: The Complete Summary*	Fast periodontal recall for perio-related SOE stations
Mosea	*MCQs in Implant Dentistry*, Mosea A, 2016	Written-paper practice; answers with short evidence notes
DSMT	*Dental Science, Materials and Technology: MCQ in Implant Dentistry*	Materials, surfaces, metallurgy, cements, impression materials
FGDP	*Training Standards in Implant Dentistry*, FGDP(UK)	Scope of practice, straightforward vs complex, referral, portfolio
App A–E	RCSEd Appendices A (Blueprint), B (Self-declaration), C (Sample SBAs), D (Sample OSCE), E (Sample SOE case)	The definitive scope document

Guidelines you should be able to name in the SOE:

- **IR(ME)R 2017** (and IRR 2017) — justification, optimisation, referrer/practitioner/operator roles
- **FGDP(UK) / Royal College of Radiologists — Selection Criteria for Dental Radiography**
- **SDCEP** — *Drug Prescribing for Dentistry*; *Management of Acute Dental Problems*; *Prevention and Management of Dental Caries*
- **SDCEP / SIGN** — *Management of Dental Patients Taking Anticoagulants or Antiplatelet Drugs*
- **SDCEP** — *Oral Health Management of Patients at Risk of Medication-Related Osteonecrosis of the Jaw (MRONJ)*
- **NICE CG64** — Prophylaxis against infective endocarditis (antibiotic prophylaxis **not routinely recommended**)
- **2017 World Workshop** classification of periodontal and peri-implant diseases and conditions (Caton et al.; Berglundh et al.)
- **EFP S3-level treatment guideline** for periodontitis (2020) and for peri-implant diseases (2023)
- **GDC Standards for the Dental Team**, especially Principles 3 (consent), 4 (records/confidentiality), 7 (competence)
- **Montgomery v Lanarkshire Health Board [2015]** — the legal test for consent
- **HTM 01-05** — decontamination in primary care dental practices

How to read this notebook

This notebook uses a small, consistent set of visual signals. Learn them once and you can find what you need on any page without reading around it.

The coloured boxes

THE CASE

Sets the scene. In the OSCE and SOE sections, this box carries the clinical scenario — the patient, the history, the findings and the radiographs. Read it twice before you read anything else on the page.

CANDIDATE INSTRUCTIONS

Tells you what you are being asked to *do*, in the wording a station would use. Identify the verb — *carry out*, *describe*, *discuss*, *explain to the patient* — because it tells you who you are talking to and what will be marked.

DIRECTION OF THOUGHT

The two minutes of thinking that should happen before you open your mouth: the framework, the sequence, and the trap in this particular scenario. This is the part of the notebook worth re-reading on the morning of the examination.

MODEL ANSWER

A full answer, written to be **spoken aloud**, at the level expected of a general dentist with an interest in implant dentistry. It is not a model of the most sophisticated answer available — it is a model of the safest defensible one, delivered with insight into your own limits.

EXAMINER CHECKLIST

1. The marking points, written as a checklist
2. Rehearse against it, mark yourself honestly, and note which points you dropped
3. The last two or three points are almost always **records, patient instructions and review** — the marks candidates most often leave unclaimed

KEY POINT

Key point. A priority, a rule of thumb, or something to commit to memory verbatim.

The other signals

Signal	Meaning
Q in a navy chip	An examiner's question in a structured oral case. The shaded box beneath it is the answer as you would say it.
Small grey type under a heading	The blueprint outcome that heading maps to, or a cross-reference to another part of the notebook.
A navy number in the question bank	The question number. **In the digital version it is a link** — click it to jump straight to the answer, and click the number again in Part 8 to come back.
Thumb tabs on the right edge	Which part you are in. They step down the page from Part 1 at the top to Part 8 at the foot, so you can find a part with the book closed.

Four ways to use the book

As a course. Work Parts 2 and 3 in order alongside the six-week plan in the introduction, using the question bank block by block as you finish each area.

As a question bank. Work a block of the written paper under timed conditions — 80 seconds per question — mark it from Part 8, then read every explanation in that block, including the ones you answered correctly.

As a rehearsal script. Take one OSCE station or one structured oral case a day and deliver it out loud, against the clock, to a colleague holding the checklist. Reading a model answer silently builds recognition; saying it builds fluency, and only fluency survives the examination.

As a reference. Use the thumb tabs and the contents list at the front of each part. Part 3 (numbers and classifications) and Part 4 (which book, for what) are designed to be looked things up in rather than read through.

PART 1

Examination technique

How to sit each component, the twelve commonest mistakes, and what to do the day before.

IN THIS PART

1.1 The written papers 11

1.2 The OSCE 11

1.3 The structured oral examination 11

1.4 The twelve mistakes that fail candidates 12

1.5 The day before 12

1.6 How you are actually marked — the examiner's side of the table 13

PART 1 — EXAMINATION TECHNIQUE

1.1 The written papers

- ~180 SBAs in 240 minutes = **~80 seconds each**. Do not linger.
- **Answer every question.** There is no negative marking in a standard-set paper.
- Read the **last line of the stem first** if the stem is long — it tells you what is actually being asked.
- Look for the qualifier: *most likely*, *first*, *best*, *least appropriate*, *initial*. These change the answer.
- Watch the **units** — the RCSEd sample question about surface roughness deliberately offers nanometres and micrometres.
- Where two options both look right, ask which is the **safest and most defensible at GDP level**. That is usually the intended answer.
- Flag and move on; return at the end. Do not change an answer without a clear reason.

1.2 The OSCE

- Use the **2 minutes** to build a sequence, not to worry. Write three or four words on your hand-out if allowed.
- **Narrate continuously.** A silent candidate scores nothing on a checklist.
- Work in a fixed order: *indication* → *preparation* → *procedure* → *completion* → *records and warnings*.
- If the task is physical and you are struggling, **keep describing what you are doing and why** — the checklist rewards knowledge as well as execution.
- If you finish early, **stay in the station and add**: post-operative advice, what you would record, what could go wrong, when you would review. These are almost always unclaimed marks.
- The examiner will not help you. Do not interpret silence as disapproval.
- If you make an error, say so and correct it. Recovering visibly is better than pretending.

1.3 The structured oral examination

- **Use the full 10 minutes of preparation.** Write the four domain headings and fill them. Take the notes in.
- **Open with a framed summary**, not a fact.
- **Signpost**: "There are three issues here — first…, second…, third…" Examiners can only give marks for what they can identify.
- **Give the whole option list, including no treatment**, before giving your recommendation.
- Say **SAC**, say **material risk**, say **what you would record**, say **when you would review and what you would measure**. These are the phrases that map onto the mark sheet.
- **Know your limits out loud.** "This is beyond what I would do as a general dentist with a special interest, so I would…" is a scoring answer, not an admission of weakness — but only when justified.
- If you don't know, say so and reason from principles. "I'm not certain of the exact figure, but the principle is…" scores far better than a confident invention.
- If an examiner pushes back, **listen**. Sometimes they are testing whether you will defend a correct position; sometimes they are giving you a lifeline. Reconsider genuinely, then either hold your position with reasons or change it with reasons. Do not simply capitulate, and do not become defensive.
- **Watch the clock in your own answers.** Fifteen minutes across four domains is under four minutes each. Do not spend eight minutes on the history.

1.4 The twelve mistakes that fail candidates

1. Treating the examination as a specialist examination and over-reaching into complex procedures.
2. Never mentioning **referral** — or using it to avoid everything.
3. Giving a treatment plan without giving the **options**, including no treatment.
4. Forgetting **consent** — especially the origin of graft materials, and the maintenance burden.
5. Forgetting **records** — no candidate has ever lost marks for saying what they would record.
6. Forgetting **maintenance and review** — the answer does not end when the crown is fitted.
7. Not recognising **peri-implantitis risk factors** in the history in front of them.
8. Talking about implants before **controlling disease** and establishing prognosis.
9. Failing to be **prosthetically driven** — planning from bone rather than from the wax-up.
10. Reciting a memorised list rather than **interpreting the specific case**.
11. Being silent in an OSCE station.
12. Not answering the question that was actually asked.

1.5 The day before

- Re-read **Part 3** (numbers and classifications) and the four **ITI Decision Trees**.
- Re-read the **four SOE domains** and the four-heading preparation framework.
- Re-read the three quotations at the front of this notebook about the level of the examination.
- Check your **passport** (Regulation 15.1 — it is required at every registration session).
- Sleep. You know more than you think you do.

APPENDIX — Quick self-declaration and eligibility checklist

Confirm before the closing date (Regulation 4, Appendix B):

☐ Primary dental qualification acceptable to Council
☐ **36 months** post-qualification experience (or part-time equivalent within 6 years)
☐ **15 cases** with surgery personally performed or under supervision
☐ **2 cases** with ≥1 year follow-up
☐ **4** single-unit replacements
☐ **5** multiple implants in different quadrants **or** 2 implants supporting a 3-unit bridge
☐ **≥1** edentulous case
☐ **≥5** cases with evidence of minor augmentation
☐ **≥30 hours** implant CPD **with reflections** in the past 18 months
☐ **Quality improvement project** in implant dentistry in the past 18 months
☐ Signed declaration emailed to **implantdentistry@rcsed.ac.uk** by the closing date
☐ Passport (or acceptable photographic ID) ready for every registration session

Failure to submit the signed statement makes you ineligible to sit.

1.6 How you are actually marked — the examiner's side of the table

Everything above is about what you say. This section is about **what happens to it once you have said it**, because the marking instruments are not secret and knowing their shape changes how you answer. The College publishes the components and the blueprint; what follows is the ordinary anatomy of marking a clinical examination of this kind, written so that you can rehearse against it.

The written papers

One mark a question, **no negative marking**, and roughly **80 seconds a question** if you divide the time by the paper. Two consequences follow, and candidates lose marks on both. First, **never leave a blank** — an eliminated-down guess is free. Second, **do not audit your own paper twice**; the second pass changes more right answers to wrong than the reverse. Flag the genuinely uncertain, answer everything, and revisit only the flags.

The OSCE: a checklist, and the "say it out loud" rule

Each station is marked against a **list of explicit points** — in this notebook, ten to a station, which is the usual shape. The examiner is not writing an impression; they are ticking things they **heard or saw**.

> **WATCH OUT**
>
> **The single rule that matters:** the mark is awarded when you **say** the thing, not when the examiner believes you know it. A step performed silently, an instrument used without being named, a force applied without being quantified, a finding noticed but not spoken — none of them scores. This is why the discussion stations, where everything must be verbalised, are harder than they look, and why narrating is not showing off but the mechanism by which you are paid.

Three habits follow directly from a checklist format:

Frame in the first thirty seconds. *"I will check the history and consent, work through X, Y and Z, and finish with what I would record and what I would tell the patient."* You have just told the examiner where each tick is coming from, and they will wait for you rather than prompt.

Give one clause per point, not one paragraph. A checklist gives no extra credit for elaboration, and elaboration is what runs the clock out before the last three points.

Reserve the final thirty seconds for records, patient instructions and review. They sit at the end of nearly every checklist and are the marks most often left unclaimed.

The structured oral: four domains, five marks each

The oral is marked on the four published domains, **five marks in each**, awarded by each examiner independently — twenty marks per examiner per case. The pass standard is conventionally set around a **mean of 3** across the domains and cases, which is worth understanding precisely: you do not have to be brilliant anywhere, but a 1 in one domain has to be bought back by a 5 somewhere else, and 5s are rare.

Mark	What it describes
5	Comprehensive, fluent and prioritised. Volunteers the material risks, the limits of their own competence and what they would record, **without prompting**. Cites evidence or guidance where it genuinely bears on the decision.
4	Sound and safe throughout, well organised, with only minor omissions that do not affect patient safety.
3	**The pass.** Safe and defensible. Reaches the right decision, though **prompting is needed** for some elements. Knows when to refer.
2	Incomplete or poorly prioritised. Gets to a workable answer only with substantial prompting, or omits an important safety element.
1	Serious gaps. Unsafe reasoning, or fails to recognise a case beyond their competence.
0	No useful response, or actively dangerous advice.

Read the difference between **3 and 5** again: it is not knowledge, it is **who raised the point**. The same content volunteered scores where the same content extracted does not. That is the whole argument for the habits this notebook keeps repeating — say the risk before you are asked for it, state your competence limit before the examiner probes it, and finish every answer with what you would record and what you would say to the patient.

What the examiner is doing while you talk

Examiners in a component of this kind work to a recognisable pattern, and each element of it is an opportunity if you see it coming.

They let you open uninterrupted for about a minute, then question briskly. The opening is the only monologue you get. After that, being moved on is normal and is not a sign you were wrong — a candidate allowed to talk at length on their strongest ground has not been tested, so an examiner who cuts you off is running the examination properly.

They prompt — and they record that they prompted. A prompted correct answer is a 3, not a 5. The neutral prompts are recognisable: *"What else?"*, *"Why that one?"*, *"What would change your mind?"*, *"What would you record?"*, *"Where does that sit on the SAC classification?"* When you hear one, you have just been told that something obvious is missing from your answer — usually risk, records, the patient's words, or the referral threshold.

They release information deliberately. A new radiograph, a result, a change in the history part-way through a case is a test of whether you will revise your position out loud. Say what it changes: *"That changes my differential."* Silence there scores nothing.

They take you to the edge of your competence and check that you notice. This is the most important sentence in this section. **Failing to recognise a case that should be referred is the commonest reason a candidate does not pass**, and it outweighs any single factual gap. The examiner is not trying to catch you out by pushing towards block grafting or full-arch work; they are checking that you stop, name the complexity, and refer. Stopping is the answer.

> **KEY POINT**
>
> **Rehearse against the instrument, not just the syllabus.** For the OSCE, work the run sheets in Part 6 out loud against a five-minute timer. For the oral, have a colleague mark you on the four domains using the descriptors above — and ask them to note, for every point you made, whether you volunteered it or they had to ask. The count of prompts is a better predictor of your result than the content of your answers.

PART 2

Core revision notes

Twelve sections mapped to Appendix A — the learning outcomes and blueprint. Read these first, and return to them between question blocks.

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PART 2 — CORE REVISION NOTES

§1 Patient assessment, history and risk

Blueprint: Clinical Interpretation & Reasoning — *History taking*. Tested **W + SCE**.
Key reading: Misch ch. "Medical Evaluation of the Dental Implant Patient"; Lindhe (systemic risk); SAC ch. 2; CCID ch. 1 Case 2.

1.1 The structure of an implant history

The examiners want to hear a **structured, comprehensive** history — not a list of questions but a reasoned sequence in which each element changes what you do next.

Element	What you ask	Why it changes the plan
Presenting complaint	In the patient's own words; how long; what triggered attendance now	Distinguishes function-driven from aesthetics-driven from pain-driven cases; sets the outcome you will be judged against
History of presenting complaint	Why the tooth was lost, when, previous treatment, previous failures	Loss from periodontitis, endodontic failure, fracture or trauma predicts residual bone and infection risk
Medical history	Systematic; **including drugs, allergies, previous radiotherapy, bisphosphonates/antiresorptives, smoking, alcohol, recreational drugs**	Absolute vs relative contraindication; MRONJ risk; healing capacity
Dental history	Attendance pattern, previous perio treatment, compliance with hygiene, previous restorative failures, orthodontics, parafunction	**Compliance is the strongest modifiable predictor of long-term implant success**
Social history	Occupation, smoking (pack-years), alcohol, stress, ability to attend, funding	Cost and time commitment are part of valid consent
Psychological / expectation	What does "success" mean to them? Have they seen images? Any previous dissatisfaction?	Unrealistic expectation is a **contraindication** in its own right
Family/cultural	Cultural attitudes to tooth loss, dietary practices, religious observance affecting appointments/fasting	Blueprint explicitly lists *"psychological, social and cultural factors"*

Exam phrase to have ready: *"I take a structured history under those headings, but I interpret it in the light of the patient's own concerns — my first question is always what they want the treatment to achieve, because that becomes the outcome measure I audit myself against."*

1.2 Medical risk — absolute vs relative

There are very few **absolute** contraindications. Learn the short list; everything else is **relative** (i.e. it modifies risk, timing, and consent, but does not by itself forbid treatment).

Absolute / near-absolute (defer or do not treat)

- **Recent myocardial infarction or CVA** (conventionally defer elective surgery ~6 months; take cardiology advice)
- **Uncontrolled/severe cardiac disease** — unstable angina, uncontrolled arrhythmia, severe valvular disease
- **Profound immunosuppression** or active haematological malignancy; neutropenia; uncontrolled bleeding disorder
- **Active intravenous bisphosphonate / antiresorptive therapy for malignancy**, or established MRONJ
- **Head and neck radiotherapy to the implant site**, particularly **≥50–60 Gy**, and especially within the first year (risk of osteoradionecrosis); requires specialist/hospital care
- **Uncontrolled diabetes** (see below) — defer until controlled
- **Active substance misuse / non-engagement**, or psychiatric illness precluding consent
- **Incomplete skeletal growth** — implants behave as ankylosed teeth and become infra-occluded as the alveolus develops. Do **not** place in growing patients; wait for cessation of growth (typically ~late teens

for females, later for males) confirmed by serial cephalometrics/hand-wrist radiographs or by clinical stability over 6–12 months.

Relative — the ones examiners love

Condition	Effect	Practical management
Smoking	Roughly **2–2.5× increased implant failure**; markedly increased risk of peri-implantitis, marginal bone loss and graft/GBR failure; impairs perfusion and neutrophil function	Cessation advice with very brief intervention; ideally stop ≥1 week pre-op and ≥8 weeks post-op; **document the discussion and that success rates are lower**
Diabetes mellitus	Well-controlled (HbA1c generally < 7%/53 mmol/mol, and certainly < 8%/64) implants perform close to normal; **poorly controlled** impairs healing, increases infection and peri-implantitis	Liaise with GP; obtain a recent **HbA1c**; morning appointments; ensure the patient has eaten and taken medication; consider antibiotic cover in poorly controlled cases and delay loading
Oral bisphosphonates (e.g. alendronate)	**Low MRONJ risk**; SDCEP: manage as "at risk", no drug holiday required routinely	Explain risk, atraumatic technique, primary closure, review healing, document
IV bisphosphonates / denosumab / antiangiogenics for malignancy	**High MRONJ risk**	Elective implants **contraindicated**; refer/liaise with oncology
Anticoagulants / antiplatelets	Bleeding risk	Follow **SDCEP**: for warfarin, check **INR within 24 h (72 h if stable) and treat if INR < 4.0**; for DOACs, consider **delaying/omitting the morning dose** for higher-bleeding-risk procedures after liaison; **do not stop antiplatelets**; use local haemostasis (sutures, packing, oxidised cellulose, tranexamic mouthwash where indicated)
Bleeding disorders	Haemophilia, von Willebrand's	Haemophilia centre liaison; factor cover
Corticosteroids / immunosuppressants	Impaired healing, infection, adrenal suppression	Liaise with physician; consider steroid cover for major procedures; meticulous asepsis
Osteoporosis	Bone *quality* affected but osseointegration generally achievable	Main issue is the **drug**, not the disease
Previous radiotherapy	Hypovascular, hypocellular, hypoxic bone	Specialist referral; hyperbaric oxygen is controversial and not a GDP decision
Sjögren's / xerostomia	Mucosal fragility, caries in remaining teeth	Implants may actually be favourable vs a mucosa-borne denture
Bruxism/parafunction	**Mechanical**, not biological, risk: screw loosening, fracture of screws/abutments/ceramics, overload	Splint, occlusal design, avoid cantilevers, wider/more implants, monolithic materials
Periodontitis (history of)	**Strongest patient-level predictor of peri-implantitis**	Must be **treated and stable** first; enrolled in supportive periodontal therapy
Pregnancy	Elective — defer	Second trimester if urgent; avoid radiographs unless justified

> **Written-paper trap:** "*The most significant risk factor for peri-implantitis is…*" → **a history of periodontitis** (with smoking and poor plaque control close behind). "*The most significant risk factor for early implant failure is…*" → **smoking / poor bone quality / lack of primary stability**, depending on the stem — read carefully whether it asks about *early* (*pre-osseointegration*) or *late* (*biological*) failure.

1.3 Local risk factors

- **Bone volume** — height, width, and the **relationship to vital structures** (§4)
- **Bone quality** — Lekholm & Zarb types I–IV; Misch D1–D4 (§5.4)
- **Soft tissue phenotype** — thin vs thick; width of keratinised mucosa
- **Infection at the site** — periapical, periodontal, residual root, retained cyst
- **Adjacent teeth** — prognosis, angulation, root proximity, restorative status
- **Occlusion and parafunction** — over-eruption, drifting, loss of space, wear facets
- **Smile line and lip dynamics**
- **Access and mouth opening** — a frequently forgotten OSCE/SOE point for second and third molars; you need vertical access for drills and handpieces

- **Oral hygiene and dexterity** — including cleansability of the planned prosthesis

1.4 Psychological, social and cultural factors *(blueprint-explicit)*

- Identify the **"difficult" aesthetic patient**: brings photographs of others' smiles; multiple previous dissatisfactions; focus on a defect invisible to you; unwillingness to accept any compromise; demands guarantees.
- Consider **body dysmorphic features** and, where suspected, decline elective aesthetic treatment and seek advice.
- **Cultural factors** are examinable: dietary and fasting practices affecting appointment timing and post-operative analgesia/antibiotic timing; attitudes to gold/metal display; the meaning of tooth loss; family involvement in decision-making; language and the **use of a professional interpreter (not a family member) for consent**.
- **Capacity**: assess decision-specific capacity; where absent, act in the patient's best interests under the relevant Mental Capacity legislation — but remember that **elective implant treatment for a patient lacking capacity is rarely justifiable**.

1.5 Structuring your answer when asked "what in the history is relevant?"

Use this four-step and you will never be caught blank:

1. **What increases the risk of failure of osseointegration?** (smoking, uncontrolled diabetes, irradiated bone, poor bone quality, infection)
2. **What increases the risk of biological complications later?** (periodontitis history, plaque control, smoking, diabetes, no maintenance, non-cleansable prosthesis, residual cement)
3. **What increases the risk of mechanical complications?** (parafunction, cantilevers, limited restorative space, opposing dentition, previous fractures)
4. **What affects consent and expectation?** (aesthetics, cost, time, occupation, previous experience, psychological factors)

§2 Examination, occlusion and space

Blueprint: Clinical Interpretation & Reasoning — *Patient examination*; the words *occlusion*, *interocclusal and intra-occlusal space*, and *aesthetic concerns* appear repeatedly and are marked in both **W** and **SCE**.
Key reading: Misch (treatment planning and occlusion chapters); CCID ch. 1 Case 4; SAC (prosthodontic classification); NID (occlusion).

2.1 Extra-oral examination

- **Facial form, symmetry, profile and skeletal classification** (I / II / III) — a Class III skeletal base changes implant angulation, prosthetic space and the achievable occlusal scheme, and is a common SOE prompt in edentulous cases
- **Lip support** — is it currently provided by a denture flange? If so, a fixed implant prosthesis **cannot** replicate it, and this must be discussed *before* treatment
- **Smile line / lip dynamics** — low, medium, **high**; high smile line is the single biggest aesthetic risk modifier
- **TMJ and muscles of mastication** — clicking, crepitus, deviation, tenderness; masseteric hypertrophy suggests parafunction
- **Maximal mouth opening** and inter-incisal access
- **Lymph nodes, salivary glands, skin, neurological function** (baseline lip/chin sensation must be recorded **before** any mandibular surgery — a medico-legal essential)

2.2 Intra-oral examination

- **Soft tissues** — mucosal health, oral mucosal lesions, frenal attachments, vestibular depth, floor-of-mouth height
- **Periodontal assessment** — BPE for screening, then **full 6-point pocket charting where BPE $\geq$ 3**, plaque and bleeding scores, mobility, furcations, recession
- **Peri-implant assessment** where implants exist — probing depths, BOP, suppuration, mobility of prosthesis vs implant, radiographic bone level
- **Caries, endodontic status, tooth wear, restorative prognosis of every tooth**
- **Edentulous ridge assessment** — width, height, contour, undercuts, knife-edge morphology, **buccal concavity** (palpate!), keratinised tissue width, ridge relationship
- **Occlusion** — see below

Coordinated approach (blueprint wording). Explain that you *"target the detailed examination from the symptoms and history"*: a patient complaining of a loose crown gets a detailed prosthetic/occlusal assessment; one complaining of discharge gets a detailed peri-implant and radiographic assessment. Don't recite a generic list — say what you are looking for and why.

2.3 Occlusal assessment — the examinable framework

Static: - Incisal relationship and Angle classification; skeletal base - **Overjet and overbite** — deep overbite restricts restorative space and loads anterior implants adversely - **Crossbite** (a crossbite in Q2/Q3 is a classic SOE complication) — determines the buccolingual position of implants and whether the occlusal scheme can be conventional - Centric relation vs maximum intercuspation; presence of a **slide** - **Over-eruption and drifting** of unopposed/adjacent teeth — the classic 46-over-erupted-into-the-space scenario - Wear facets, tooth surface loss, cracks, abfractions

Dynamic: - **Canine guidance vs group function**; working and non-working interferences - **Protrusive guidance**, anterior guidance, presence of an "envelope of function" - **Parafunction** — masseteric hypertrophy, scalloped tongue, linea alba, cervical abfraction, fractured restorations, wear patterns not explained by diet

Vertical dimension: - **OVD** (occlusal vertical dimension) and **RVD** (rest vertical dimension); **freeway space = RVD − OVD**, normally **2–4 mm** - Methods to measure freeway space (frequently asked, see OSCE 15): Willis gauge or two marks (nose/chin) with dividers or a ruler; phonetics ("Emma", "Mississippi" — closest speaking space); swallowing then relaxing; patient at rest, unsupported head, relaxed lips

2.4 Space analysis — the numbers you must know

Mesiodistal space

Requirement	Value
Implant to adjacent **tooth**	≥**1.5 mm** at the bone crest
Implant to adjacent **implant**	≥**3 mm**
Minimum mesiodistal space for a standard (Ø4.1 mm) implant	≈**7 mm**
Minimum for a narrow (Ø3.3 mm) implant	≈**5.5–6 mm**
Two adjacent implants (Ø4.1)	≈**14 mm**

Why: the **saucerisation/lateral bone remodelling** around the implant neck (~1.0–1.5 mm) will consume the interproximal bone; if two remodelling craters overlap, the interproximal crest drops and the **papilla is lost**.

Tarnow's papilla rules (learn these; classic SBA material): - Distance from **contact point to bone crest**: ≤**5 mm** → **papilla present ~100%**; 6 mm → ~56%; **7 mm** → **~27% - Inter-implant** horizontal distance <3 mm → mean crestal bone loss ~1.04 mm; ≥3 mm → ~0.45 mm

Buccolingual space

Requirement	Value
Facial bone thickness after placement	≥**1.5–2 mm** (aesthetic zone: ≥2 mm)
Palatal/lingual bone	≥1 mm
Total ridge width for a Ø4.1 mm implant	≈**6.5–7 mm**

Vertical / restorative space

Restoration	Minimum interocclusal (restorative) space
Single **screw-retained** crown (from implant platform to opposing)	≈**5–6 mm** (4 mm abutment/interface + ~2 mm ceramic occlusal thickness)
Cement-retained crown	≈**7–8 mm** (needs ≥4–5 mm abutment height for retention)
Bar-retained overdenture	≈**12–15 mm**
Locator/stud overdenture	≈**8–10 mm**
Fixed full-arch (hybrid) prosthesis	≈**12–15 mm**

If there is insufficient space: options are enameloplasty/occlusal adjustment or restoration of the over-erupted opposing tooth, orthodontic intrusion, elective devitalisation and crown-lengthening/shortening of the antagonist, increasing the OVD (only within the freeway space and with a Dahl/appliance trial), or **alveoloplasty/crest reduction at surgery** to create prosthetic space. In the edentulous case, alveoloplasty is often the correct answer.

Bone height relative to vital structures

Structure	Safety margin
Inferior alveolar canal	≥**2 mm** above the superior border of the canal
Mental foramen / anterior loop	≥**2 mm** distal, and beware an anterior loop of up to 3–4 mm (sometimes more) — plan ≥**5 mm** anterior to the radiographic foramen if you have no CBCT
Maxillary sinus floor	Leave intact, or elevate; transcrestal SFE where residual height **>6 mm and a flat sinus floor**
Nasal floor	≥1 mm
Incisive (nasopalatine) canal	Do not engage; may require enucleation and grafting (a specialist decision)
Adjacent root	≥1.5 mm

2.5 Aesthetic assessment and aesthetic risk

Aesthetic risk assessment (ERA — ITI): low / medium / high across: medical status; smoking; **patient aesthetic expectations**; **lip line**; gingival biotype/phenotype; shape of tooth crowns; infection at the site; bone level at adjacent teeth; restorative status of neighbouring teeth; **width of the edentulous span**; **soft tissue anatomy**; **bone anatomy of the alveolar crest**.

High-risk features to name aloud in an SOE: high lip line, thin scalloped phenotype, triangular crown form, bone loss on the adjacent tooth, multiple adjacent missing teeth, active infection, and high patient expectation.

> **A classic anterior scenario — a young patient in a public-facing occupation missing a 22:** the triangular crown form of 23 vs rectangular 21, narrow ridge, and the patient's occupation are all aesthetic risk elevators — say them explicitly. **Triangular crowns have a more coronal contact point and a longer papilla to fill, so black-triangle risk is high.**

§3 Investigations

> **Blueprint:** *"Assess the need and justification for additional and appropriate investigations… radiographs, study casts, blood tests… in planning treatment"* and *"the role and importance of diagnostic wax-ups"* — the latter tested in **all three** components.
> **Key reading:** Misch (radiographic evaluation); FGDP/RCR Selection Criteria; ITI TG vol 11 (digital).

3.1 Justifying imaging — IR(ME)R language

Every radiographic answer should contain three words: **justification, optimisation, limitation**.

- **Justification** — the exposure must do more good than harm, for *this* patient, answering a *specific* question you cannot answer another way.
- **Optimisation (ALARP)** — smallest field of view, appropriate resolution, rectangular collimation, correct kV/mA, thyroid protection where appropriate.
- **Limitation** — dose constraints; do not repeat what already exists.

Roles under IR(ME)R: the **referrer** (supplies clinical data), the **IR(ME)R practitioner** (justifies), the **operator** (carries out any practical aspect). In practice a GDP may be all three — say so, and say you are **entitled and trained** (adequate training under the regulations) to act in those roles.

Every image must be reported — a written radiological report of the whole image, not just the region of interest, entered in the notes. This is an easy checklist mark in an OSCE.

3.2 Which image, when

Image	Use in implant dentistry	Limitations
Periapical (paralleling, with holder)	Baseline and follow-up **bone levels**; adjacent tooth pathology; verifying component seating; assessing residual cement	2-D only; no buccolingual information; geometric distortion if not parallel
Bitewings	Caries, restorative status, crestal bone in posterior sextants	Not for implant planning
Panoramic (OPT/DPT)	Screening the whole jaw; **gross vertical bone height**; sinus and canal position; pathology; over-eruption; also used to identify **bruxism-related wear and condylar changes**	Magnification (≈1.2–1.3×, not uniform), **no buccolingual dimension**, ghosting, superimposition
CBCT	**Buccolingual width, ridge morphology, buccal concavity, proximity to IAN/mental loop/sinus, bone density approximation, planning guided surgery, assessing existing implants when 2-D is inconclusive**	Higher dose than 2-D; artefact from metal (beam hardening/streak); **must be justified case-by-case — never "routine"**; the whole volume must be reported
Study casts, mounted	Space analysis, occlusal analysis, arch relationship, diagnostic wax-up, surgical/radiographic guide fabrication	Needs facebow + interocclusal record for anything beyond simple cases
Intraoral scan / digital workflow	Casts, wax-up, guide design, restoration design; superimposition with CBCT for planning	Scan-body accuracy across long spans; needs training
Clinical photographs	Consent, aesthetics, records, medico-legal, monitoring	Consent for images required
Blood tests	HbA1c (diabetes control), FBC, INR, vitamin D, bone profile where indicated; **liaise with the GP rather than requesting blindly**	Not routine
Bone mapping / ridge mapping	Old-fashioned but valid: LA + calibrated probe through the mucosa to plot ridge width	Invasive, crude — CBCT has largely replaced it

When is CBCT justified? When the answer will change what you do: suspected proximity to the IAN or sinus, suspected buccal concavity (especially the posterior mandible with a **lingual undercut**), assessing the volume for augmentation, planning guided surgery, atypical anatomy, and assessing an implant with unexplained symptoms where periapicals are inconclusive.

Radiographic interpretation drill for the SOE. Report in this order and you will not miss anything: (*1*) type of image, date, quality/diagnostic acceptability; (2) the whole image — teeth, restorations, caries, periapical status, bone levels, sinus, canal, condyles, any incidental finding; (*3*) the region of interest — vertical height, mesiodistal width, crestal morphology, angulation, relationship to structures; (*4*) what I still cannot tell (buccolingual width, soft tissue) and what further imaging or investigation I would justify.

3.3 The diagnostic wax-up — the highest-yield single topic

Blueprinted to* **W + OS-P + SCE** *and used as RCSEd's own published sample OSCE station (Appendix D).

Purpose (say all six): 1. Determines whether the **desired tooth position** is achievable — i.e. is the case **prosthetically driven**? 2. Assesses **space** in three dimensions, including interocclusal space and the restorative space at the planned platform level 3. Evaluates the **occlusal scheme** and vertical dimension before committing 4. Communicates with the patient — converted to a **trial/mock-up or provisional** to test appearance and phonetics and to support **valid consent** 5. Becomes the blueprint for the **radiographic guide** and then the **surgical guide** — the mechanism by which planning is transferred to surgery 6. Communicates with the **laboratory** and the rest of the team

Practical steps for requesting one (this is the OSCE checklist): 1. Consent, and take **clinical photographs** 2. **Primary impressions** in stock trays / **intraoral scan**; special tray if needed → **definitive impressions** in an appropriate material 3. **Interocclusal record** at the correct vertical dimension; **facebow** transfer where the occlusal scheme is being altered 4. **Mount** on a semi-adjustable articulator 5. Record **shade, midline, incisal plane, smile line, lip support, occlusal plane** 6. Written **laboratory prescription** — patient identifier, your details, date required, **exactly what you want**: teeth to be waxed, desired tooth form/size/position, whether to alter the vertical dimension and by how much, duplicate in clear acrylic for a **radiographic guide** with radiopaque markers (gutta-percha/barium sulphate 20–30%), request a **vacuum-formed matrix/putty index** for chairside mock-up and for the provisional 7. Specify **return date**, materials, and that the case be **disinfected**; comply with the **Medical Devices Regulations** (custom-made device statement) and record the lab's details 8. On return: **verify** the wax-up in the mouth (mock-up), check with the patient, adjust, re-photograph, then convert to the guide

Model sentence: *"The wax-up is not a laboratory nicety — it is how I decide whether this is a restoratively driven case that I can treat, and it is the physical link between my plan and my drill."*

3.4 Radiographic and surgical guides

Guide type	Description	Use
Radiographic guide	Duplicate of the wax-up in clear acrylic with radiopaque markers (gutta-percha, barium sulphate, metal spheres of known diameter for calibration)	Worn during CBCT/OPT to relate planned tooth position to bone
Non-limiting (tooth-position) guide	Shows planned tooth position only	Simple cases; keeps you restoratively driven
Partially limiting	Indicates entry point and angle (e.g. a drilled channel)	Most GDP-level cases
Completely limiting (fully guided)	Sleeve-guided drills; static guided surgery from merged CBCT + scan	Accuracy at the cost of flexibility; needs sufficient mouth opening; **you must know the concept, not the software detail**
Support	**Tooth-supported** (most accurate) > **mucosa-supported** (needs fixation pins) > **bone-supported**	Partially vs fully edentulous

Accuracy expectations for static guided surgery (approximate, from systematic reviews): mean angular deviation ~2–4°, entry point ~1 mm, apex ~1.3–1.6 mm. **Therefore always keep a safety margin of ≈2 mm from vital structures even with a guide.**

Limitations to state: cost; laboratory/scan time; error accumulates through impression, scan, merge, manufacture and seating; **cannot compensate for a poorly fitting or unstable guide**; restricted irrigation and heat generation within sleeves; limited access posteriorly; and a guide does not remove the need to know the anatomy.

§4 Surgical anatomy relevant to implant placement

> **Blueprint:** *"Apply knowledge of the anatomy of the mandible, maxilla and related structures and how this may influence implant treatment"* (W).
> **Key reading:** Misch (anatomy and avoidance of complications); *The Mental Foramen and Nerve: Clinical and Anatomic Factors*.

4.1 Mandible

Inferior alveolar nerve (IAN) and canal - Runs from the mandibular foramen (lingula) forward, usually lingually then centrally then buccally, to divide into the **mental** and **incisive** branches. - **Safety margin ≥2 mm** above the canal roof. Add the imaging error: OPT magnification and the 2 mm rule together are why **CBCT is justified when the planned length approaches the canal**. - Bifid canals occur in a small percentage — a specific reason to look at the CBCT rather than assume.

Mental foramen and the anterior loop - Typically located **apical to, or between, the premolars**, most often at/near the second premolar apex. - The **anterior loop** — the portion of the nerve that runs anterior to the foramen before turning back — is present in a substantial proportion of patients and commonly measures **1–3 mm, occasionally up to ~5 mm or more**. - **Practical rule:** without CBCT, keep **≥5 mm anterior** to the anterior border of the radiographic mental foramen; with CBCT, measure it. - **Incisive canal branch** continues anteriorly and can cause bleeding and, occasionally, altered sensation — the anterior mandible is *not* automatically "safe".

Lingual anatomy — the genuinely dangerous part - **Submandibular fossa (lingual undercut)** at the posterior mandible: the ridge crest can be lingually inclined over a concavity, so a drill following the apparent long axis perforates the lingual plate. **Always palpate the lingual concavity and consider CBCT.** - **Sublingual artery** (from the lingual artery) and **submental artery** (from the facial artery) anastomose in the anterior floor of mouth and enter through the **lingual foramina** of the genial region. Perforation of the lingual plate anteriorly can cause a rapidly expanding **floor-of-mouth haematoma with airway obstruction** — a genuine emergency (see §11.2). - **Lingual nerve** — at risk from lingual flap reflection and from lingually placed incisions/releasing incisions in the third-molar/retromolar region. - **Mylohyoid ridge and muscle attachment**, **genial tubercles**, **mental spines**.

Other mandibular considerations - **Cortical thickness**: the anterior mandible has dense cortical bone (D1/D2) — excellent primary stability, but a **higher risk of overheating**; use sharp drills, copious irrigation, intermittent pressure, and consider cortical bone tapping/counter-sinking. - **Retromolar and external oblique ridge** — the usual intraoral donor site for particulate autogenous bone at GDP level (a **block** graft from the ramus is a *complex* procedure and out of the examination's scope for planning). - **Buccal shelf, mandibular canal in the atrophic mandible** — in severe atrophy the canal may be at/near the crest, and the mandible is at risk of **iatrogenic fracture**.

4.2 Maxilla

- **Maxillary sinus** — pneumatisation increases after posterior tooth loss; the floor may scallop between roots. **Septa (Underwood's septa)** occur in a significant minority and increase the risk of membrane perforation during sinus elevation.
- **Schneiderian membrane** — normal thickness ~0.8–1 mm; thickened >2 mm suggests sinus pathology; **pre-existing sinus disease and ostium obstruction are contraindications to elevation until treated** (ENT referral).
- **Ostiomeatal complex** — drainage is superior/medial; graft material or blood obstructing the ostium causes sinusitis.
- **Posterior superior alveolar artery** and its **intraosseous anastomosis** with the infraorbital artery — runs in the lateral wall, typically **15–20 mm from the alveolar crest**; the main bleeding risk in a lateral window (out of scope to perform, in scope to know).
- **Nasopalatine (incisive) canal** — behind the central incisors; enlarges with age and resorption; implants must avoid it (fibrous tissue → no integration).
- **Greater palatine foramen/artery** — the donor site limit for connective tissue and free gingival grafts; the **greater palatine neurovascular bundle** lies ~7–17 mm from the CEJ of the palatal teeth depending on vault height; **stay within ~2 mm of the CEJ-adjacent zone and within 5–7 mm apical to the gingival margin** to harvest safely.
- **Nasal floor and anterior nasal spine**.
- **Bone quality** — the posterior maxilla is typically **D3/D4** (type III/IV): thin cortex, sparse trabeculae → lowest primary stability, longest healing, highest failure rate of any site.
- **Buccal plate in the anterior maxilla — thin (<1 mm) in the large majority of sites**, and it is **bundle bone** dependent on the periodontal ligament, so it resorbs after extraction. This single fact underlies ridge preservation, immediate placement risk, and mid-facial recession.

4.3 Ridge changes after extraction *(learn the numbers)*

- Post-extraction resorption is **greater horizontally than vertically**, and **greater on the buccal than the palatal/lingual**.
- Classical figures (Schropp; Tan et al. systematic review): **~50% of ridge width lost at 12 months**, of which **two-thirds occurs in the first 3 months**; horizontal loss ≈ **3.8 mm**, vertical loss ≈ **1.2 mm** at 6 months.
- **Bundle bone** (the ligament-dependent inner socket wall) is lost regardless of technique — **ridge preservation reduces but does not eliminate resorption** (typically halves horizontal loss).
- Consequence: the ridge migrates **palatally/lingually and apically**, which is why late-placed anterior implants often need GBR and why the emergence profile must be built prosthetically.

4.4 Anatomy of the peri-implant tissues

(*Cross-reference §5.3.*)

- Junctional/barrier epithelium ~2 mm + connective tissue ~1–1.5 mm = **biologic width / supracrestal tissue attachment ≈ 3–4 mm** around implants (slightly greater than around teeth).
- **No cementum, no periodontal ligament, no Sharpey's fibre insertion**: connective tissue fibres run **parallel** to the implant surface (circular and longitudinal), not perpendicular.
- Vascular supply is **reduced** — derived from the periosteum/supraperiosteal vessels only, with no PDL plexus. Hence reduced defence, faster progression of inflammation, and **peri-implantitis lesions extend closer to the bone crest** than periodontitis lesions.
- **Keratinised mucosa ≥2 mm** is associated with less plaque, less inflammation, less recession and greater patient comfort during brushing — an increasingly examinable point, and the rationale for a free gingival graft or an apically positioned flap.

§5 Osseointegration, bone biology and healing

> **Blueprint:** *"Demonstrate understanding of the phenomenon of osseointegration and the differences between implant systems"* (W).
> **Key reading:** Lindhe ch. on osseointegration and peri-implant tissues; Misch (bone density, biomechanics); NID.

5.1 Definitions you must be able to quote

Osseointegration (Brånemark, 1977/1985): *"a direct structural and functional connection between ordered living bone and the surface of a load-carrying implant."*

Osseointegration (Albrektsson & Zarb, histological/clinical): *"a process whereby clinically asymptomatic rigid fixation of alloplastic materials is achieved, and maintained, in bone during functional loading."*

Practical clinical definition: an implant that is **immobile**, has **no peri-implant radiolucency**, **no persistent pain, infection, neuropathy or paraesthesia**, and stable bone levels.

Fibrous encapsulation = failure of osseointegration; occurs with excessive micromotion, overheating, infection, or an implant placed into fibrous tissue (e.g. into the incisive canal).

5.2 Albrektsson's six factors for osseointegration *(a classic SBA)*

1. **Implant material** (biocompatibility — commercially pure titanium/Ti alloy, zirconia)
2. **Implant design** (macro-geometry: shape, thread form)
3. **Implant surface** (micro/nano-topography, chemistry)
4. **Status of the host bed** (quality, quantity, health, previous irradiation)
5. **Surgical technique** (atraumatic, minimal heat, adequate irrigation, primary stability)
6. **Loading conditions** (timing and magnitude; avoidance of excessive micromotion)

Critical thresholds: - **Heat: bone necrosis at ~47 °C for 1 minute** (Eriksson & Albrektsson). Hence graded drilling, sharp single-use/limited-use drills, copious external $\pm$ internal irrigation with sterile saline at $\leq$ room temperature, intermittent pressure, low speed for dense bone. - **Micromotion: >50–150 μm** at the interface prevents osseointegration and produces fibrous repair. This is the biological basis of the immediate-loading criteria (**insertion torque >35 Ncm and/or ISQ >70, no direct occlusal contact, no parafunction** — see ITI Decision Tree).

5.3 The healing cascade at the implant interface

1. **Haemostasis and protein adsorption** (seconds–minutes): plasma proteins adsorb onto the surface; the surface chemistry determines which proteins and in what conformation — this is why hydrophilic surfaces (e.g. SLActive) accelerate the early phase.
2. **Inflammatory phase** (days 1–3): platelet degranulation $\rightarrow$ growth factors (PDGF, TGF-β, VEGF); neutrophils then macrophages.
3. **Proliferative phase / osteoconduction** (days 3–14): migration of osteogenic cells along the surface; **woven bone** formation. *Contact osteogenesis* (bone forms **on** the implant surface, de novo, from cells that migrated to it — favoured by roughened surfaces) vs *distance osteogenesis* (bone forms **from** the old bone surface toward the implant — typical of machined surfaces).
4. **Remodelling / maturation** (weeks–months): woven bone replaced by **lamellar bone**, then load-adapted remodelling.

The stability dip. Total stability = **primary (mechanical)** + **secondary (biological)**. Primary stability falls as the bone at the interface is resorbed during remodelling; secondary stability rises as new bone forms. The **trough is at approximately 2–4 weeks** — the period of greatest vulnerability, and the reason "early loading" is defined as $\geq$1 week but <2 months, and the reason a wobble at 3 weeks is ominous.

Peri-implant soft tissue seal (see also §4.4): a hemidesmosomal attachment of the barrier epithelium plus a parallel-fibred connective tissue cuff. It is a **weaker seal** than the periodontium and is **disrupted by repeated abutment disconnection** — the biological argument for the "**one abutment, one time**" concept.

5.4 Bone quality and quantity classifications

Lekholm & Zarb (1985) — bone quality

Type	Description
I	Almost entirely homogeneous compact bone
II	Thick cortical layer surrounding a core of dense trabecular bone
III	Thin cortical layer surrounding dense trabecular bone of favourable strength
IV	Thin cortical layer surrounding a core of **low-density** trabecular bone

Lekholm & Zarb — bone quantity (resorption of the edentulous ridge)

A minimal resorption · **B** moderate residual ridge resorption · **C** advanced resorption, only basal bone remains · **D** some resorption of basal bone · **E** extreme resorption of basal bone

Misch bone density (with approximate Hounsfield units)

Misch	Description	Typical site	HU (approx.)
D1	Dense cortical	Anterior mandible, atrophic	>1250
D2	Porous cortical + coarse trabecular	Anterior mandible, posterior mandible	850–1250
D3	Thin porous cortical + fine trabecular	Anterior maxilla, posterior mandible	350–850
D4	Fine trabecular, minimal cortex	Posterior maxilla	150–350
(D5)	Immature, non-mineralised (grafted)	—	<150

Clinical consequences: - **D1/D2** — excellent primary stability; **heat is the enemy**; consider countersinking/tapping, slow speed, sharp drills; earlier loading possible. - **D3/D4** — poor primary stability; **under-prepare (osteotomy narrower than the implant)**, use tapered/aggressive-thread implants, osteotomes/osseodensification, longer healing (posterior maxilla conventionally **6 months**, or 3 months for modern surfaces in D3), consider more/wider implants, avoid immediate loading. - **D4 has the highest reported failure rate of any bone type.**

5.5 Success and survival criteria

Albrektsson et al. (1986) — success criteria (still the standard exam answer): 1. The individual, unattached implant is **immobile** when tested clinically 2. **No peri-implant radiolucency** on an undistorted radiograph 3. **Vertical bone loss <0.2 mm annually** after the first year of loading 4. **No persistent** pain, discomfort, infection, neuropathies, paraesthesia or violation of the mandibular canal 5. **Success rate of 85% at 5 years and 80% at 10 years** as a minimum

Marginal bone remodelling in the first year: classically $\leq$1.5 mm in year one, then $\leq$0.2 mm/year. Modern platform-switched, bone-level implants often show considerably less.

Distinguish carefully: - **Survival** = the implant is still in the mouth (says nothing about health) - **Success** = survival + defined bone/soft tissue/prosthetic criteria met - **Failure** — **early** (before or at abutment connection; failure to integrate: micromotion, overheating, infection, poor bone, smoking) vs **late** (after successful integration: peri-implantitis, overload, fracture)

Typical figures for the written paper: implant survival ~**95–97% at 5 years** and ~**90–95% at 10 years**; but **technical/biological complication rates are high** — roughly **one-third of patients experience some complication over 5 years**; ceramic chipping, screw loosening and mucositis are the commonest.

§6 Implant materials, design and surfaces

> **Blueprint:** *"Different implant designs and surfaces and how they influence provision of treatment"*; *"different dental materials and their role in implant dentistry, e.g. impression materials, cements, implant component materials"* (W).
> **Key reading:** DSMT (materials MCQ book); Misch; CCID ch. 2.

6.1 Implant body materials

Commercially pure titanium (cpTi), grades 1–4 - Graded by **oxygen, iron, carbon and nitrogen content**: increasing grade = increasing oxygen = increasing strength but decreasing ductility. **Grade 4 is the strongest cpTi** and the most common for implant bodies. - Biocompatibility comes from the spontaneous **titanium dioxide (TiO_2) passivation layer**, a few nanometres thick, which reforms in milliseconds if scratched. Damage to this layer (e.g. by a steel curette) impairs biocompatibility — hence **titanium, plastic or carbon-fibre instruments** for debriding implants. - True titanium **allergy is very rare**; where suspected, **zirconia** is the alternative.

Titanium alloy — Ti-6Al-4V (grade 5) - ~4× the tensile strength of grade 1; used for narrow implants, abutments and screws. Concerns about **vanadium and aluminium ion release** led to **Ti-6Al-7Nb** and to **titanium–zirconium alloys (Roxolid®, TiZr ~13–17% Zr)**, which allow reliably strong **narrow-diameter (≈3.3 mm)** implants.

Zirconia (yttria-stabilised tetragonal zirconia polycrystal, Y-TZP) - White, so avoids grey show-through with thin phenotypes; low plaque affinity; good soft tissue response. - **Transformation toughening**: a crack triggers tetragonal→monoclinic transformation with ~4% volume expansion that closes the crack tip. - **Low-temperature degradation (ageing)** in moist environments is the trade-off; **cannot be bent, welded or easily adjusted**; predominantly **one-piece** designs, which limits prosthetic flexibility and prevents submerged healing. - Evidence base is shorter-term than titanium — **be cautious in the SOE**: "an option with promising but less mature evidence."

Historic/obsolete (know the names only): subperiosteal frames, blade/plate-form implants, transmandibular (staple) implants, ceramic (alumina) implants, hydroxyapatite-**coated** implants (coating delamination/dissolution).

6.2 Surface topography and treatments

Roughness classification (Albrektsson & Wennerberg), by Sa:

Class	Sa	Examples
Smooth	<0.5 μm	polished
Minimally rough	0.5–1.0 μm	machined/turned
Moderately rough	**1.0–2.0 μm**	**SLA, TiUnite, OsseoSpeed, most modern implants**
Rough	>2.0 μm	plasma-sprayed (TPS), HA-coated

Moderately rough (1–2 μm) is optimal: best balance of bone-to-implant contact and rate of integration against plaque accumulation and peri-implantitis risk. (*Note the RCSEd sample SBA: "micro-textured" surfaces have a roughness of **1–5 micrometres** — read the units carefully; the examiners deliberately offer nanometre distractors.*)

How surfaces are made: - **Subtractive**: acid-etching (HCl/H_2SO_4), sandblasting/grit-blasting (Al_2O_3, TiO_2), **SLA = sandblasted, large-grit, acid-etched**, anodisation (TiUnite), laser ablation - **Additive**: titanium plasma spray (TPS), hydroxyapatite coating, calcium-phosphate nanoparticle deposition - **Hydrophilic/chemically modified**: SLActive (stored in isotonic saline), OsseoSpeed (fluoride-modified) — faster early integration, useful in D3/D4 and where early loading is intended

Bone-to-implant contact (BIC) is greater on moderately rough surfaces; do not confuse **BIC** (histological) with **ISQ** (clinical stiffness measure).

6.3 Macro-design

Feature	Options and consequences
Shape	**Parallel-walled (cylindrical)** — simple osteotomy, good in D1/D2; **tapered (root-form)** — better primary stability in soft bone and immediate sockets, allows placement between converging roots and away from concavities
Thread	V-thread (mainly compressive), square/power thread (best for compressive load transfer), buttress/reverse buttress (tensile). Deeper, wider-pitched, more aggressive threads improve stability in **soft** bone; finer threads in dense bone
Thread pitch	Smaller pitch = more threads per unit length = greater surface area, slower insertion
Diameter	Narrow ≈3.0–3.5 mm (lateral incisors, lower incisors, narrow ridges); standard ≈4.0–4.5 mm; wide ≈5.0–6.0 mm (molars, immediate molar sockets, poor bone). **Diameter increases surface area more than length does, and resists bending moments far better**
Length	Standard ≈8–13 mm; **short implants ≤8 mm**; extra-short ≈4–6 mm. Modern evidence: **short implants in the posterior jaws perform comparably to longer implants with augmentation at 3–5 years, with fewer biological complications** — an important, defensible, GDP-level answer
Neck	Machined/polished collar (soft-tissue level, e.g. tissue-level implants) vs roughened to the top (bone level). Microthreads at the neck may reduce crestal bone loss
Platform switching	Abutment narrower than the implant platform → the microgap and inflammatory infiltrate are moved **medially away from the crest** → **reduced crestal bone loss** (typically ~0.3–0.5 mm less)

6.4 Implant–abutment connection

Connection	Features
External hex (original Brånemark)	Long track record; low anti-rotational height; **higher rates of screw loosening**; micro-movement at the joint
Internal hex / internal octagon	Better load distribution, deeper engagement
Internal conical ("Morse taper", 8–16°)	Friction-fit; **best seal, least micro-leakage, least screw loosening, best load distribution**; may exhibit abutment "settling"/sinking
Conical + indexed (conical connection with anti-rotation)	Current standard for single units
Tissue-level vs bone-level	Tissue-level places the microgap supracrestally (advantage: less crestal remodelling, easier cementation control); bone-level allows better aesthetic emergence and is more flexible in the aesthetic zone

The microgap and micro-leakage. Any two-piece implant has a microgap that harbours bacteria; the closer it is to bone, the more crestal remodelling. This drives: platform switching, conical connections, tissue-level designs in non-aesthetic sites, and "one abutment, one time".

6.5 Prosthetic component materials

- **Abutments:** titanium (strongest, versatile), **titanium base + zirconia/lithium disilicate superstructure ("Ti-base hybrid")** — current mainstream, gold (cast/UCLA, largely historic), zirconia (aesthetic; **avoid a direct zirconia-on-titanium implant interface without a Ti-base** because of wear and titanium particle release).
- **Screws:** titanium alloy or gold alloy; gold-coated/DLC-coated screws reduce friction and improve preload. **Preload** = the clamping force generated by tightening; **torque** applied is largely lost to friction. **Settling/embedment relaxation** loses ~2–10% of preload in the first minutes — hence **re-torque after 10 minutes**.
- **Frameworks:** cast cobalt-chromium, milled titanium, milled zirconia, PEEK (provisional/hybrid), fibre-reinforced.
- **Veneering ceramics:** feldspathic, leucite-reinforced, lithium disilicate (e-max), zirconia (monolithic — least chipping), composite/acrylic (repairable, shock-absorbing, but wears).

- **Impression materials: polyvinyl siloxane (addition silicone)** — best elastic recovery and dimensional stability; **polyether** — hydrophilic, very rigid (excellent for splinted open-tray impressions, but difficult to remove with undercuts); *avoid alginate and impression compound for definitive implant impressions*. **Do not use latex gloves with addition silicone** (sulphur inhibits the set).
- **Cements:** see §10.6 — the safest are **radiopaque** ones you can see and remove.

6.6 Differences between implant systems — how to answer

The blueprint asks you to understand *"the differences between implant systems"*. A strong answer covers:

1. **Body**: material (Ti grade, TiZr, zirconia), shape, thread design, surface treatment
2. **Connection**: external/internal/conical, indexing, platform switching
3. **Level**: bone-level vs tissue-level
4. **Prosthetic portfolio**: range of abutments, angulated screw channels, availability of narrow/short options, digital library availability
5. **Evidence and support**: is there **long-term peer-reviewed data (≥5 years)**? Is the manufacturer stable and likely to supply components in 10 years? (*The FGDP standards explicitly require you to satisfy yourself the manufacturer "is sufficiently stable and of adequate stature to be able to provide component parts for the foreseeable future."*)
6. **Practical**: cost, training and support, compatibility with your laboratory, availability of spare/rescue components

> **SOE answer to "why do you use your system?"** — *"Because it has long-term published data, a full prosthetic range including narrow and short implants, a stable manufacturer who will still supply parts in ten years, and my laboratory and my mentor both work with it. If a patient of mine moves practice, another clinician can service it."* That answer scores in domain 2 and domain 4 simultaneously.

§7 Treatment planning, case complexity and consent

Blueprint: the whole of *Clinical decision making and treatment planning including consent* (W + SCE, some OS-P). This is the largest single block of marks in the SOE.
Key reading: SAC Classification (learn thoroughly); Misch (treatment planning); ITI TG vol 1 & 3; CCID.

7.1 The treatment planning sequence

Present it as a **phased plan** — examiners mark for *sequencing*:

Phase	Content
1. Emergency/relief of pain	Extirpation, drainage, extraction of hopeless teeth, temporary restoration
2. Disease control (initial/hygiene phase)	OHI, smoking cessation, diet, caries control, non-surgical periodontal therapy, endodontics, provisional restorations, **removal of hopeless teeth**
3. Re-evaluation	Re-assess periodontal response, plaque and bleeding scores, prognosis of each tooth. **Do not proceed to implants until disease is controlled and the patient has demonstrated compliance**
4. Definitive/reconstructive	Implant placement ± augmentation; definitive restorations; orthodontics where needed
5. Maintenance	Individualised recall, peri-implant monitoring, radiographic baseline and follow-up

Prognosis of the remaining dentition must be settled before implants are planned — the blueprint says: *"arrive at a prognosis of the remaining teeth and the impact of these on the treatment options."* A questionable tooth adjacent to a planned implant is a planning decision, not an afterthought: either treat it to a good prognosis, or extract it and plan the space as a unit.

7.2 Options for tooth replacement — always give all of them

For **any** missing tooth, be able to give the full option list with pros and cons in 60 seconds:

Option	Advantages	Disadvantages
Do nothing / accept the space (with monitoring)	No cost, no biological price; often appropriate for a shortened dental arch	Drifting, over-eruption, aesthetics, function
Resin-bonded bridge (RBB), cantilever	Minimally invasive, no LA often needed, reversible-ish, excellent survival for single anterior units (~80–90% at 10 yrs for cantilever designs), cheap, quick	Debonding, metal show-through, limited in heavy occlusion/parafunction
Conventional bridge	Fixed, immediate, restores contour	Destructive to abutments (biological cost), risk of abutment pulp necrosis/caries; needs sound abutments
Removable partial denture	Cheap, replaces soft tissue, non-surgical, can add teeth later	Plaque retention, comfort, function, patient acceptance, caries/perio risk
Implant-supported crown/bridge	Preserves adjacent teeth, maintains bone, high survival, independent	Surgery, cost, time, biological/mechanical complications, maintenance, contraindications
Orthodontic space closure	Biological, no prosthesis	Time, cost, may not be feasible; changes occlusion
Autotransplantation	Biological, maintains alveolus, grows with the patient	Case-specific (typically premolar to incisor site in a young patient); technique-sensitive

Examiner's favourite: *"Why not just place an implant?"* — the marks are in the reasoning: age and growth status, cost, medical status, the patient's own priorities, the condition of adjacent teeth (an unrestored virgin tooth strongly favours an implant or a minimal-prep RBB), the number of adjacent missing teeth, and the risk profile.

7.3 The SAC classification — the language of the examination

SAC = Straightforward / Advanced / Complex, applied *separately* to the **surgical** and **prosthodontic** aspects of a case (ITI/Dawson & Chen).

How it works: start from the **normative** classification for the generic case type, then apply **modifying factors** which can raise (or occasionally lower) the classification.

General modifiers (apply to all cases): - Contraindications (medical/absolute) - Smoking - Growth considerations - **Patient's aesthetic expectations** - Compliance/oral hygiene - Adjacent teeth periodontal status

Site/surgical modifiers: bone volume (adequate / deficient horizontally / deficient vertically), anatomical risk (proximity to IAN, sinus, adjacent roots), aesthetic risk zone, soft tissue quality and quantity, complexity of the augmentation required, need for provisionalisation, extraction socket status, infection.

Prosthodontic modifiers: occlusal scheme and parafunction, interarch space, span, need to alter vertical dimension, aesthetic demands, provisionalisation, loading protocol, need for hard/soft tissue augmentation for prosthetic reasons.

Typical normative classifications (learn these examples):

Situation	Surgical	Prosthodontic
Single posterior tooth, adequate bone, adequate space	**S**	**S**
Single anterior tooth, adequate bone, low lip line, thick phenotype	**S–A**	**S–A**
Single anterior tooth, **high lip line, thin phenotype, deficient bone**	**C**	**C**
Immediate placement + immediate provisional in the anterior maxilla	**A–C**	**A–C**
Extended edentulous space in the aesthetic zone	**A–C**	**A–C**
Edentulous mandible, 2-implant overdenture	**S–A**	**S–A**
Edentulous maxilla, fixed full-arch	**C**	**C**
Any case requiring **major augmentation** (block graft, lateral sinus lift, vertical augmentation)	**C**	—

> **Use it out loud:** *"On the SAC classification I would grade this as surgically Advanced-to-Complex and prosthodontically Advanced, because of the high lip line, the thin phenotype and the need for simultaneous contour augmentation. That places it beyond what I would undertake as a general dentist with an interest in implants without mentoring, so I would either refer or co-manage."* — that single sentence hits blueprint items on complexity, referral and self-limitation.

7.4 Referral — when and how

Refer (or co-manage) when: - Major augmentation is needed (block graft, **lateral window sinus elevation**, vertical ridge augmentation, distraction) - Full-arch fixed rehabilitation, or an altered vertical dimension across the whole arch - High aesthetic risk you cannot mitigate (high lip line + thin phenotype + bone loss on adjacents) - Complex medical history (antiresorptives for malignancy, irradiated jaws, bleeding disorders, immunosuppression) - Nerve proximity that cannot be managed with a safe margin - Management of a failing implant beyond your competence, or nerve injury - Patient factors: severe anxiety needing sedation/GA you cannot provide; psychological concerns - **Any time the case exceeds your training, experience or equipment (GDC Principle 7.2)**

A good referral letter contains: patient identifiers; your details; the presenting complaint and the patient's expectation; relevant medical, dental and social history; your clinical findings and diagnoses; investigations already done (**with the radiographs/CBCT attached** so they are not repeated — an IR(ME)R point); what you have already discussed and consented; **precisely what you are asking for** (opinion / treatment / shared care); who will provide the restorative phase and the maintenance; and urgency.

7.5 Consent — the second-biggest source of SOE marks

Valid consent requires: capacity + information + voluntariness, and it is a **process, not a signature**.

Montgomery v Lanarkshire (2015) replaced the Bolam test for information disclosure. The clinician must take reasonable care to ensure the patient is aware of **any material risks** involved in the recommended treatment and of **any reasonable alternative or variant treatments**. A risk is *material* if **a reasonable person in the patient's position would attach significance to it, or if the clinician is or should reasonably be aware that this particular patient would**.

Application: a 1% risk of altered lip sensation may be immaterial to one patient and decisive to a saxophonist or a professional actor. **You must ask what matters to them.**

Content of implant consent (checklist — an easy OSCE/SOE structure): 1. **Diagnosis** and why treatment is being proposed 2. **All reasonable options including no treatment**, with their risks, benefits, prognosis and costs 3. **The proposed procedure**, stage by stage, including number of visits, healing times and provisional arrangements 4. **Risks — general surgical:** pain, swelling, bruising, bleeding, infection, wound dehiscence, need for further surgery 5. **Risks — site-specific:** altered/lost sensation of lip, chin, tongue (**temporary and permanent**, quote your figures), damage to adjacent teeth/roots, sinus perforation and sinusitis, oro-antral communication, mandibular fracture (atrophic mandible), floor-of-mouth haemorrhage 6. **Risks — failure:** failure to integrate (~2–5%), late failure, need for removal and replacement, possibility that the site cannot be re-implanted 7. **Risks — aesthetic:** recession, grey show-through, black triangles, mismatch of shade/contour, papilla loss, **and that aesthetics can deteriorate over years as the face changes** 8. **Risks — biological:** peri-implant mucositis and peri-implantitis, the requirement for lifelong maintenance 9. **Risks — mechanical:** screw loosening, ceramic chipping, screw or abutment fracture, implant fracture, wear of overdenture attachments; **the need for replacement of components at intervals** 10. **Grafting materials** — the blueprint explicitly requires consent for *"the use of grafting materials and barriers"*: **name the origin** (autogenous, allograft — human donor, xenograft — usually **bovine or porcine**, alloplast — synthetic) and discuss religious, cultural and personal objections, and the theoretical (extremely low) risk of disease transmission 11. **Costs**, including maintenance and future component replacement, and what happens if it fails 12. **Alternatives if it fails**, and who is responsible for aftercare 13. **Time**: total treatment duration, and that they must attend maintenance 14. **Written information** given, **cooling-off period**, opportunity to ask questions, and **check understanding by asking them to summarise back** 15. **Record it**: what was discussed, what was given, what the patient said, and the date; re-confirm consent at each surgical stage

> **Quotable:** *"I check understanding by asking the patient to tell me, in their own words, what we have agreed and what could go wrong. If they can't, I haven't consented them — I've just talked at them."*

7.6 Drug therapy and safe prescribing

> **Blueprint:** *"Prescribe drugs including analgesics, local anaesthetics and antibiotics and justify their use"*; *"apply national guidelines"*; *"drug interactions"* and *"suggest appropriate management for drug interactions"* — the last of these is blueprinted to **W + OS-P + SCE**.

Local anaesthetics (know the table — an OSCE station in itself)

Agent	Typical formulation	Max dose (healthy adult)	Notes
Lidocaine (lignocaine) 2% with adrenaline 1:80,000	2.2 mL cartridge = 44 mg lidocaine + 27.5 μg adrenaline	**4.4 mg/kg, max 300 mg** (≈7 cartridges for a 70 kg adult)	Workhorse; amide
Articaine 4% with adrenaline 1:100,000 or 1:200,000	2.2 mL = 88 mg	**7 mg/kg, max ~500 mg**	Superior bone diffusion → good **buccal infiltration in the mandible**; the debated association with **paraesthesia after IDB** means many clinicians avoid it for blocks; not for children <4
Prilocaine 3% with felypressin 0.03 IU/mL	2.2 mL = 66 mg	**6 mg/kg, max 400 mg**	Useful when adrenaline is undesirable; **methaemoglobinaemia** in high dose; avoid felypressin in pregnancy (oxytocic effect)

Agent	Typical formulation	Max dose (healthy adult)	Notes
Mepivacaine 3% plain	2.2 mL = 66 mg	**4.4 mg/kg, max 300 mg**	Short procedures, adrenaline-free
Bupivacaine 0.5% with adrenaline	long-acting	2 mg/kg, max 150 mg	Post-operative analgesia after longer surgery; slow onset

Techniques: infiltration (buccal/palatal/lingual), inferior alveolar nerve block, long buccal, mental/incisive block, greater palatine and nasopalatine blocks (for palatal harvesting), intraligamentary and intraosseous as supplements. **For implant surgery in the posterior mandible, some clinicians deliberately avoid an IDB** so that the patient can report an electric-shock sensation if the drill approaches the nerve — a genuinely examinable point; the counter-argument is inadequate anaesthesia. Know both sides.

Side effects/complications: failure, intravascular injection (palpitations, tachycardia), haematoma, trismus, facial nerve palsy (misplaced IDB into the parotid), needle fracture, prolonged anaesthesia/paraesthesia, allergy (**true amide allergy is rare — most "allergy" is to the preservative or is a vasovagal/adrenaline response**), methaemoglobinaemia (prilocaine, benzocaine), toxicity (perioral tingling, tinnitus, drowsiness, convulsions, cardiorespiratory depression).

Drug interactions relevant to implant dentistry *(blueprint requires you to manage them)*

Interaction	Consequence	Management
Adrenaline + non-selective β-blockers (propranolol)	Unopposed $\alpha \rightarrow$ hypertensive crisis, reflex bradycardia	Limit adrenaline dose, aspirate, consider prilocaine/felypressin or plain agent
Adrenaline + tricyclic antidepressants	Potentiated pressor response	Limit dose; avoid adrenaline-containing cord/retraction agents
Adrenaline + cocaine/amfetamines	Severe hypertension, arrhythmia	Do not treat; defer
Warfarin + metronidazole, erythromycin/clarithromycin, fluconazole/miconazole (incl. topical gel)	↑ INR, bleeding	Avoid; use alternative; check INR
Warfarin + NSAIDs / aspirin	↑ bleeding, GI risk	Use paracetamol ± codeine
Statins (simvastatin/atorvastatin) + clarithromycin/erythromycin/fluconazole	Rhabdomyolysis	Avoid macrolide/azole, or withhold statin on specialist advice
Methotrexate + NSAIDs	↑ methotrexate toxicity	Avoid NSAIDs
Lithium + NSAIDs	↑ lithium toxicity	Avoid NSAIDs
ACE inhibitors / diuretics + NSAIDs	Renal impairment, ↑BP	Avoid or limit
Amoxicillin + allopurinol	Rash	Note
Metronidazole + alcohol	Disulfiram-like reaction	Warn explicitly and record
Combined oral contraceptive + antibiotics	(Only enzyme-inducing antibiotics such as rifampicin genuinely reduce efficacy; **routine advice changed** — but advise about diarrhoea/vomiting)	Give accurate, current advice
Bisphosphonates/denosumab	MRONJ	See SDCEP; assess risk, atraumatic surgery, review
Warfarin / DOAC	Bleeding	See SDCEP anticoagulant guidance (§1.2)
Clopidogrel/DAPT	Bleeding	Do **not** stop; local measures
Immunosuppressants, systemic steroids	Infection, adrenal crisis	Liaise, consider cover

Antibiotics

- **Prophylaxis for implant placement:** the evidence (Cochrane) supports a **single pre-operative dose of amoxicillin 2–3 g orally, 1 hour before surgery** to reduce early implant failure; **NNT ≈ 25**. Post-operative courses are **not** supported for straightforward placement.
- **When to consider a post-operative course:** extensive augmentation/GBR with membranes, sinus floor elevation, immunocompromise, established infection. Justify it — do not give it reflexively.
- **Infective endocarditis: NICE CG64 — antibiotic prophylaxis is not routinely recommended** for dental procedures. Consult the cardiologist for patients at the highest risk, and record the discussion.
- **Antimicrobial stewardship**: right drug, right dose, right duration, documented indication; penicillin allergy → clindamycin or clarithromycin (check local guidance and SDCEP).

Analgesia

- First line **ibuprofen 400 mg qds** (with food; avoid in asthma sensitive to NSAIDs, peptic ulceration, renal impairment, anticoagulation, pregnancy 3rd trimester) ± **paracetamol 1 g qds** — combined is superior to either alone.
- Codeine has poor evidence in dental pain and significant side effects; use sparingly.
- **Pre-emptive analgesia** (dose before the LA wears off) and post-operative ice, head elevation and soft diet.
- Consider a short course of **dexamethasone** to reduce swelling after extensive surgery (know it exists; justify it).

Sedation *(blueprint: "awareness of the different types of sedation available and the associated regulation")*

- **Inhalation sedation (nitrous oxide/oxygen)** — anxiolysis, rapid recovery, patient can drive home after recovery; needs scavenging.
- **Intravenous conscious sedation (midazolam)** — titrated; **flumazenil** must be immediately available; escort required; the patient must remain **conscious, able to maintain their own airway and respond to verbal command**.
- **Oral/transmucosal sedation** — less controllable.
- **General anaesthesia** — hospital setting only.
- **Regulation:** the *Standards for Conscious Sedation in the Provision of Dental Care* (IACSD) — training, team, equipment, monitoring (pulse oximetry, BP), immediate-life-support training for the whole team, audit and clinical governance. **GA in primary dental care is not permitted.**

§8 Surgical placement — decision making and procedural knowledge

Blueprint: *Decision making in relation to treatment procedures and basic procedural knowledge/skills — surgical placement, handling of tissues and complications.* **Tested in all three components.** Flap design, atraumatic elevation, site preparation, one/two stage, suturing and asepsis all carry the W + OS-P + SCE triple marking.
Key reading: Misch (surgical protocols); ITI TG vols 1, 3, 10; Lindhe; CCID chs 5–7.

8.1 Infection control and aseptic technique

(*Blueprinted W + OS-P + SCE — treat it as guaranteed content.*)

- **Environment:** designated clean surgery, cleaned and disinfected surfaces, minimal traffic, doors closed, no unnecessary equipment on the field.
- **Sterilisation and decontamination:** compliant with **HTM 01-05** (or local equivalent) — instruments cleaned (ultrasonic/washer-disinfector), inspected, **vacuum autoclave (type B) for hollow/lumened instruments**, correctly packaged, dated, stored, and the cycle records retained.
- **Personnel:** surgical scrub (2–5 min for the first case), sterile gown, sterile gloves (double-gloving optional), mask, cap, eye protection.
- **Patient:** pre-operative **chlorhexidine 0.2% mouthrinse for 1 minute**, perioral skin disinfection, sterile drapes.
- **Field:** sterile handpiece and tubing, **sterile saline irrigation**, single-use or validated drills, implant components handled with sterile instruments and **not touched by gloved fingers** or non-titanium instruments (protect the oxide layer).
- **Waste and sharps** handled correctly; needle re-sheathed with a device; instrument count.
- **Documentation:** batch/lot numbers and expiry of the implant, abutment, graft material and membrane recorded in the notes and on a **patient implant card** — this is both a governance requirement and a medical-device traceability requirement.

KEY POINT

See also Set-piece 3 at the start of Part 7, which gives a full table of incision and flap designs by clinical situation, the atraumatic elevation script, the periosteal releasing incision technique, the matching closure, and the complications attributable to flap design — written as a spoken SOE answer.

8.2 Flap design

Principles (say them in this order): 1. **Access** adequate for the procedure — including for a bur, a membrane, or a graft 2. **Base wider than the apex** to preserve blood supply 3. Incisions over **sound bone**, not over the defect or the graft, and closure **not over a membrane edge** 4. Avoid **vital structures** (mental nerve, lingual nerve, greater palatine artery) 5. **Preserve papillae** where aesthetics matter 6. **Tension-free closure** — the single biggest determinant of graft success 7. **Full thickness (mucoperiosteal)** where you need bone access; split thickness where you are repositioning soft tissue

Design options:

Design	Use
Crestal (mid-crestal) incision, no releasing incisions	Straightforward placement with adequate keratinised tissue; least morbidity
Crestal with 1 or 2 vertical relieving incisions	Where more access or coronal advancement is required; place reliefs at the **line angles**, not over the papilla or over the graft
Palatally/lingually displaced crestal incision	To move keratinised tissue buccally at second-stage or to gain buccal tissue
Buccally displaced	Rarely — to bring keratinised tissue palatally
Papilla-sparing (papilla-preserving)	Aesthetic zone; avoids papilla recession
Envelope (sulcular) flap	Small elevations; no vertical incisions

Design	Use
Flapless	Adequate bone width, sufficient keratinised tissue, guided placement; **less morbidity, faster healing, but no visualisation of the bone — only for straightforward cases with CBCT/guide**
Apically positioned flap	To increase keratinised tissue at second-stage, or in resective peri-implantitis surgery
Roll flap / pouch	Small buccal contour augmentation at second stage

Atraumatic elevation (a blueprint phrase — use the exact vocabulary): sharp incision to bone in one firm stroke; a **sharp periosteal elevator** working *against bone, not against tissue*; elevate from a corner or from the vertical incision; keep the flap **fully thick and undisturbed**; **retract with a broad retractor resting on bone**, never crushing the flap edge; keep the flap **moist with saline gauze**; minimise the duration of retraction; handle with tissue forceps at the base, not the margin.

Periosteal releasing incision — to advance the flap for tension-free primary closure over a graft: score the periosteum horizontally at the base of the buccal flap with a sharp blade held nearly parallel to the flap, then blunt-dissect. Risk: bleeding, flap perforation, mental nerve injury if too deep in the premolar region.

8.3 Site preparation — the principles

1. **Determine the position first** from the guide/wax-up, not from the bone.
2. **Sequential (graded) drilling** from small to final diameter, **at 800–1500 rpm** (system-dependent), **with copious sterile irrigation**, **intermittent ("pumping") pressure**, **sharp drills** with a controlled number of uses.
3. Maintain **<47 °C**: heat generation increases with drill bluntness, drill diameter, speed, pressure, depth, dense cortical bone and inadequate irrigation.
4. **Direction indicators / parallel pins** after the pilot drill: check mesiodistal and buccolingual angulation, parallelism with adjacent implants/teeth, and take a **periapical radiograph with the pin in situ** if the canal or an adjacent root is close.
5. **Under-preparation** in D3/D4 (skip the final drill, or use osteotomes/osseodensification) to raise insertion torque; **over-preparation/tapping/counter-sinking** in D1 to avoid excessive torque and pressure necrosis.
6. **Irrigate and inspect the osteotomy walls** — a probe or depth gauge should meet bone in all directions; perforation must be detected before, not after, the implant is placed.
7. **Insert the implant** at low speed (≈15–30 rpm) with the driver, then finish with a hand ratchet, monitoring **insertion torque**.

8.4 Three-dimensional implant position — the rules that generate SBAs

Buser's "comfort and danger zones" for the anterior maxilla:

Dimension	Rule
Mesiodistal	≥1.5 mm from an adjacent tooth; ≥3 mm from an adjacent implant; centre the implant in the space
Orofacial (buccopalatal)	The implant shoulder should be ≈**1–2 mm palatal to the point of emergence** of the adjacent teeth (the "comfort zone" is palatal); leaves ≥2 mm of buccal bone
Coronoapical	Shoulder ≈**3 mm apical to the planned mid-facial gingival margin** (≈1 mm apical to the CEJ of the adjacent teeth), to allow an emergence profile
Angulation	Long axis directed through the cingulum of the planned crown (anterior) or the central fossa (posterior); **screw access should emerge palatally/occlusally**

Insertion torque: typically **25–45 Ncm**. **>35 Ncm** is the conventional threshold for immediate restoration/loading. Torque **>50 Ncm** risks compression necrosis of the crest in dense bone.

Implant stability quotient (ISQ) by resonance frequency analysis (Osstell): scale **1–100**; **>70 = high stability** (immediate loading candidate); **60–69 = medium**; **<60 = low** (do not load early). A **falling** ISQ over successive measurements is the strongest early warning of failure.

8.5 One-stage vs two-stage; and the timing protocols

One-stage (non-submerged) vs two-stage (submerged) — *blueprinted W + OS-P + SCE*

	One-stage (healing abutment placed at surgery)	Two-stage (cover screw, primary closure, second-stage surgery later)
Indications	Good primary stability, adequate bone and keratinised tissue, no simultaneous graft or only minor GBR, non-aesthetic or straightforward aesthetic case, tissue-level implants	Poor primary stability, simultaneous augmentation requiring protection, **membrane placed**, thin/compromised soft tissues, need to graft soft tissue at second stage, patient wearing a mucosa-borne denture over the site
Advantages	One surgical episode, no second-stage discomfort, no repeated disruption of the mucosal seal, allows earlier soft tissue maturation, cheaper	Protects the graft/membrane from exposure and infection, protects from micromotion under a denture, allows soft-tissue manipulation at uncovering
Disadvantages	Exposure risk for the graft; micromotion under a denture	Second procedure, longer treatment, extra cost, disrupts the seal

Second-stage (uncovering) options: simple crestal incision or tissue punch (fastest, but removes keratinised tissue — avoid where KM is scarce); **apically positioned flap** or **roll flap** to gain buccal keratinised tissue/contour; connective tissue graft simultaneously if the phenotype is thin.

> KEY POINT
>
> **See also Set-piece 1 at the start of Part 7**, which separates placement timing from loading protocol, gives the advantages and disadvantages of each of Types 1–4, and reproduces the combination grid with its ITI evidence grading.

Placement timing after extraction (Hämmerle/ITI consensus — memorise)

Type	Timing	Situation
Type 1 — Immediate	Same day as extraction	Socket walls intact, ≥**1 mm facial wall**, no infection, primary stability achievable apically/palatally, ideal 3-D position achievable, thick phenotype, low aesthetic risk
Type 2 — Early with soft tissue healing	**4–8 weeks**	Soft tissue closure achieved, infection resolved; the commonest compromise in the aesthetic zone
Type 3 — Early with partial bone healing	**12–16 weeks**	Significant socket healing; useful where more bone fill is needed
Type 4 — Late	≥**6 months**	Fully healed ridge; often requires GBR because of resorption

Loading protocols (ITI/Gallucci consensus — memorise)

Protocol	Timing
Immediate restoration	Restoration fitted within 1 week, **out of occlusion**
Immediate loading (Type A)	Restoration **in occlusion** within 1 week
Early loading (Type B)	Between **1 week and 2 months**
Conventional loading (Type C)	**>2 months** (classically 3 months mandible, 6 months maxilla)

Criteria for immediate loading (all must be met): 1. **Insertion torque >35 Ncm** or **ISQ >70** 2. **No direct occlusal contacts** (immediate restoration) / controlled contacts 3. **No parafunction** 4. Good bone quality and quantity, adequate primary stability, no simultaneous major graft 5. Patient compliance with a soft diet

Evidence grading used in the ITI decision tree — worth quoting: **SCV** (scientifically and clinically validated) > **CWD** (clinically well documented) > **CD** (clinically documented) > **CID** (clinically insufficiently documented). *Conventional loading of a late-placed implant is SCV; immediate placement with immediate*

loading is only CD, and immediate placement in a compromised socket with early loading is CID. This is exactly the sort of evidence-grading answer that scores in SOE domain 2.

8.6 Immediate placement — the details that separate candidates

Advantages: fewer surgical episodes, shorter overall time, patient satisfaction, uses the socket as a guide, may preserve soft tissue architecture. **Disadvantages/risks: mid-facial recession** (the principal risk; higher with thin phenotype and a buccally positioned implant), unpredictable bone fill, difficulty achieving ideal position (the drill skids off the palatal wall), reduced primary stability, cannot fully prevent buccal bundle-bone resorption, **higher aesthetic complication rate**, technique-sensitive.

Technique points: - **Atraumatic extraction** — periotomes, luxators, physics forceps, sectioning of multi-rooted teeth; **preserve the buccal plate at all costs** - **Debride the socket thoroughly** and confirm the buccal wall is intact by probing - Engage the **palatal wall and 3–5 mm of bone apical to the socket** for primary stability; start the osteotomy on the palatal wall, not in the socket centre - Position the implant **palatally**, leaving a **≥2 mm horizontal (jumping) gap** buccally - **Graft the gap** with a slowly resorbing xenograft (deproteinised bovine bone mineral) — reduces horizontal ridge change - Consider a **connective tissue graft** to thicken the phenotype - Manage the soft tissue: immediate provisional (screw-retained, out of occlusion) shapes the emergence; or a custom healing abutment; or a socket-seal/free gingival "punch" graft - **Do not place immediately into a site with acute infection/suppuration**; chronic apical granuloma can be managed with thorough debridement but is higher risk

8.7 Suturing — materials and techniques *(blueprinted W + OS-P + SCE)*

Materials

Suture	Resorbable?	Filament	Notes
Silk (3/0, 4/0)	No	Braided (multifilament)	Cheap, excellent handling and knot security, **but wicks bacteria** — avoid over grafts; remove at 7 days
Polyamide/nylon (Ethilon)	No	Mono	Low tissue reactivity, poor knot security (needs more throws), stiff
ePTFE (Gore-Tex, Cytoplast)	No	Mono	**Best choice over GBR/membranes** — minimal plaque, minimal wicking, excellent tissue response
Polypropylene (Prolene)	No	Mono	Inert, slippery
Polyglactin 910 (Vicryl)	Yes (~56–70 days mass absorption; tensile strength ~2–3 weeks)	Braided	Good handling; **Vicryl Rapide** loses strength in ~7–10 days
Polyglycolic acid	Yes	Braided	Similar
Poliglecaprone (Monocryl)	Yes	Mono	Low reactivity, ~1–2 weeks strength
Catgut / chromic gut	Yes	—	Largely obsolete; high tissue reaction

Sizes: implant surgery generally **4/0 or 5/0** (5/0–6/0 in the aesthetic zone and for microsurgical/CTG work); 3/0 for flap closure under tension or in the posterior mandible. **Reverse-cutting needle** for mucosa; 3/8 circle for accessible sites, 1/2 circle for posterior/limited access.

Techniques

Technique	Use
Simple interrupted	Universal; failure of one does not lose the closure
Interrupted mattress — horizontal	Everts wound edges; distributes tension; good over grafts
Interrupted mattress — vertical	Everts and adapts deep and superficial layers; used for tension-free primary closure over GBR
Continuous (running)	Speed over long spans; one failure loses all
Continuous locking (blanket)	Better tension control over long flaps

Technique	Use
Sling / suspensory	Suspends the flap around a tooth or healing abutment without pulling on the opposite flap — ideal for papilla-preserving and apically positioned flaps
Figure-of-eight	Over extraction sockets to retain graft material
Periosteal suture	Anchors and stabilises a repositioned flap

Rules to state: pass from **mobile to fixed** tissue; from **thin to thick**; enter perpendicular, **3 mm from the margin**, taking equal bites of each side; place the knot **away from the incision line**; **approximate, do not strangulate** (tension causes ischaemic necrosis and dehiscence — the leading cause of membrane exposure); **remove non-resorbable sutures at 7–10 days** (or 10–14 days over augmentation).

8.8 Post-operative care

Give it verbally *and* in writing and record that you did.

- **Bite on gauze** 20–30 minutes; expect some ooze on the first day
- **Cold compress** for the first 24 hours (20 min on/off), head elevated when sleeping
- **Analgesia** — ibuprofen ± paracetamol; take the first dose before the LA wears off
- **Antibiotics** if prescribed — complete the course; **metronidazole and alcohol warning**
- **Chlorhexidine 0.2% mouthrinse** twice daily from day 1 for 1–2 weeks (do not rinse vigorously on day 1); **no brushing of the wound** but continue elsewhere
- **Soft diet, avoid chewing on the site**; avoid hot food/drink on day 1
- **No smoking**, no alcohol, no strenuous exercise for 48–72 hours
- **Denture wear** — remove for 24–48 hours, then relieve generously and reline with a soft liner; **do not wear over a graft** if possible
- **Sinus precautions** where relevant: no nose-blowing, sneeze with the mouth open, no straws, no smoking, no diving/flying if advised
- **What is normal** (swelling peaks 48–72 h, bruising, mild ooze) and **what is not** — return/contact if: bleeding not controlled by pressure, increasing pain after day 3, fever, spreading swelling, difficulty swallowing or breathing (emergency), numbness persisting beyond the expected LA duration, or the graft/membrane becoming exposed
- **Emergency contact number and review appointment** (suture removal 7–10 days)

§9 Minor hard and soft tissue augmentation

Blueprint: *"Evaluate the appropriateness of **minor** augmentation procedures and their execution", "describe the different types of bone defects and suggest appropriate means to manage these", "differentiate between the different types of soft and hard tissue augmentation materials", "apply an understanding of the role of barriers", "limitations… and the timing".* All blueprinted **W + OS-P + SCE**.
Remember the scope: minor augmentation is in; **block onlay grafts, lateral-window sinus elevation and vertical augmentation are out of scope to plan or perform** — in scope to recognise and refer.
Key reading: ITI TG vol 7; Lindhe; CCID ch. 4–5; ITI Decision Trees.

9.1 What counts as "minor" at GDP level

In scope: alveolar ridge preservation/socket grafting; simultaneous horizontal GBR of a dehiscence or fenestration; grafting the jumping gap at immediate placement; **transcrestal (osteotome/Summers) sinus floor elevation** with residual height **>6 mm and a flat sinus floor**; ridge split/expansion of modest degree (arguably advanced); connective tissue and free gingival grafts; roll flaps.

Out of scope (recognise and refer): autogenous **block onlay** grafts; **lateral window** sinus elevation; **vertical** ridge augmentation; guided bone regeneration for large defects requiring titanium-reinforced membranes/mesh; distraction osteogenesis; Le Fort I with interpositional graft; nerve lateralisation.

The **ITI Decision Tree for the atrophic posterior maxilla** is a clean way to organise this: favourable interarch relationship + insufficient height → **SFE alone**; inadequate crest width as well → SFE + horizontal augmentation; severe vertical resorption → **vertical ridge augmentation ± SFE**; severe vertical resorption + inadequate width → horizontal *and* vertical augmentation, or **Le Fort I with interpositional graft**. Only the first branch, and only in its transcrestal form, is GDP territory.

9.2 The biology: GBR and the PASS principle

Guided bone regeneration (GBR) uses a **barrier membrane** to exclude the fast-migrating epithelial and connective tissue cells from the defect, creating and maintaining a space in which slower osteoprogenitor cells can regenerate bone.

PASS principles for predictable regeneration (Wang & Boyapati) — quote them: - **P**rimary wound closure (tension-free) - **A**ngiogenesis — blood supply (decorticate/perforate the cortical plate to open marrow spaces) - **S**pace maintenance/creation - **S**tability of the wound and the graft (fixation, tacks, screws, or a stable graft/membrane construct)

Graft mechanisms (know the three words): - **Osteogenesis** — the graft contains living cells that form bone. **Only autogenous bone.** - **Osteoinduction** — the graft induces host mesenchymal cells to differentiate into osteoblasts (BMPs). **Autogenous and (variably) DFDBA.** - **Osteoconduction** — the graft acts as a scaffold along which bone grows from the host. **All graft materials.**

KEY POINT

See also Set-piece 2 at the start of Part 7, which expands this into the four categories with their full advantages, disadvantages and consent points, and adds a structured account of the **complications of bone grafting** by donor site, recipient site, early, late and material-specific.

Type	Examples	Properties	Notes

9.3 Bone graft materials — the comparison table

Type	Examples	Properties	Notes
Autogenous ("gold standard")	Intraoral: chin, ramus/external oblique ridge, tuberosity, bone from the osteotomy, bone scrapers/collectors, bone traps	Osteogenic + osteoinductive + osteoconductive; no disease transmission; **fastest incorporation**	**Donor site morbidity**, limited volume, **most rapid and unpredictable resorption**; particulate autogenous from a scraper is the realistic GDP source
Allograft (human donor, processed)	**FDBA** (freeze-dried bone allograft), **DFDBA** (demineralised — exposes BMPs, more osteoinductive, less space-maintaining), cortical/cancellous chips	Osteoconductive ± osteoinductive	Ethical/religious considerations; consent must state the human origin; theoretical disease transmission (essentially eliminated by processing); tissue-bank regulation
Xenograft (another species)	**Deproteinised bovine bone mineral (DBBM)** — Bio-Oss®; porcine; equine; **anorganic**	Osteoconductive only; **very slow/negligible resorption** → outstanding **volume and contour stability**	Bovine/porcine origin — **must be disclosed** (religious, dietary, vegetarian objections); the standard material for jumping gaps, ridge preservation and contour augmentation
Alloplast (synthetic)	β-tricalcium phosphate, hydroxyapatite, biphasic calcium phosphate, bioactive glass, calcium sulphate, polymers	Osteoconductive; unlimited supply; no disease/ethical issue	Variable resorption; generally less well documented than DBBM in implant sites
Biologics/adjuncts	Autologous platelet concentrates (**PRF/PRP**), enamel matrix derivative, **rhBMP-2**, rhPDGF	Growth factors; soft tissue healing benefit best documented for PRF	Evidence variable; **know what they are and that the evidence for hard-tissue gain is limited**

Composite grafting (autogenous + DBBM, typically 1:1) is the common clinical compromise: autogenous supplies osteogenic cells, xenograft supplies volume stability.

9.4 Barrier membranes

Membrane	Examples	Pros	Cons
Resorbable — natural collagen	Porcine/bovine type I & III collagen (Bio-Gide®)	No second surgery, excellent tissue integration, **exposure is better tolerated** (may still heal), haemostatic	Poor space maintenance (collapses), **unpredictable resorption (~4 weeks–6 months)**, animal origin — consent
Resorbable — synthetic	PLA/PGA copolymers	No second surgery, predictable degradation	Acidic degradation products may cause inflammation; stiffness variable
Non-resorbable — dPTFE / ePTFE	Cytoplast™, Gore-Tex®	Excellent space maintenance; **high-density PTFE can be left exposed** intentionally over sockets	Requires removal; ePTFE exposure → infection and graft loss
Non-resorbable — titanium-reinforced PTFE / titanium mesh	—	Best space maintenance for large/vertical defects	**Advanced–complex; refer**

The commonest complication is membrane exposure, caused by tension at closure, a suture line over the membrane, denture pressure, or smoking. Manage exposure by: chlorhexidine, no brushing, close review; if a **collagen** membrane and no infection, it often heals; if **ePTFE** or infected, **remove the membrane early** to save what bone has formed.

"Ridge preservation" vs "socket seal": a graft placed in the socket **plus** a barrier/seal (membrane, collagen sponge/plug, free gingival punch graft, or a customised healing abutment). It **reduces but does not abolish** ridge resorption — typically halving horizontal loss (≈1.5–2 mm saved). **Indications:** thin or

damaged buccal plate, aesthetic zone, a site to be implanted later, and where the patient cannot proceed immediately. **Not indicated** routinely where immediate placement is planned and achievable, or where the site will be left edentulous with no prosthesis.

9.5 Bone defect classification and management

Around an implant at placement:

Defect	Description	Management
Dehiscence	Bone missing from the crest down the implant face, communicating with the crest	Simultaneous GBR: decorticate, particulate graft (composite), collagen membrane, tension-free closure, **usually two-stage/submerged**; expect ~4–6 months
Fenestration	A "window" of exposed implant surface with intact bone at the crest	GBR as above; often more predictable than a dehiscence
Horizontal (buccal) deficiency	Insufficient ridge width	If ≥1–2 mm implant coverage achievable → simultaneous GBR; if not → **staged GBR** first, or ridge split, or refer
Vertical deficiency	Insufficient height	**Refer** — vertical augmentation is complex and unpredictable; consider **short implants** as the GDP-level alternative
Combined	—	Refer

Peri-implantitis defects (Schwarz classification) — used in the ITI peri-implantitis decision tree: - **Class I** — intrabony (with sub-types Ia dehiscence; Ib dehiscence + semicircular; Ic dehiscence + circumferential; Id semicircular buccal + lingual; **Ie circumferential**) - **Class II** — supracrestal/horizontal bone loss **Also used clinically:** CID (**c**ircumferential **i**ntrabony **d**efect), DD (dehiscence defect), CD (combined defect), CSD (combined semicircular defect), ID (intrabony defect) — as in the ITI decision tree, where **reconstructive therapy is reserved for defects with an intrabony component** (Types 1 CID, 3 CD, 5 ID) and **resective therapy for non-aesthetic sites** with Types 4 CSD and 5 ID.

Extraction socket classification (Elian): - **Type 1** — intact facial soft tissue **and** intact facial plate (best for immediate placement) - **Type 2** — soft tissue present but the **facial plate is damaged/missing** - **Type 3** — both soft tissue and facial plate deficient (worst; ridge preservation or staged approach)

9.6 Transcrestal (osteotome) sinus floor elevation — the one you may do

Indications: posterior maxilla, **residual bone height >6 mm** (some say ≥5 mm), **flat sinus floor**, need for only **3–4 mm of gain**, adequate ridge width, healthy sinus. **Contraindications:** height ≤6 mm, oblique/steep sinus floor, septa, sinus pathology, ostium obstruction, previous sinus surgery, heavy smoker → **lateral window instead — which means refer**.

Sequence (Summers): flap → osteotomy to **1–2 mm short of the sinus floor** → concave osteotomes of increasing diameter with **gentle malleting** to green-stick fracture and elevate the floor with a plug of graft/bone → confirm membrane integrity with the **Valsalva manoeuvre** (no air movement/bubbling; the membrane should not move) → place the implant → close. **Modern alternatives:** hydraulic elevation, sequential drills with a stopper, piezo-surgery.

Complications: Schneiderian membrane perforation (commonest; if small, a collagen membrane may seal it, otherwise abandon and re-plan), benign paroxysmal positional vertigo from malleting, graft migration into the sinus, sinusitis, oro-antral communication, loss of the implant into the sinus.

Post-operative sinus precautions and a course of antibiotics ± decongestant are usually appropriate.

> **Alternative to sinus grafting for the GDP: short implants.** Say this in the SOE — *"Given a residual height of 6 mm with a flat floor I would consider either a transcrestal elevation with simultaneous placement, or a short (6 mm) implant with a favourable crown-to-implant ratio, which has comparable medium-term survival with fewer complications and would keep the case within my competence."*

9.7 Soft tissue augmentation

Procedure	Purpose	Notes
Free gingival graft (FGG)	**Increase the width of keratinised mucosa** (target ≥2 mm)	Epithelialised graft from the palate; heals by secondary intention at the donor site; colour mismatch ("patch"); most predictable for KM gain
Subepithelial connective tissue graft (SCTG)	**Increase soft tissue thickness/volume**, improve colour and contour, treat recession	Harvested from the palate (single/double incision, trapdoor); gold standard for phenotype modification; can be combined with a coronally advanced flap
Soft tissue substitutes	Xenogeneic collagen matrix, acellular dermal matrix	Avoid a second surgical site; slightly less volume gain than autogenous
Roll flap / pouch technique	Small buccal contour gain at second stage	Uses the palatal tissue over the implant
Apically positioned flap	Increase KM at uncovering; also resective peri-implantitis surgery	Preserves keratinised tissue rather than excising it
Vestibuloplasty / frenectomy	Deepen the sulcus, remove a pull on the margin	Often before overdenture provision

Timing: phenotype modification is easiest and most predictable **at implant placement or at second stage**, not after the prosthesis is fitted. **Donor site care:** palatal stent/plate, haemostatic dressing, avoid the greater palatine bundle, warn about discomfort and bleeding — the palate is the most painful part of the whole procedure for most patients.

9.8 Ridge split / expansion

Indication: ridge width ≥**3 mm** with cancellous bone between two cortical plates, adequate height, **maxilla far more predictable than mandible** (the mandible's dense cortex fractures rather than expands). **Technique:** crestal incision, careful (often partial-thickness) flap to preserve the buccal periosteal blood supply, mid-crestal corticotomy with a thin disc/piezo, vertical releases, expansion with chisels/osteotomes/expanders, implant placement, graft the gap, membrane, close. **Risks:** fracture or loss of the buccal segment, malposition, dehiscence. **A recurring SOE line of questioning** — *"short implant risks; would you split?"* — — a good answer compares ridge split, GBR (staged or simultaneous), a narrow-diameter implant, a short implant, and the non-implant options, then states which you would do **and why**, referencing your own competence.

9.9 The limitations to state whenever asked about augmentation

1. **Vertical augmentation is far less predictable than horizontal.**
2. Autogenous grafts resorb, sometimes substantially and unpredictably.
3. **Regenerated bone is not native bone** — lower density, slower to integrate, longer healing (typically **4–6 months** for simultaneous GBR, **6–9 months** for staged).
4. **Membrane exposure is the commonest complication** and reduces the gain.
5. **Smoking markedly reduces success.**
6. Every graft adds **cost, time, morbidity and risk**, and must be **consented for specifically**, including the material's origin.
7. Sometimes the right answer is **not to graft**: a shorter or narrower implant, a different implant position, a different prosthetic design, or a non-implant option.

§10 Restoring implants — prosthodontics and occlusion

Blueprint: *Decision making in relation to treatment procedures — restoring implants and managing complications* (the majority of these lines are marked **W + OS-P + SCE**).
Key reading: Misch (prosthetic sections); CCID ch. 3; ITI TG vols 2, 4, 11; DSMT.

10.1 The prosthetic options

Prosthesis	Indication	Notes
Single crown	Single tooth	Screw- or cement-retained; anti-rotational connection essential
Implant-supported FDP (bridge)	Multiple adjacent units	**Never connect an implant rigidly to a natural tooth** for routine cases (differential mobility → intrusion of the tooth, screw loosening, cement failure)
Cantilever	Where an implant cannot be placed	One unit only, out of the aesthetic/heavy-load zone, avoid in parafunction; increases mechanical complications
Implant-retained overdenture	Edentulous, especially mandible	**Two implants in the mandibular canine regions is the accepted minimum standard of care for the edentulous mandible (McGill and York consensus statements)**
Implant-supported fixed full arch	Edentulous	**Out of scope** for planning in this examination
Implant-assisted RPD	Selected partially dentate cases	Uncommon; know it exists

Attachment systems for overdentures:

Attachment	Features
Ball/stud	Simple, cheap, resilient; wear of the matrix; needs parallelism
Locator®	Low profile (needs less vertical space), various retention inserts, tolerates divergence up to ~20–40°, easy to change nylon inserts
Bar (Dolder, milled)	Splints the implants, distributes load, good for divergent implants and poor bone; needs **more vertical space (12–15 mm)**, harder to clean, more expensive, more laboratory steps
Magnets	Least retentive, easy to place/clean, useful with poor dexterity; corrosion/wear
Telescopic/conus	Rigid, excellent but costly

10.2 Impression techniques

Technique	How	When
Open tray (direct, pick-up)	Long guide screws protrude through a window in a custom tray; screws unscrewed before removal; the coping is picked up in the impression	**Multiple implants, non-parallel implants, deep implants** — the most accurate for splinted/multiple units
Closed tray (indirect, transfer)	Transfer copings remain in the mouth; the impression is removed and the copings re-seated into the impression with analogues	Single units, limited mouth opening; risk of re-seating error
Splinted	Copings joined with acrylic (sectioned and re-joined to compensate polymerisation shrinkage), pattern resin, or floss + composite	≥3 implants, full arch — **improves accuracy**
Digital / intraoral scan with scan bodies	No physical impression; direct to CAD	Excellent for single units and short spans; **accuracy across long edentulous spans remains the limitation**

Technique	How	When
Abutment-level vs implant(fixture)-level	Abutment-level simplifies the lab work and preserves the seal; implant-level allows full customisation of the emergence profile	Choose implant-level in the aesthetic zone

Material choice: polyether or addition silicone (PVS) — high rigidity, dimensional stability, elastic recovery. Custom tray, correct adhesive, uniform thickness, verify the coping seating **radiographically** where deep. **Also record:** shade, jaw registration, opposing arch, facebow where needed, and photographs.

10.3 Jaw registration

- Needed whenever there is **no stable, reproducible intercuspal position**, when the vertical dimension is being altered, and for all extensive/edentulous cases.
- Use **rigid record bases with an accurate fit** (ideally screwed to the implants for edentulous cases), wax rims adjusted for lip support, incisal level, occlusal plane and midline, and a **low-distortion registration material** (e.g. addition silicone bite registration, or a rim locked with pattern resin).
- **Facebow** to relate the maxilla to the hinge axis for anything more than a single unit in an established occlusion.
- **Common error:** displacing a resilient mucosa or a flexible base and building in an error you then blame on the laboratory.

10.4 Abutment selection *(blueprinted W + OS-P + SCE)*

Decide in this order: 1. **Retention type** — screw or cement (see §10.6) 2. **Level** — implant-level (customisable emergence) vs abutment-level (multi-unit abutments for full-arch/angled cases) 3. **Angulation** — straight vs pre-angled; or an **angulated screw channel (ASC)** to keep a screw-retained design when the implant is buccally inclined 4. **Height** — sufficient for retention (cement) and to bring the margin to the right level; the **subgingival margin depth is the key aesthetic and cement-related decision** — keep it **≤1 mm subgingival, ideally at or just below the margin**, and follow the scalloped gingival contour 5. **Material** — titanium (posterior, high load), **Ti-base + zirconia/lithium disilicate** (aesthetic), full zirconia (avoid direct zirconia-titanium interfaces at the implant connection), gold (historic) 6. **Stock vs custom (CAD-CAM)** — custom gives a correct emergence profile, correct margin level and better soft tissue support; stock is cheaper and faster 7. **Emergence profile** — developed with the provisional and then copied by a customised impression coping

10.5 Screw-retained vs cement-retained *(blueprinted W + OS-P + SCE — expect it)*

	Screw-retained	Cement-retained
Retrievability	**Excellent** — the decisive advantage	Difficult/destructive
Cement risk	**None**	**Residual cement is a major cause of peri-implantitis**
Minimum space	Less (~4–5 mm)	More (needs abutment height ~5 mm)
Aesthetics	Access hole may show if the implant is buccally angulated (solved by ASC abutments)	Better where angulation is unfavourable
Passive fit	Demands a passive framework; misfit is transmitted	Cement layer compensates for minor misfit
Occlusion	Access hole in the occlusal table can weaken porcelain/cause chipping	Uninterrupted occlusal surface
Cost/complexity	Fewer components	Extra abutment step

Rules of thumb: screw-retained wherever the angulation allows, and **especially where the margin would be >1–2 mm subgingival**; cement-retained only where the emergence is unfavourable and the margin can be kept at or above the mucosal margin.

10.6 If you must cement — how to avoid residual cement

1. Keep the **margin at or above the mucosal margin** wherever possible (≤1 mm subgingival).
2. Use a **radiopaque cement** so you can see what you have left behind.
3. Prefer **weak/temporary cements** (e.g. zinc oxide eugenol-free temporary cement, or resin-modified glass ionomer) — retrievability matters more than maximal retention.
4. **Minimise the volume**: apply cement only to the coronal half of the intaglio; use an **extra-oral replica/analogue technique** (seat the crown on a duplicate abutment first to extrude excess), or a Teflon/silicone spacer.
5. **Remove excess meticulously** with floss, scalers and air/water; **take a radiograph** to confirm.
6. **Do not use eugenol-containing cement** with resin-based restorations.

10.7 Implant occlusion — "implant-protective occlusion"

Because there is **no periodontal ligament**, implants have: - **Axial movement of only ~3–5 μm** (teeth: ~25–100 μm) - **Lateral movement of ~10–50 μm** (teeth: ~56–108 μm) - **No proprioception from a PDL** → higher tactile threshold, reduced protective reflex, and slower/absent detection of overload - **Movement pattern:** teeth show an immediate, non-linear, two-phase movement; implants show gradual, linear movement with no initial phase

Therefore the occlusal scheme is designed to protect the implant:

Rule	Detail
Light contacts in ICP	Implant restorations should contact **lightly** — the classic instruction is that shimstock (8–12 μm) holds under **heavy** clenching but not under light tapping; i.e. **contact ~25–30 μm lighter than the adjacent teeth**
Contacts directed along the long axis	Centralise contacts over the implant; **narrow the occlusal table** and reduce cusp inclines
No lateral/excursive contacts on implants where teeth can guide	Use **canine guidance on natural teeth**; if the canine is an implant, use **group function** to distribute load
Eliminate working and non-working interferences	Especially non-working (balancing) interferences
Reduce cantilevers and offsets	Cantilever length and buccolingual offset generate the largest bending moments
Avoid connecting implants to teeth	Differential mobility
Cusp height/angle	Flatter cusps reduce lateral force
In parafunction	Splint (hard acrylic full-coverage), more/wider implants, no cantilevers, monolithic zirconia rather than layered ceramic, accept a higher complication risk and consent for it
In an overdenture	Bilateral balanced or lingualised occlusion for a complete overdenture, as for conventional dentures

Progressive loading (Misch): in soft bone (D3/D4), gradually increase load — provisional in acrylic, out of/light occlusion, then progressive contacts over weeks — to allow bone to mature and remodel. Know the concept; the evidence is modest.

10.8 Screw mechanics — preload, torque, loosening

- **Preload** is the tensile clamping force in the screw that holds the joint together. It is generated by torque, but **~90% of applied torque is consumed by friction**.
- **Settling (embedment relaxation)**: micro-roughness flattens under load, losing ~2–10% of preload → **always re-torque after ~10 minutes**.
- **Causes of screw loosening:** inadequate torque (the **most likely cause of a loose single crown**, per the RCSEd sample SBA the answer given is *excessive occlusal loads* — read the stem carefully: the published sample question with a **12-month-old, symptomless, gradually loosening** screw-retained crown is marked **B, excessive occlusal loads**), excessive/eccentric occlusal load, cantilever, non-passive fit, worn or reused screws, absent anti-rotational feature (single unit on an external hex), lack of re-torque.
- **Management of a loose screw:** remove the restoration, inspect and **replace the screw with a new one**, check for component damage and for **implant fracture**, identify and correct the *cause* (occlusion, cantilever, fit), re-torque to the manufacturer's value, re-torque after 10 minutes, seal the access with PTFE tape + composite, review. **Never simply retighten and send the patient away.**
- **Fractured screw retrieval:** ultrasonic scaler used counter-clockwise on the fragment, an explorer/probe, a proprietary screw-removal kit, or a purpose-made reverse-thread drill; protect the internal threads; if the threads are destroyed the implant may be lost.

10.9 Laboratory communication *(blueprinted W + OS-P + SCE)*

Every prescription must contain: - Patient identifier and your name, address and GDC number; date sent and **date required** - **Precise description of the work** — units, tooth notation, implant system, connection, platform diameter, abutment type (stock/custom, angulation, material), retention (screw/cement), framework material, veneering material, occlusal material - **Shade** (with photographs, ideally with a shade tab in shot), characterisation, translucency, stump shade - **Occlusal scheme** required, and the vertical dimension - **Emergence profile** instructions, margin position, contact points, contour for cleansability - Whether a **try-in** is required (framework/verification jig/bisque) - Statement of **disinfection** on despatch and on return - **Custom-made device statement** under the medical devices regulations, retained in the notes; and record the laboratory's details and the batch/lot numbers of the components returned

Verification jig / verification of fit: for multi-unit frameworks, use a sectioned-and-rejoined jig or a screw-test (**Sheffield/one-screw test**: tighten only the most distal screw and check the fit at the other end — a gap means the framework is not passive).

10.10 Managing a mis-fitting prosthesis or framework *(a named blueprint line, marked OS-P)*

1. **Confirm the misfit objectively:** visual gap, the **one-screw (Sheffield) test**, tactile check with a probe, **radiograph** (a periapical taken with a parallel technique — the classic sign is a wedge-shaped gap between abutment and implant), patient symptoms (pressure, pain on tightening).
2. **Identify the cause:** interference from soft tissue or bone over the platform; a component not fully seated; incorrect analogue orientation; impression distortion; inaccurate cast; laboratory error; framework distortion during firing; the wrong component.
3. **Do not force it.** Never torque a non-seating restoration — this causes screw fracture, bone loss and pain.
4. **Remedy according to cause:** remove interference; re-seat; **section the framework and re-solder/laser weld**; take a **new impression with a splinted open-tray technique or a verification jig**; remake.
5. **Explain to the patient**, apologise for the delay, and record everything. Do not blame the technician in front of the patient.

10.11 Digital workflows *(blueprinted W + OS-P + SCE)*

Know the chain and its error sources: **intraoral scan** → (**merge with CBCT via DICOM/STL superimposition**) → **virtual wax-up** → **guide design** → **3-D printed/milled guide** → **guided placement** → **scan body** → **CAD abutment/crown** → **milled/printed restoration**.

Advantages: patient comfort, no impression material, immediate assessment and re-scan, digital archiving, communication with the laboratory and patient, restoratively driven planning, potential accuracy for single units. **Limitations:** cost, learning curve, accuracy over long edentulous spans, scan-body wear/accuracy, metal artefact in CBCT merging, deep subgingival margins scan poorly, and the **guide cannot compensate for surgical judgement**.

§11 Complications and their management

Blueprint: *Interpretation and problem solving of clinical outcomes — Patient safety; "Suggest appropriate strategies for managing complications that may occur during a surgical procedure"; "Recognize clinical situations which are unsafe or could lead to an adverse event".*
Key reading: Misch (complications — the most complete single source); ITI TG vol 8; CCID (complications chapter).

11.1 A framework for any complication question

Answer **every** complication question in the same five steps. Examiners are marking a *process*, not a fact.

1. **Recognise** — what are the signs and symptoms, and what is the differential?
2. **Immediate management** — stop, protect the airway/haemostasis/patient safety, relieve pain
3. **Definitive management** — what will I do, and over what timescale?
4. **Refer?** — to whom, how urgently, and what will I say?
5. **Prevent, record and learn** — how would I stop this happening again; what do I record; do I need to report it (significant event analysis, duty of candour, indemnity)?

Say "duty of candour" out loud. When something goes wrong you must **tell the patient promptly, apologise, explain what happened and what you will do**, and record it. An apology is not an admission of liability. This is explicitly in GDC Standards and is a domain-4 marking point.

11.2 Intra-operative complications

Complication	Recognition	Management	Prevention
Haemorrhage — soft tissue	Brisk bleeding from flap/vessel	Pressure, ligation/diathermy, suture, local haemostatics (oxidised cellulose, collagen, tranexamic acid)	Flap design avoiding named vessels
Floor-of-mouth haemorrhage (lingual plate perforation anteriorly; sublingual/submental arteries)	Rapidly expanding swelling of the floor of mouth, tongue elevation, **dysphagia, dyspnoea, stridor**	**AIRWAY EMERGENCY**: sit the patient up, call 999, secure the airway, direct pressure/bimanual compression, transfer immediately; do not send home	CBCT, respect the lingual concavity, angle the drill lingually-aware, palpate the lingual plate
Nerve injury — IAN/mental	Bleeding from the osteotomy, patient reports electric shock, post-op anaesthesia/paraesthesia/dysaesthesia	**If recognised during surgery: remove or shorten the implant immediately.** If post-op: review within 24 h, **map and document the area with a diagram**, prescribe a short course of steroids ± NSAID (evidence limited), remove/back out the implant within 24–36 h if it is impinging, **refer urgently to oral/maxillofacial surgery within days if not resolving — do not "wait and see" for months**	2 mm margin, CBCT, direction indicators + radiograph, drill stops, avoid IDB in selected cases
Lingual nerve injury	Altered tongue/taste sensation	As above; usually from flap/incision, not the drill	Avoid lingual releasing incisions; careful retraction
Perforation of the lingual/buccal plate	Loss of tactile resistance, probe passes through	If small and the implant is stable → graft ± membrane; if the implant is unstable or malpositioned → **remove, graft, re-plan**	Ridge assessment, CBCT
Sinus membrane perforation	Valsalva positive, bubbling, visible membrane tear	Small (<5 mm): collagen membrane patch, continue cautiously; large: **abandon, close, re-plan in 3–6 months**	Height >6 mm and flat floor for transcrestal; check for septa

Complication	Recognition	Management	Prevention
Oro-antral communication	Air/fluid passage	<2 mm: suture and sinus precautions; larger: buccal advancement flap ± referral; antibiotics + decongestant	As above
Implant displaced into the sinus	Implant not retrievable at the crest	Do not probe blindly; **refer for retrieval** (endoscopic/Caldwell-Luc) — leaving it risks sinusitis/fungal ball	Assess bone height; do not force
Damage to an adjacent root	Drill deviates; radiograph confirms	Reposition the osteotomy; monitor the tooth (vitality, radiograph); may need endodontics; inform the patient	Guide, direction indicator + radiograph, 1.5 mm margin
Lack of primary stability	Implant spins in the osteotomy	Place a wider/longer implant if bone allows; graft and abandon the site (**submerge and re-enter in 4–6 months**); never load	Under-prepare in soft bone; assess D-classification
Overheating / bone necrosis	Smoking/charring, bone smell, later loss	Abandon that osteotomy; re-site or delay	Sharp drills, irrigation, intermittent pressure, low speed in dense bone
Mandibular fracture (severely atrophic mandible)	Crack, mobility, malocclusion, pain	Stabilise, **urgent maxillofacial referral**	Do not place standard implants in a mandible <7–8 mm high — refer
Instrument/component aspiration or ingestion	Coughing, choking, or a lost component	**Airway first**: back blows/abdominal thrusts if obstructing; if swallowed and asymptomatic → chest and abdominal radiograph; if aspirated → **urgent hospital referral for bronchoscopy**	**Throat pack/gauze screen, floss ligature on small components, high-volume suction, four-handed dentistry**
Emphysema	Sudden crepitant swelling with air-driven handpiece	Stop, antibiotics, observe; usually self-limiting	**Never use an air-turbine/air-driven handpiece for implant surgery** — use an electric surgical motor

11.3 Early post-operative complications

- **Pain, swelling, bruising** — expected; peak at 48–72 h. Increasing pain after day 3 suggests infection or dry socket-type problem.
- **Bleeding** — pressure, review anticoagulation and local haemostasis.
- **Infection / abscess** — drain, irrigate, antibiotics, review; consider removing an implant that is mobile.
- **Wound dehiscence / membrane exposure** — chlorhexidine, no brushing, close review; remove ePTFE early; a collagen membrane may still heal.
- **Failure to integrate (early failure)** — mobile implant, pain on percussion/loading, peri-implant radiolucency. **Remove the implant, curette and irrigate the site, allow 2–3 months of healing (± graft), then re-implant** — usually with a wider implant. Warn the patient that a second failure at the same site is more likely.
- **Prolonged altered sensation** — see §11.2; document, map, review at 1 week, 1 month, and refer if not improving.

11.4 Late biological complications

Covered in §12 (peri-implant mucositis and peri-implantitis). Others:

- **Retrograde peri-implantitis** — apical radiolucency around an otherwise integrated implant, typically from residual pathology in the site or an adjacent endodontically involved tooth. Treat the source; apical surgery/decontamination or removal.
- **Soft tissue hyperplasia / peri-implant polyp** — usually from an ill-fitting or over-contoured prosthesis, or a poorly seated abutment.
- **Mucosal recession and grey show-through** — thin phenotype, buccal implant position, titanium abutment. Manage with a connective tissue graft, a zirconia or pink-ceramic abutment, prosthetic recontouring, or accept and monitor; in a badly malpositioned implant the definitive answer is **removal**.

- **Sinusitis** after sinus procedures.

11.5 Mechanical/technical complications

Complication	Cause	Management
Screw loosening (commonest technical complication)	Inadequate preload, overload, cantilever, non-passive fit, no anti-rotation	See §10.8 — new screw, correct the cause, re-torque, re-torque again at 10 min
Screw fracture	Fatigue from repeated loosening, overload	Retrieve the fragment (ultrasonic, screw-removal kit); replace; **correct the cause**
Abutment fracture	Overload, parafunction, narrow abutment	Replace; consider a stronger material/wider implant/splinting
Ceramic chipping/fracture (commonest complication of implant FDPs)	Layering porcelain, parafunction, framework support, thermal stresses	Polish/repair, or remake in **monolithic zirconia**; adjust occlusion; splint
Framework fracture	Inadequate bulk, misfit	Remake
Implant fracture (rare, ~0.2–1%)	Fatigue, narrow implant under heavy load, bone loss increasing the lever arm, untreated screw loosening	**Implant must be removed** (trephine) or the fractured portion left submerged ("sleeper") if removal would be destructive
Wear/loss of retention of overdenture attachments	Normal wear	Replace inserts/matrices — **consent for this maintenance cost from the outset**
Denture base fracture	Insufficient bulk over attachments	Reinforce with a metal framework
Loss of cement retention	Weak cement, short abutment, taper	Re-cement after checking fit and occlusion — or convert to screw-retained

11.6 Aesthetic complications

Causes are almost always **planning and positioning**, not the ceramic. Recognise:

- Implant too **buccal** → recession, grey show-through, thin tissue
- Implant too **shallow** → visible metal collar, poor emergence
- Implant too **deep** → deep pocket, difficult hygiene, cement risk, "ridge lap" contour
- Implants too **close together** → papilla loss, black triangles
- **Contact point too far from the crest** (>5 mm) → black triangle
- Failure to graft the **buccal contour** → concavity beneath the crown

Management: soft tissue grafting, prosthetic camouflage (pink ceramic, modified contours, a longer crown), orthodontic adjunct, or **explantation and re-treatment**. A candidate who recognises that "this implant is in the wrong place and no restoration will fix it" is scoring highly, provided they also explain the conversation with the patient, the costs, and who should do the re-treatment.

11.7 Managing complaints and dissatisfaction *(blueprinted to SCE)*

1. **Listen** — uninterrupted, in a private setting, with time set aside.
2. **Acknowledge and apologise** for the patient's experience (**duty of candour**).
3. **Establish the facts** — examine, take records/radiographs, review the notes and the consent.
4. **Explain** in plain language what has happened and what the options are; separate what is a complication from what is an error.
5. **Offer a remedy** — repair, remake, refund, referral for a second opinion.
6. **Document** everything contemporaneously; **inform your indemnity provider early**.
7. **Follow the practice complaints procedure**, respond within the stated timescales, and tell the patient about the relevant ombudsman/regulator if unresolved.
8. **Learn** — significant event analysis, change of protocol, and re-audit.

> **Never:** become defensive, blame the laboratory or the previous dentist in front of the patient, alter the records, or promise an outcome you cannot deliver.

§12 Peri-implant health and disease, maintenance and outcomes

> **Blueprint:** *Quality and maintenance*; *Complications and Clinical Outcomes*. Marked **W + SCE** throughout.
> **Key reading:** Lindhe; ITI TG vol 13 and the **ITI Decision Tree "Treating Peri-Implant Mucositis and Peri-Implantitis"**; 2017 World Workshop (Berglundh et al.); EFP S3 guideline; Termeie; PCS.

12.1 The 2017 World Workshop definitions — learn them verbatim-ish

Peri-implant health - **No** clinical signs of inflammation: no erythema, no swelling, **no bleeding on gentle probing** (**≤0.25 N**), no suppuration - **No** increase in probing depth compared with previous examinations - **No** bone loss beyond the initial crestal remodelling - *Health can exist around implants with reduced bone support.*

Peri-implant mucositis - **Bleeding on gentle probing** (the primary diagnostic criterion), ± erythema, swelling, suppuration - **Probing depth may be increased** because of swelling or reduced resistance - **No bone loss beyond initial crestal remodelling - Caused by plaque; reversible if the biofilm is controlled** (the key message)

Peri-implantitis - Bleeding and/or suppuration on gentle probing - Increased probing depth compared with previous examinations - Bone loss beyond crestal bone level changes resulting from initial bone remodelling

In the absence of previous examination data (the usual clinical reality), peri-implantitis is diagnosed when all of: 1. **Bleeding and/or suppuration on gentle probing** 2. **Probing depth ≥6 mm** 3. **Bone level ≥3 mm apical to the most coronal portion of the intraosseous part of the implant**

Peri-implant soft and hard tissue deficiencies — the fourth category: recession, lack of keratinised mucosa, ridge deficiency, papilla loss.

Prevalence (weighted means, be able to quote approximate figures): peri-implant mucositis ≈ **43%** of patients, peri-implantitis ≈ **22%** of patients (with wide reported ranges, ~20–47% and ~9–47%, depending on case definition — say that the case definition drives the number, which is an insightful examiner-pleasing point).

12.2 Risk factors and risk indicators

Strong evidence: history of **periodontitis**, **poor plaque control / no supportive maintenance**, **smoking** (dose-dependent). **Moderate/emerging:** poorly controlled diabetes, **residual submucosal cement**, **absence of keratinised mucosa (<2 mm)**, prosthesis design preventing access for cleaning (over-contoured, ridge-lap, deep margins), occlusal overload (contested as a primary cause but a co-factor), implant surface characteristics, genetic factors, alcohol.

12.3 Diagnosis in practice

- **Probe gently (0.2–0.25 N)** with a plastic or metal probe — probing is safe and essential; the "do not probe implants" idea is obsolete
- Record **probing depths at 6 sites**, **BOP**, **suppuration**, **mucosal recession**, **width of keratinised mucosa**, plaque score
- **Mobility** = terminal (loss of integration) — check the *implant*, not the prosthesis, by **removing the restoration** if in doubt
- **Radiograph**: standardised, paralleling technique, compared with a **baseline taken at prosthesis fitting** (this is why the baseline radiograph matters so much)
- **Assess the prosthesis**: fit, contour, cleansability, residual cement, screw integrity, occlusion
- **Assess the patient**: periodontal status of remaining teeth, plaque control, smoking, diabetes, dexterity, motivation

12.4 Treatment — follow the ITI decision tree structure

Step 0 — Examination and cause-related therapy

Assess periodontal status; assess and **modify potential risk factors** (smoking, diabetes); assess the **prosthesis for cleansability** and modify or remake its contours if required; assess the patient's ability to perform plaque control; **assess patient expectations**; obtain previous records and radiographs.

Step 1 — Diagnosis and explanation

Explain the disease and the treatment; **discuss costs and possible adverse effects, e.g. soft tissue recession** (this exact wording is in the ITI tree — quote it). For peri-implantitis, decide explicitly whether it is **rational to keep the implant** or **irrational** (in which case plan removal from the outset).

Step 2 — Non-surgical therapy (for both mucositis and peri-implantitis)

- **Mechanical debridement** — removal of biofilm and calculus with **titanium, PEEK/plastic or carbon-fibre curettes**, ultrasonics with plastic-sheathed tips, **glycine/erythritol air-polishing** (better access to threads than curettes)
- **Modify prosthesis contours** where they prevent access
- **Individualised oral hygiene instruction** — interdental brushes sized to the site, single-tufted brushes, superfloss, powered brushes
- **Adjunctive measures** (optional, modest evidence): air-polishing, local antimicrobials (chlorhexidine chip/gel, minocycline microspheres), antimicrobial photodynamic therapy, probiotics, systemic antibiotics **only in selected peri-implantitis cases**
- **Re-assess at approximately 1 month** with peri-implant probing

Step 3 — Re-assessment outcomes

- **Resolution** (no BOP/suppuration, no increased PD) → **supportive therapy**
- **Persistent mucositis** (persistent BOP) → further non-surgical therapy, re-motivate, re-examine the prosthesis
- **Peri-implantitis with BOP and PD ≤5 mm** → further non-surgical therapy and close monitoring
- **Persistent peri-implantitis — suppuration and/or BOP with PD ≥6 mm** → **surgical treatment**

Step 4 — Surgical treatment

Common surgical steps: **mucoperiosteal flap; removal of inflamed granulation tissue; decontamination of the implant surface** (chemical — hydrogen peroxide, chlorhexidine, saline; mechanical — ultrasonic, titanium brush; laser — Er:YAG); **adequate flap adaptation**; **adequate post-operative care** — antiseptics ± peri-operative systemic antimicrobials.

Then choose the approach by **defect morphology**:

Approach	Indication (ITI tree)
Open-flap debridement	Types 1 CID, 2 DD, 3 CD, 4 CSD, 5 ID — i.e. any morphology, where the aim is access and decontamination only
Reconstructive therapy (bone graft/substitute ± membrane ± soft-tissue graft)	Defects **with an intrabony component**: Type 1 CID, Type 3 CD, Type 5 ID
Resective therapy (bone removal ± **implantoplasty**, apically positioned flap)	**Non-aesthetic areas**: Type 4 CSD, Type 5 ID
Implant removal	Extent of the defect too great, or the aesthetic outcome would be severely compromised. Then either **spontaneous healing** or **hard and/or soft tissue grafting**

Implantoplasty = smoothing the exposed rough implant surface with burs/polishers to make it plaque-resistant. Use high suction and protect the tissues from titanium particles.

Step 5 — Supportive therapy

Regular monitoring every 3–5 months: professional biofilm removal and oral-hygiene reinforcement. (*That interval is straight from the ITI tree — quote it.*)

When to remove an implant

Mobility (loss of integration), fracture of the implant, progressive bone loss unresponsive to treatment, bone loss beyond ~50–60% of the implant length, uncontrollable infection, severe malposition making a satisfactory restoration or hygiene impossible, or a patient who cannot maintain it. Methods: reverse-torque (counter-torque ratchet) — least destructive; piezo-surgery; trephine — most destructive.

12.5 Maintenance, records and outcome monitoring

A written maintenance programme is part of the treatment plan, not an afterthought — and this is precisely the sort of thing a candidate forgets when the examiner asks "and then what?"

Recall interval: risk-based; commonly **3–6 months** for implant patients, **3–4 months** where there is a history of periodontitis, smoking, or previous peri-implant disease. **The ITI tree specifies 3–5 monthly after treating peri-implant disease.**

At every implant recall: - Update medical history, smoking status, symptoms - **Plaque and bleeding scores**; peri-implant probing at 6 sites; **BOP and suppuration**; recession; keratinised mucosa - Check **prosthesis**: mobility, screw integrity, wear, chipping, contour, cleansability, occlusion - **Professional biofilm removal** with implant-safe instruments; polish with a non-abrasive paste - **Re-motivate and re-instruct** with site-specific aids - **Radiographs**: baseline at prosthesis fitting, then **at 1 year**, then **only when clinically indicated** (change in probing depth, BOP/suppuration, symptoms) — an IR(ME)R justification point - Record everything, including the **date, findings, advice given and the next review**

Parameters used to monitor outcomes (a blueprint phrase): implant survival; **marginal bone level change** against baseline; probing depths and BOP; suppuration; soft tissue level/recession; papilla index; prosthetic complication events; patient-reported outcome measures (**PROMs**: OHIP-14, satisfaction, aesthetics, function, comfort); and function/aesthetics assessed against the **pre-operative plan and the patient's stated expectations**.

Records — what must be in the notes for every implant case: medical history and updates; consent discussion and written information given; radiographs and reports; the treatment plan and any modifications with reasons; **the surgical record — implant system, diameter, length, site, batch/lot number, insertion torque, graft and membrane products with lot numbers, closure, sutures**; post-operative instructions given; the prosthetic record — abutment type and torque, cement type, occlusion; the maintenance plan; every review; and any complication with what you did and said. Give the patient an **implant record/patient card**.

12.6 Quality improvement and audit *(entrance requirement and blueprint content)*

Audit cycle: identify the topic → **set the standard (with a source)** → collect data → compare with the standard → **implement change** → **re-audit** to close the loop. **Examples of implant-relevant audits:** completeness of implant surgical records; radiographic justification and reporting; consent documentation; peri-implant probing at recall; rate of early implant failure; antibiotic prescribing against guidance; sterilisation records; patient-reported outcomes. **Significant event analysis** for adverse events; **clinical governance** components: audit, risk management, education and training, evidence-based practice, patient involvement, information governance, staffing.

Have ready for the SOE: your own QI project in 60 seconds — problem, standard, method, result, change, re-audit.

12.7 Professionalism, confidentiality and the team

- **Confidentiality and data protection** — UK GDPR/Data Protection Act; records retained (commonly 11 years for adults, or until age 25 for children, per NHS guidance — know that a defined retention period exists); patients may request access; images require specific consent; be careful with messaging apps and lab communications.
- **Communicating the plan to the wider team** (blueprint line): written treatment plan shared with the hygienist/therapist for maintenance; a clear laboratory prescription; the referring/receiving clinician; the GMP where medical liaison is needed; and the patient's own copy.
- **Working within competence (GDC Principle 7)**, mentoring, indemnity for implant dentistry specifically, verifiable CPD with reflection.
- **Evidence-based practice** — be able to describe the hierarchy (systematic review/meta-analysis of RCTs > RCT > cohort > case-control > case series > expert opinion), and to say why a particular claim is or is not well supported. *"Defend a treatment plan derived using an evidence-based approach"* is a literal blueprint line.

PART 3

High-yield numbers, classifications and lists

Everything in the notebook that has to be known as a number, a name or a list. Read this section daily in the last ten days.

IN THIS PART

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2.2 Physiological and biomechanical numbers 60
2.3 Classifications to have at your fingertips 61
2.4 Drugs — quick recall 61

PART 3 — HIGH-YIELD NUMBERS, CLASSIFICATIONS AND LISTS

Revise this part daily in the last ten days. Almost every number here has appeared in a written paper somewhere.

2.1 Distances and dimensions

Parameter	Value
Implant–tooth (crestal)	≥1.5 mm
Implant–implant	≥3 mm
Buccal bone thickness (aesthetic)	≥2 mm (minimum 1.5 mm)
Palatal/lingual bone	≥1 mm
Ridge width for a Ø4.1 mm implant	≈6.5–7 mm
Mesiodistal space, standard implant	≈7 mm
Mesiodistal space, narrow implant	≈5.5–6 mm
Safety margin from the IAN	≥2 mm
Distance anterior to the mental foramen (no CBCT)	≥5 mm
Anterior loop length	commonly 1–3 mm, up to ~5 mm
Implant shoulder below the planned mid-facial gingival margin	≈3 mm
Implant shoulder palatal to the emergence of adjacent teeth	≈1–2 mm
Contact point to crest for a papilla	**≤5 mm = ~100%; 6 mm = 56%; 7 mm = 27%**
Restorative space, screw-retained crown	≈5–6 mm
Restorative space, cement-retained crown	≈7–8 mm
Restorative space, stud/Locator overdenture	≈8–10 mm
Restorative space, bar overdenture / fixed hybrid	≈12–15 mm
Freeway space (normal)	2–4 mm
Biologic width / supracrestal tissue attachment around implants	≈3–4 mm
Keratinised mucosa target	≥2 mm
PSA artery in the lateral sinus wall	≈15–20 mm from the crest
Transcrestal SFE indication	residual height **>6 mm**, flat floor

2.2 Physiological and biomechanical numbers

Parameter	Value
Bone necrosis threshold	**47 °C for 1 minute**
Micromotion threshold preventing osseointegration	**50–150 μm**
Insertion torque, typical	25–45 Ncm
Insertion torque for immediate loading	**>35 Ncm**
ISQ high stability	**>70** (medium 60–69; low <60)
Implant axial movement	3–5 μm (tooth 25–100 μm)
Implant lateral movement	10–50 μm (tooth 56–108 μm)
Optimal implant surface roughness (Sa)	**1.0–2.0 μm (moderately rough)**
TiO_2 passivation layer	a few nanometres, reforms in milliseconds
Marginal bone loss, year 1	≤1.5 mm (Albrektsson); thereafter ≤0.2 mm/year
Loss of preload from settling	~2–10% → **re-torque at 10 minutes**
Ridge width lost at 12 months post-extraction	~50% (two-thirds in the first 3 months)
Horizontal / vertical ridge loss at 6 months	≈3.8 mm / ≈1.2 mm
Probing force for peri-implant probing	**0.2–0.25 N (gentle)**
Peri-implantitis (no baseline) criteria	BOP/suppuration + **PD ≥6 mm + bone loss ≥3 mm**
Prevalence, mucositis / peri-implantitis (patient level)	≈43% / ≈22%
Implant survival, 5 yr / 10 yr	≈95–97% / ≈90–95%
Smoking effect on failure	≈2–2.5×

2.3 Classifications to have at your fingertips

Bone quality: Lekholm & Zarb I–IV · Misch D1–D4 (+HU) **Bone quantity:** Lekholm & Zarb A–E **Placement timing:** Type 1 immediate · Type 2 early soft-tissue (4–8 wk) · Type 3 early partial bone (12–16 wk) · Type 4 late ($\geq$6 mo) **Loading:** Immediate restoration/loading (Type A, <1 wk) · Early (Type B, 1 wk–2 mo) · Conventional (Type C, >2 mo) **Evidence grading (ITI):** SCV > CWD > CD > CID **Case complexity: SAC** — Straightforward / Advanced / Complex, applied separately to surgical and prosthodontic **Extraction socket:** Elian Type 1 / 2 / 3 **Peri-implantitis defects:** Schwarz Class I (a–e) intrabony, Class II supracrestal; ITI tree types CID / DD / CD / CSD / ID **Surface roughness:** smooth <0.5 · minimally rough 0.5–1.0 · **moderately rough 1.0–2.0** · rough >2.0 μm **Periodontal (2017 World Workshop):** Stage I–IV (severity/complexity) and Grade A/B/C (rate of progression); *e.g. "generalised periodontitis stage IV grade C"* — a failing dentition loosely described as **"grade 4C"** is, in current language, **Stage IV, Grade C generalised periodontitis Aesthetic risk (ITI ERA):** 12 factors, low/medium/high **Albrektsson's 6 factors for osseointegration**; **Albrektsson's 5 success criteria**; **PASS principles**; **osteogenesis/osteoinduction/osteoconduction**

2.4 Drugs — quick recall

Lidocaine 2% + adr 1:80,000	44 mg/cartridge; max 4.4 mg/kg (300 mg)
Articaine 4%	88 mg/cartridge; max 7 mg/kg (~500 mg)
Prilocaine 3% + felypressin	66 mg/cartridge; max 6 mg/kg (400 mg)
Antibiotic prophylaxis for placement	**Amoxicillin 2–3 g single dose, 1 h pre-op** (NNT $\approx$ 25)
Infective endocarditis	**NICE CG64 — not routinely recommended**
Warfarin	Treat if **INR <4.0**, checked within 24 h (72 h if stable)
Analgesia	Ibuprofen 400 mg qds $\pm$ paracetamol 1 g qds
Chlorhexidine	0.2% rinse, 1 minute pre-op; bd post-op 1–2 weeks
Metronidazole	**Alcohol warning**; interacts with warfarin
Midazolam reversal	**Flumazenil**

PART 4

The literature: books, papers and numbers

A topic-by-topic map of the standard texts, the papers and names worth quoting, and the survival and complication figures — so that you always know where to read next, and what to cite.

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4.4 The numbers: survival, success, longevity and prognosis 68

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PART 4 — THE LITERATURE: WHICH BOOK, WHICH PAPER, WHAT NUMBERS

The commonest mistake in preparing for this examination is reading too widely. You do not need to read every book on the list. Use this table to go straight to the right source for the topic you are weak on.

4.1 Topic → source map

If you are weak on…	Read this first	Then this
The scope and level of the exam	RCSEd **Appendix A — Learning Outcomes & Blueprint**; Regulations §2 and §18	FGDP(UK) *Training Standards in Implant Dentistry*
Medical assessment and risk	**Misch** — "Medical Evaluation of the Dental Implant Patient"	**CCID** ch. 1 Case 2 (Medical Considerations); SDCEP guidance
Anticoagulants, MRONJ, prescribing	**SDCEP** — *Anticoagulants, MRONJ, Drug Prescribing for Dentistry*	BNF; NICE CG64
Examination, occlusion, space analysis	**Misch** — treatment planning and "Available Bone / Force Factors" chapters	**CCID** ch. 1 Case 4 (Occlusal/Anatomical Considerations)
Radiography and IR(ME)R	FGDP(UK)/RCR **Selection Criteria for Dental Radiography**	**CCID** ch. 1 Case 5 (Radiographic Interpretation); **Misch** radiographic chapters
The diagnostic wax-up and guides	RCSEd **Appendix D** (the sample OSCE station)	**Misch** — prosthetic planning; **ITI TG vol 11** (digital)
Surgical anatomy	**Misch** — anatomy and complications chapters	*The Mental Foramen and Nerve: Clinical and Anatomic Factors* (the paper supplied)
Osseointegration and peri-implant biology	**Lindhe** — *Clinical Periodontology and Implant Dentistry*, the implant biology section	**NID** (condensed summaries); **OIR**
Bone density and biomechanics	**Misch** — bone density (D1–D4), force factors, implant design	**DSMT**
Ridge changes after extraction	**Lindhe** — Araújo & Lindhe on socket healing and bundle bone	**ITI TG vol 10** (extraction sockets)
Implant materials, surfaces, metallurgy	**DSMT** — *Dental Science, Materials and Technology: MCQ in Implant Dentistry*	**CCID** ch. 2 (Implant Design); **Misch**
Case complexity and risk assessment	**SAC Classification, 2nd edn** — read it cover to cover; it is short	**ITI TG vol 1** (aesthetic risk assessment)
Aesthetic zone planning	**ITI TG vol 1**	**SAC**; **CCID** aesthetic cases
Loading protocols	**ITI TG vol 2**; the **ITI Decision Tree on immediate placement**	Gallucci et al. consensus
Posterior sites, short implants	**ITI TG vol 3**	**CCID** ch. 2 Case 5 (Short Implants)
Extraction sockets and immediate placement	**ITI TG vol 10**; **ITI Decision Tree: Immediate Placement**	**Lindhe**
Ridge preservation and GBR	**ITI TG vol 7**; **CCID** ch. 5 (Ridge Site Preparation — 8 cases)	**Lindhe**; **Termeie** (regeneration Q&A)
Sinus floor elevation	**ITI TG vol 5**; the two **ITI Decision Trees** (sinus technique; atrophic posterior maxilla)	**Misch**
Soft tissue management	**CCID** ch. 4 (FGG, SCTG, vestibuloplasty/frenectomy)	**Lindhe**; **PCS**
Prosthetic design, abutments, retention	**CCID** ch. 3 (Abutment Design; Screw-Retained; Restorative Materials)	**ITI TG vol 11**; **Misch**
Implant occlusion	**Misch** — implant-protective occlusion	**NID**
Overdentures and the edentulous patient	**ITI TG vol 4**	**Misch**
Complications — biological and technical	**ITI TG vol 8**	**Misch** — complications chapters
Peri-implant disease	**ITI TG vol 13**; the **ITI Decision Tree on mucositis/peri-implantitis**	**Lindhe**; 2017 World Workshop (Berglundh et al.); EFP S3 guideline
Periodontal underpinning	**Termeie** — *Periodontal Review Q&A*	**PCS** — *Periodontics: The Complete Summary*; **Lindhe**
Written-paper practice	**Mosea** — *MCQs in Implant Dentistry*	**DSMT**; **Termeie**; this notebook Parts 5 and 8
Case-based rehearsal for the SOE	**CCID** — *Clinical Cases in Implant Dentistry*	This notebook Part 7

If you are weak on…	Read this first	Then this
Consent, records, professionalism	**GDC** *Standards for the Dental Team* (Principles 1, 3, 4, 7)	*Montgomery v Lanarkshire* summary; FGDP standards

4.2 A realistic reading strategy

If you have three months: - Read **SAC** in full (a weekend). It is the single highest-return book for the SOE. - Read the four **ITI Decision Trees** until you can reproduce them from memory. They are one page each and they structure four whole SOE answers. - Work through **CCID** chapters 1–5 as SOE rehearsal — read the case, close the book, answer the four domains aloud, then read the discussion. - Use **Misch** as a reference, not a read-through. Look up what you don't know. - Do **Mosea** and **DSMT** MCQs for the written paper, plus the 800 in this notebook. - Read the **2017 World Workshop** peri-implant definitions until you can state them verbatim.

Before anything else, whatever time you have: read **"Where the marks are"** at the start of Parts 6 and 7, and the **three set-pieces** at the start of Part 7. They will stop you revising the wrong component.

If you have three weeks: - Part 3 of this notebook (numbers and classifications) every day. - The four ITI Decision Trees. - **OSCE stations 51–58, rehearsed out loud against the clock (2 minutes reading, 5 minutes speaking).** - All 800 SBAs here — a block a day, marked against Part 8. - As many of the 50 SOE cases as you can, **spoken aloud, timed at 15 minutes**, ideally with a colleague playing examiner. - The blueprint (Appendix A) — read the actual wording; the examiners write questions from those sentences.

If you have three days: - Part 3 (numbers and classifications). - The SAC classification summary in §7.3. - The peri-implant disease definitions and the ITI peri-implantitis decision tree. - The four-domain SOE preparation framework, and the nine phrases in Part 7. - The calibration analyses at the head of Parts 6A and 7A. - Skim the complete answer key in Part 8 and re-read every explanation for a question you got wrong.

4.3 The papers behind the techniques: who to name, and for what

There is a mark most candidates never claim. It is given when you attach a **name** to a technique or a classification and say, in one sentence, **what that person or group actually contributed**. It costs four extra words in a sentence you were going to say anyway, and it converts a competent answer into an evidence-based one.

The tables below are grouped by territory: the name, the contribution, and the sentence to say. Learn the third column — the first two exist so that it makes sense.

> **KEY POINT**
>
> **Name the contribution, not the citation.** *"Following Araújo and Lindhe's work on bundle bone…"* scores. *"There was a paper in 2005…"* does not, and invites a question you cannot answer. If you are unsure of a year, leave it out — the contribution is what is examined, and a wrong year undermines everything else you say.

Osseointegration and its definition

Name(s)	Contribution	What to say in the oral
Brånemark and the Gothenburg group	Discovered and named **osseointegration** (first patient **1965**); the machined, submerged, two-stage protocol	*"Osseointegration is a direct structural and functional connection between ordered living bone and the surface of a load-carrying implant. That definition excludes fibrous encapsulation, and my irrigation, drill speed, primary stability and control of micromotion all exist to achieve it rather than repair."*
Schroeder and the Swiss/Bern line (origin of the **ITI**)	Independently showed direct bone apposition, naming it **functional ankylosis**; the **one-stage transmucosal** implant	*"Two groups reached the same biology independently. Schroeder's contribution to how I work is the one-stage transmucosal protocol — which is why a single-stage healing abutment is a legitimate default, not a shortcut."*

Name(s)	Contribution	What to say in the oral
Albrektsson, with Brånemark, Hansson and Lindström (**1981**)	The **six factors influencing osseointegration**: material, design, surface, host bed, surgical technique, loading	*"Three of the six are chosen at planning, one is the patient's biology, and two — surgical technique and loading — are entirely mine. When an implant fails, that tells me where to look."*
Albrektsson & Zarb (with Worthington and Eriksson, **1986**)	The **success criteria**: immobility, no radiolucency, under **0.2 mm** annual bone loss after year one, no pain or neuropathy, **85% at five years and 80% at ten**	*"I would distinguish survival from success — an implant can survive and still fail against Albrektsson and Zarb's criteria."*
Buser on surface topography	**Moderately rough** sandblasted, acid-etched surfaces increase bone-to-implant contact and removal torque; later, hydrophilic surfaces	*"Moderately rough surfaces are why healing times have shortened from Brånemark's three and six months — but roughness is a trade-off, because the same topography is harder to decontaminate if peri-implantitis develops."*

Bone quality and quantity

Name(s)	Contribution	What to say in the oral
Lekholm & Zarb (**1985**)	Bone **quantity A–E**; bone **quality 1–4**	*"I would call this quantity C, quality 3 — a thin cortex around dense trabecular bone. That drives my drilling protocol, whether I under-prepare, and whether immediate loading is even on the table."*
Misch	Bone density **D1–D4**, tied to **drilling protocol, healing time and force management**	*"I use D1 to D4 because it comes with instructions: D1 needs heat control and tapping, D4 needs under-preparation, bone condensing and a longer healing period — and it has the highest failure rate."*
Cawood & Howell (**1988**)	Classification of the **edentulous jaw**, Classes I–VI	*"Class IV is a knife-edge ridge — a horizontal problem. Class V and VI are flat or depressed — vertical problems, far less predictable, and I would refer them."*

Extraction socket biology and ridge change

Name(s)	Contribution	What to say in the oral
Araújo & Lindhe (**2005**)	The coronal buccal wall is largely **bundle bone**, is **ligament-dependent**, and resorbs when the tooth is removed	*"The buccal plate depends on the periodontal ligament, so it resorbs when the tooth comes out — which is why an immediate implant does not preserve it."*
Schropp and colleagues (**2003**)	Dimensional change over 12 months: roughly **50% of width lost**, about **two-thirds in the first three months**	*"Half the ridge width goes in the first year, two-thirds of that in the first three months. A patient who wants a year to think about it needs to be told what the year costs them."*
Chappuis and colleagues (**2013**)	**Buccal bone thickness in the anterior maxilla** — most sites have a facial wall of **1 mm or less**, and thin-wall sites lose markedly more vertical bone	*"I cannot assume the facial wall is thick; in most anterior sites it is a millimetre or less, so I need a CBCT. That measurement decides between immediate and early placement here."*
Cardaropoli, **Botticelli** and the Lindhe group	The **stages of socket healing**, and that the **marginal gap** around an immediate implant fills with bone while the **outer contour still resorbs**	*"The inside of the socket is not the problem — it fills. The outside is. That is why I graft the jumping gap for contour, not for osseointegration."*

Placement timing and loading

Name(s)	Contribution	What to say in the oral
Hämmerle, Chen & Wilson (ITI consensus, **2004**)	The **Type 1–4 placement classification**, defined by the **state of socket healing**, not elapsed days	*"Type 2, early placement at four to eight weeks. The classification is by the state of the socket, not the calendar — a socket still inflamed at eight weeks is not a Type 2 site."*
Gallucci and colleagues (**ITI consensus** on loading)	**Immediate / early / conventional** loading definitions, the **restoration versus loading** distinction, and the **evidence-graded combination grid**	*"Placement timing and loading are separate decisions. Conventional loading is scientifically and clinically validated with any placement type; immediate placement with immediate loading is only clinically documented, and I would say so rather than present them as equivalent."*
Buser and colleagues on **early placement**	Established **Type 2 early placement with simultaneous contour GBR** in the aesthetic zone	*"Early placement with simultaneous contour augmentation is the workhorse aesthetic-zone protocol — infection resolved, soft tissue healed enough for tension-free closure, most horizontal resorption not yet complete."*
The **ITI consensus line on immediate placement**	Immediate placement is **not bone-preserving**; **site-selected**; highest aesthetic complication rate	*"Intact socket, facial wall at least a millimetre, no acute infection, thick phenotype, achievable primary stability, correct three-dimensional position. If any one is missing I place early instead."*

Guided bone regeneration

Name(s)	Contribution	What to say in the oral
Hom-Lay Wang (with Boyapati, **2006**) — the **PASS principles**	The four requirements for predictable regeneration. **P**rimary wound closure protects the graft from the oral environment while it is most vulnerable; **A**ngiogenesis supplies blood, growth factors and osteoprogenitor cells, which is exactly why the buccal plate is decorticated; **S**pace maintenance preserves a volume for bone to occupy, via a membrane that will not collapse, a slowly resorbing graft, or tenting screws; **S**tability of the clot prevents micromotion, achieved by fixation, tacks, or containment within bony walls	*"I would think about this graft through Wang's PASS principles, and then say which of the four is at risk **in this case**. Here it is primary closure, because the flap will be under tension — so I would perform a periosteal releasing incision and test the flap passively before placing the membrane. Naming the threatened principle is worth more than reciting all four."*
Nyman, Karring, Gottlow and colleagues	The **barrier principle** — excluding faster-growing epithelium and connective tissue lets the slower cell population repopulate the wound	*"The membrane excludes the cells that would otherwise win the race. It is not there to hold the graft in."*
Dahlin, with Nyman, Linde and colleagues (**1988**)	Showed experimentally that a **barrier membrane over a bone defect** produces bone regeneration	*"This is why a membrane is not optional over a dehiscence — without a barrier the defect fills with soft tissue and the graft becomes an encapsulated scaffold rather than bone."*
Buser on the **staged approach**	Graft first and place later where simultaneous placement cannot achieve correct position or stability; augmentation is **prosthetically driven**	*"If I cannot place in the correct prosthetic position with the bone that will be there, I stage — graft, wait six to nine months, place into a healed ridge. You augment to the position the restoration needs."*
The biology of **membrane exposure**	Exposure converts a closed regenerative wound into a contaminated one and **substantially reduces bone gain**; the **commonest complication** of the procedure	*"Exposed collagen without infection I manage conservatively with chlorhexidine and no brushing; exposed or infected PTFE I remove early to salvage the bone already formed."*

Soft tissue and aesthetics

Name(s)	Contribution	What to say in the oral
Tarnow, with Magner and Fletcher (**1992**)	**Contact point to bone crest** and papilla fill: at **5 mm or less** the papilla was almost always present, falling substantially at 6 mm and further at 7 mm	*"When a patient asks about the black triangle, my answer is a measurement, not a reassurance — the papilla is nearly always present at 5 mm or less and unpredictable beyond that."*
Tarnow, with Cho and Wallace (**2000**), and later work on adjacent implants	**Inter-implant distance** under **3 mm** gives significantly more crestal bone loss; soft tissue height between two implants is **lower** than between an implant and a tooth	*"At least 3 mm between implants, 1.5 to 2 mm to a tooth. The papilla between two adjacent implants is the hardest tissue to create, so I would ask whether one implant with a pontic gives a better result than two."*
Grunder and colleagues	The **buccal bone requirement** — approximately **2 mm** of facial bone to maintain the facial soft tissue level	*"I would graft the buccal aspect even when the implant is stable. That is a prosthetic and aesthetic requirement, not a stability one."*
Kan & Rungcharassaeng	**Phenotype** as the determinant of **mid-facial recession**, especially after immediate placement, continuing over years	*"Thin phenotypes recede more after immediate placement and go on doing so, so here I would place early and consider a connective tissue graft at the same visit."*
Fürhauser and colleagues (**2005**)	The **Pink Esthetic Score** — a reproducible index for the peri-implant soft tissue	*"I would assess the result with an objective index rather than an opinion — papillae, level, contour, colour, texture, alveolar process — so I can record and compare."*
Belser and colleagues (**2009**)	The **modified PES and White Esthetic Score**, adding objective assessment of the **crown**	*"The white component scores the crown itself — form, outline, colour, texture, translucency. Together they are the vocabulary for documenting a result, or explaining why I am not satisfied with one."*

Peri-implant disease

Name(s)	Contribution	What to say in the oral
Berglundh and colleagues, **2017 World Workshop**	The **case definitions** of peri-implant health, **mucositis** and **peri-implantitis**, and of tissue deficiencies	*"Mucositis is bleeding on gentle probing without bone loss beyond initial remodelling; peri-implantitis adds bleeding or suppuration, increased probing depth and progressive bone loss. Without baseline records, 3 mm or more of bone loss with bleeding and probing of 6 mm or more supports the diagnosis."*
Papapanou, **Tonetti** and **Sanz** (same workshop)	**Staging and grading of periodontitis** — stage by severity and complexity, grade by rate of progression and risk factors	*"I would stage and grade the periodontitis before discussing implants at all — the grade gives me the rate of progression and the risk profile."*
Derks & Tomasi (**2015**)	Meta-analysis of **prevalence**, showing figures depend heavily on the **case definition and threshold**	*"Mucositis in roughly 20 to 65% of patients and peri-implantitis in roughly 1 to 47%, weighted means near 43% and 22%. The spread reflects different thresholds between studies, and that is the limitation I would state."*
Lang & Berglundh	The **mucositis-to-peri-implantitis model** — mucositis is the precursor and is **reversible**; the peri-implant lesion is less well contained than the periodontal one	*"Mucositis is the reversible stage and the point at which I can still change the outcome, so it is what I screen for at every review."*
Wilson, and **Linkevicius** and colleagues	**Residual excess cement** as a cause of peri-implant disease, far more likely and less detectable with **deep subgingival margins**	*"This is why I use screw retention where the access channel allows. If I must cement, the margin goes no deeper than about 1 mm subgingivally, I extrude the excess on a copy abutment extra-orally, and I radiograph afterwards."*
Heitz-Mayfield and **Renvert**	The evidence for **treatment**: non-surgical decontamination first, its limited predictability in established disease, surgical access, and **maintenance**	*"Non-surgical therapy resolves mucositis reliably but is much less predictable in established peri-implantitis, so I would treat, reassess at six to eight weeks, and refer for surgery if it persists."*
The **EFP S3-level guideline** (**2023**)	A stepwise, graded guideline for **prevention and treatment** of peri-implant diseases	*"Risk factor control, non-surgical decontamination, reassessment, surgery if needed, supportive care throughout. Saying I follow a stepwise protocol matters more than naming any one instrument."*

The edentulous mandible

Name(s)	Contribution	What to say in the oral
The **McGill consensus** (**2002**) — **Feine** and colleagues	The **two-implant overdenture** should be the **first-choice standard of care** for the edentulous mandible	*"The evidence for satisfaction and function over a conventional lower denture is strong and consistent, and I would open with that."*
The **York consensus** (**2009**) — **Thomason** and colleagues	**Reaffirmed** McGill, and added that patients should be **informed** the option exists	*"Not offering the option, or not recording that it was discussed, is a consent failing rather than a clinical preference."*

Sinus floor elevation and augmentation

Name(s)	Contribution	What to say in the oral
Boyne & James (**1980**)	First published the **lateral window** elevation with autogenous marrow grafting	*"What has changed since is the graft material, not the principle — the Schneiderian membrane is elevated and the space beneath it becomes bone."*
Tatum	Developed and popularised the **lateral window** clinically, and the crestal approach	*"Membrane integrity is the whole procedure, and perforation is the commonest intra-operative complication."*
Summers (**1994**)	The **osteotome** transcrestal technique through the implant osteotomy	*"Where residual height is adequate and the gain needed is modest I would consider a transcrestal approach — less invasive, but blind. I would refer a lateral window and would not attempt an atrophic posterior maxilla."*
The **ITI and consensus position on graft materials**	Survival in the grafted sinus is **comparable across materials**, including bone substitute alone; the graft is a space-maintaining **scaffold**	*"I choose on morbidity, cost and consent rather than a claim of superiority. What changes the outcome is residual bone height, membrane integrity and implant stability."*

Consent and the law

Name(s)	Contribution	What to say in the oral
Montgomery v Lanarkshire Health Board (**2015**, UK Supreme Court)	Replaced the professional standard for disclosure with a **patient-centred test of material risk** — any risk a **reasonable person in the patient's position**, or **this patient**, would attach significance to, plus **reasonable alternatives**	*"The test is not what a body of dentists would disclose but what this patient would consider material, including reasonable alternatives and no treatment. Here that means the origin of graft materials, the lifetime maintenance burden and cost, the risk of peri-implant disease, and that the aesthetic result may not be perfect — and I must record the discussion, because a consent form is not consent."*
GDC, *Standards for the Dental Team*	**Principle 1** patients' interests first; **Principle 3** valid consent; **Principle 4** records; **Principle 7** work within your knowledge, skills and experience	*"Principle 7 governs the referral decision. Working within my competence is a professional obligation, not a personal preference, so referring a vertical augmentation is the standard being applied — and I would record the reasoning."*

4.4 The numbers: survival, success, longevity and prognosis

WATCH OUT

Read this before you memorise anything below. These are **indicative figures for oral examination purposes**. Every one varies with the case definition, the population, the threshold applied and the length of follow-up, and reported values across systematic reviews often differ by more than the figure itself. Quote them as **ranges with a time frame**, and be ready to name the limitation. *"Around 95 to 98% at five years in systematic reviews,*

though survival is not success and these are largely specialist-treated cohorts" scores. *"96.4%"* invites a question you will not survive.

Implant and prosthesis survival

Restoration	Implant survival, 5 years	10 years	Source type	How to say it
Single crown	approx. **95–98%**	approx. **90–95%**	Systematic reviews of prospective cohorts	*"Around 95 to 98% at five years, 90 to 95% at ten. The crown survives slightly less well than the implant, and most cohorts are specialist-treated — so a good prognosis, not a guarantee."*
Implant-supported fixed partial denture	approx. **95%** implants and prostheses	approx. **90–93%** implants; **85–90%** prostheses	Systematic reviews of prospective and retrospective cohorts	*"At ten years roughly nine in ten survive — but only about half the patients have been free of any complication over that period, which is the figure the patient actually needs."*
Full-arch fixed prosthesis	approx. **95–98%**	approx. **90–95%**	Systematic reviews, largely specialist cohorts	*"High implant survival with a substantial technical maintenance burden — and losing one implant in a full-arch reconstruction is a far larger problem than losing a single implant."*
Implant overdenture (two implants, mandible)	approx. **95–98%**	approx. **90–95%**	Systematic reviews, long-term cohorts, consensus statements	*"Implant survival is comparable to fixed work, but the prosthesis needs regular maintenance. I would consent for maintenance rather than for survival."*

KEY POINT

Three distinctions the examiners listen for. Implant survival is the implant still being in the mouth. **Prosthesis survival** is the restoration still in service — always the lower number over time, and the same implants may be carrying a second prosthesis. **Success** adds criteria: no pain or infection, no radiolucency, bone loss within limits, and an acceptable aesthetic and functional result. An implant can survive, fail against the success criteria, and be carrying its third crown. Say which one you mean.

Technical and biological complications over five years

Complication	Indicative 5-year rate	Source type	How to say it
Screw loosening	approx. **5–10%**, higher on single molars and cantilevers	Systematic reviews of prosthodontic complications	*"One of the commonest technical complications, higher with a wide occlusal table or a cantilever. Modern connections and correct torque have reduced it, not abolished it."*
Ceramic or veneer chipping	approx. **3–6%** metal-ceramic; **higher** for veneered zirconia	Systematic reviews; varies markedly with material	*"Highly material-dependent, which is why monolithic materials have become popular posteriorly."*
Abutment or screw fracture	**around 1% or less**	Systematic reviews	*"Uncommon — but the complication with the worst consequence, because a fractured screw within the implant may end the restoration."*

Complication	Indicative 5-year rate	Source type	How to say it
Loss of retention / decementation	approx. **5%**, higher with short or tapered abutments	Systematic reviews of cemented restorations	*"Around 5% over five years, and one argument for screw retention where the access channel allows — the others being retrievability and avoiding residual cement."*
Overdenture attachment maintenance — insert or matrix replacement, activation, relines, denture repair	**The highest of any complication in implant dentistry.** Most patients need some attachment maintenance within five years; figures vary very widely by attachment type	Systematic reviews and long-term prospective cohorts	*"Attachment maintenance is the commonest complication in the whole of implant dentistry. The implants are very likely to last; the attachments, denture teeth and fit surface are consumables. That is planned maintenance, not failure — and rates vary too widely between systems for me to quote one figure."*

Peri-implant disease prevalence

> **WATCH OUT**
>
> **Do not quote a single number here.** Reported prevalence depends almost entirely on the **case definition and bone loss threshold** used, and on whether the figure is per patient or per implant. Pooling studies with different thresholds produces very wide ranges — that spread *is* the finding, and saying so is the marked answer.

Condition	Reported range	Weighted mean where reported	How to say it
Peri-implant mucositis	roughly **20–65%** of patients	around **40–45%**	*"Common, reversible, and what I am screening for at review."*
Peri-implantitis	roughly **1–47%** of patients	around **20–22%**	*"Anywhere from about 1% to nearly 50%, weighted mean around 20%. That range is not sloppiness in the literature — it reflects genuinely different case definitions and thresholds, which is why the 2017 definitions matter. I would give the range and the reason for it."*

Marginal bone loss

Measure	Figure	Where it came from	How to say it
First year after loading	up to about **1.5 mm** traditionally accepted; modern platform-switched connections commonly well under **1 mm**	The Albrektsson and Zarb criteria, derived from **machined, externally connected, two-stage** implants	*"The classical figure comes from machined implants with an external connection — modern conical connections with platform switching typically lose considerably less."*
Annually thereafter	under **0.2 mm** per year	Same source	*"What matters clinically is whether the loss is **progressive** on serial radiographs taken reproducibly. One radiograph gives me a level; two give me a direction."*
The practical rule	Compare against a **baseline radiograph** at prosthesis fitting	2017 World Workshop position on diagnosis	*"Without a baseline I cannot separate initial remodelling from disease, and I am reduced to the 3 mm fallback threshold."*

Success of the procedures themselves

Procedure	Indicative outcome	Source type	How to say it
Guided bone regeneration (horizontal)	Survival in augmented sites **comparable to native bone**, around **95%** or better at five years; **membrane exposure** the commonest complication	Systematic reviews of cohorts and controlled studies	*"Comparable to native bone — with the caveat that these are mostly experienced operators and selected defects, and the complication that changes the result is exposure."*
Vertical augmentation	**Less predictable, markedly higher** complication rate than horizontal	Systematic reviews; consistent direction of finding	*"The least predictable augmentation procedure. Outside what I would undertake, and I would refer."*
Ridge preservation	**Reduces** dimensional change, does **not abolish** it — typically **1–2 mm less horizontal loss** and around **1 mm or more less vertical loss** than unassisted healing	Meta-analyses of randomised trials	*"It **limits** resorption, it does not prevent it. I consent the patient that the ridge will be better than it would otherwise have been, not that it stays as it is, and that a proportion of sites still need augmentation later."*
Transcrestal (osteotome) sinus elevation	approx. **95–98%** at three to five years; gain of **3–4 mm**; generally needs residual height around **5 mm or more**; perforation less frequent than lateral but **undetected** when it occurs	Systematic reviews of cohorts	*"Good medium-term survival, but the technique is blind — a perforation may go unrecognised. That is why residual height and a modest required gain are the selection criteria."*
Lateral window sinus elevation	approx. **95%** or better at three to five years; **membrane perforation** the commonest complication, reported over a wide range, commonly cited **10–25%** and higher in some series	Systematic reviews and consensus reports	*"Highly operator- and anatomy-dependent, particularly with antral septa. I would refer it."*
Immediate vs early vs conventional placement, aesthetic zone	**Survival is similar**; the difference is **aesthetic** — immediate placement carries the highest risk of **mid-facial recession** in thin phenotypes and thin buccal walls	Systematic reviews and consensus statements	*"Survival is the wrong outcome to compare them on. The difference is the aesthetic complication rate, which is why I select on phenotype and buccal wall thickness rather than on the patient's preference for fewer visits."*

Risk factors, and how much they change the numbers

Risk factor	Effect on the numbers	How to say it
Smoking	Roughly **doubles** failure risk; greater effect in the **maxilla** and in **grafted** sites; also raises peri-implantitis and dehiscence	*"I would treat a smoker, but after a documented cessation discussion, and I would record that they were told their risk is higher."*
Uncontrolled diabetes	**Control matters more than the diagnosis** — well-controlled shows little or no increase in failure; poorly controlled means delayed healing and more peri-implant disease	*"I would ask for a recent HbA1c and liaise with the general medical practitioner. Well-controlled diabetes is not a contraindication; poorly controlled, I would defer until control improves."*
History of periodontitis	Roughly a **two- to fivefold** increase in peri-implantitis risk, greatest with severe or unstable disease	*"Not a contraindication, but among the strongest risk factors. I would establish periodontal stability first, consent specifically to the raised risk, and set a shorter maintenance interval from the outset."*
Parafunction	Increases **technical** complications — screw loosening, chipping, component fracture — rather than biological failure	*"A technical risk, so I design for it: screw retention, a narrower occlusal table, shallow cuspal inclines, no interferences, a nightguard — and consent to a higher maintenance rate."*
Poor plaque control	A consistent risk indicator for mucositis and peri-implantitis; the restoration must also be **cleansable**	*"If the patient cannot clean the restoration I designed, that is my failure — so plaque control before placing, and an emergence profile that can actually be cleaned."*

Risk factor	Effect on the numbers	How to say it
Lack of supportive maintenance	Absence of supportive therapy is associated with a **substantially higher** rate of peri-implantitis — commonly reported as **two- to threefold**	*"Maintenance is not an add-on I offer; it is part of the treatment, and I make that explicit at consent, including the cost."*

How to quote a number in the oral

A quoted figure should contain four things:

1. **The range**, not a point estimate — unless the number is a definition or a threshold, in which case be exact.
2. **The follow-up period** — "at five years", "at ten years". A survival figure without a time frame is meaningless.
3. **What was actually measured** — implant survival, prosthesis survival, patient-level prevalence, millimetres of horizontal change.
4. **One limitation** — the case definition, the population, the operator experience, the follow-up, or the heterogeneity between studies.

And one prohibition: **never invent a precise figure.** If you do not know, say what you do know. *"I am not certain of the exact figure, but the principle is…"* is a scoring answer. A confident wrong decimal is not.

DIRECTION OF THOUGHT

Three good answers, and one bad one.

Good — survival. *"Systematic reviews of single implant crowns report implant survival of around 95 to 98% at five years and roughly 90 to 95% at ten. That is implant survival — crown survival is a little lower, because the crown may be remade over the same implant. The main limitation is that most cohorts are specialist-treated and selected, so I would present this as a good prognosis rather than a guarantee."*

Good — peri-implantitis prevalence. *"I would give a range. Peri-implantitis is reported in about 1% to nearly 50% of patients, with meta-analysed means around 20%, and mucositis in roughly 20 to 65%. That spread is because studies use different case definitions and bone loss thresholds, and report per patient or per implant — which is exactly why the 2017 World Workshop definitions were produced."*

Good — ridge preservation. *"Meta-analyses of randomised trials show ridge preservation reduces horizontal ridge loss by roughly one to two millimetres compared with unassisted healing, over healing periods of four to six months. Crucially it limits change rather than preventing it, so I would consent the patient that the ridge will still change and that a proportion of sites still need augmentation at placement."*

Bad. *"Implants have a 98.2% success rate."* Three failures in one sentence: no time frame, no source, and *success* used where *survival* was meant — and the false precision invites the examiner to ask where the figure came from. There is no recovery from that answer once it has been given.

4.5 One-line summaries: the papers worth carrying into the room

Section 4.3 gave you the **names**. This section gives you the **papers themselves**, compressed to one line each, in the form an examiner will accept: *who, what they showed, and the number that came out of it*.

Read this section the way it is written — not as a reading list but as a **spoken resource**. In an oral you will almost never be asked "what did the paper say?"; you will be asked "how do you know?", and the answer that scores is one clause of attribution followed by the clinical consequence. *"Chen and Buser's review of post-extraction sites found recession of more than a millimetre at the mid-facial mucosa is substantially more frequent after immediate placement, which is why I would place this one at four to eight weeks."* That is the whole technique.

KEY POINT

Three rules for using this section. 1. Learn the *consequence* column; the citation is only there to license it. **2.** If you are not certain of a year, give the author and the finding and leave the year out — a wrong year invites a question you will lose. **3.** Where a figure is a range across studies, say the range and say why it is a range. Precision you do not have is the fastest way to be caught.

On the source of this section. These summaries were consolidated with the help of *Notes in Implant Dentistry* (Chu and Tang, 1st edn, 2021), a freely distributed postgraduate study note that condenses each paper it covers into a single statement, alongside the standard texts and consensus documents already listed in §4.1. **The citation is always the underlying paper, not the note.** Papers dealing with work outside this examination's scope — full-arch immediate-function protocols, block grafting, lateral window sinus surgery, ridge splitting, biologics — have been left out deliberately; where they appear below, they appear as the boundary at which a general dentist with an interest in implants refers.

Diagnosis, prognosis and the language you are marked on

Paper	What it established	What it means at the chair
Tonetti, Greenwell and Kornman (2018) — staging and grading	The **stage** (I–IV) describes severity and complexity of management; the **grade** (A–C) describes the rate of progression and the risk of further loss	*"I would stage from interproximal attachment loss at the worst site and from tooth loss due to periodontitis, and grade from bone loss as a percentage of age, smoking and glycaemic control."*
Berglundh and colleagues (2018) — 2017 World Workshop peri-implant case definitions	Peri-implant **health**, **mucositis** and **peri-implantitis** defined; bleeding on gentle probing is the pivot, progressive bone loss beyond initial remodelling makes the diagnosis	The single most examinable definition set in the component. Learn the three, in order.
Lindhe and Lang — prognosis of the individual tooth	Separates **"irrational to treat"** (recurrent perio-endo lesion, attachment loss to the apex, vertical root fracture, caries into the canal) from **"doubtful"** (class II–III furcation, angular defect, bone loss beyond two-thirds of the root)	Gives you defensible language for the extraction decision instead of "hopeless".
Misch and colleagues (2008) — ICOI Pisa consensus health scale	Four categories rather than two: **success**, **satisfactory survival**, **compromised survival**, **failure**, with bone loss thresholds attached	*"An implant is not simply in or out — I would place this one in the compromised-survival group and say why."*
The Glossary of Prosthodontic Terms	The formal definitions of implant, abutment and the prosthesis types	Where the wording of a definition question comes from. Worth reading once.

Healing, the peri-implant tissues, and why they are not periodontal tissues

Paper	What it showed	Consequence
Araújo and Lindhe (2005) — dimensional ridge alterations after extraction	The **buccal bundle bone is ligament-dependent** and resorbs once the tooth is removed; the buccal wall loses more height than the lingual	The reason immediate placement does not preserve the buccal plate. Say it in those words.
Araújo and colleagues (2005) — implants into fresh sockets, dog model	Placing an implant into the socket **did not prevent** the buccal wall resorbing	Kills the "immediate placement holds the bone" answer outright.
Schropp and colleagues (2003)	About **half the ridge width is lost in twelve months** — of the order of 5–7 mm horizontally — most of it in the first three months, with the crest moving palatally	The number behind the ridge-preservation decision.

Paper	What it showed	Consequence
Davies (1998) — mechanisms of endosseous integration	Three phases: **osteoconduction** (fibrin retention and cell migration to the surface), **de novo bone formation**, then **remodelling**	Explains why surface topography and clot stability matter, and why you do not disturb a site at two to four weeks.
Ivanovski and Lee (2017–2018)	Peri-implant connective tissue fibres run **parallel** to the surface rather than inserting; the seal is adaptive, less vascular, and a weaker barrier	Why peri-implantitis progresses faster and why probing behaves differently.
Berglundh, Bosshardt and Lang on the junctional epithelium	The epithelial attachment sits at a bacterial interface and is the first line of defence	Underpins the whole maintenance argument.
Judgar and colleagues (2014)	Two-piece implants showed a **thicker connective tissue component** and a greater biologic width than one-piece	The microgap has a biological price; relevant to platform choice and depth.
Wennström and Derks (2012)	Less than **2 mm of keratinised mucosa** was associated with higher plaque and bleeding scores, though evidence on bone loss was weaker	*"I would say the width matters for comfort and plaque control, and that the evidence for it preventing bone loss is weaker than people claim."*
Boynueğri and colleagues (2013); Lin and colleagues (2018)	Sites with adequate keratinised tissue showed lower inflammatory markers; a width of **at least 2 mm** is the usual threshold quoted	The augmentation-timing question: soft tissue is easier before, not after, the problem.

Medical risk — the figures examiners ask for by name

Paper	The figure	How to use it
Chen and colleagues (2013) — meta-analysis	**Smoking and radiotherapy** significantly increase failure; diabetes and osteoporosis did not reach significance in that analysis	Lets you rank risks honestly rather than listing them flatly.
Chrcanovic and colleagues (2017)	Failures **cluster in individuals** — a minority of patients carry a large share of all failures	*"Failure is patient-level as much as site-level, so a previous failure changes my consent conversation."*
Chappuis and colleagues (2018) — medication-related failure	Failure rates reported around **4.3% with proton pump inhibitors** and **7.5% with SSRIs**	Two drug classes that are easy to miss in an ordinary history.
AAOMS position paper (2014)	MRONJ in patients on **long-term oral bisphosphonates ~0.1%**, rising to about **0.21%** with longer exposure; risk far higher with intravenous antiresorptives for oncology	Lets you separate the low-risk osteoporosis patient from the oncology patient.
Gaudin and colleagues (2015)	MRONJ after extraction about **3.2%** where the drug was given for oncological indications, and very much lower for osteoporosis	The number that makes the referral threshold defensible.
Madrid and Sanz (2009)	Patients on oral anticoagulants with **INR 2–4** who continued their medication did not bleed significantly more	*"I would not stop the warfarin; I would check a recent INR and use local measures."*
Hwang and Wang (2006)	Named the small list of genuine **absolute** contraindications — recent infarct or stroke, valve surgery, profound immunosuppression, active malignancy, uncontrolled bleeding disorder	Most "contraindications" are relative. Saying so, and then managing the risk, is the scoring answer.
Lee and colleagues (2012)	**Residual pockets of 5 mm or more with bleeding** after periodontal treatment were a risk indicator for peri-implantitis, with a markedly higher rate at implant level in treated periodontitis patients	The reason "stability before implants" is a hard rule, not a preference.
Morris and colleagues (2000)	Type 2 diabetes showed more failures, but the effect was marginal	Control matters more than the diagnosis.

Placement timing and the immediate-placement argument

Paper	What it showed	Consequence
Buser, Chappuis, Belser and Chen (2017) — when immediate, when early, when late	Sets out **Type 1 immediate / Type 2 early with soft tissue healing (4–8 weeks) / Type 3 early with partial bone healing (12–16 weeks) / Type 4 late**, and argues Type 2 as the workhorse of the aesthetic zone	The spine of any placement-timing answer.
Chen and Buser (2009, 2014) — post-extraction outcomes	Survival is similar across types, but **recession of the mid-facial mucosa of more than 1 mm is more frequent, and outcomes more variable, after Type 1**	Survival is the wrong outcome measure in the aesthetic zone. Say that.
Cosyn and colleagues (2012, 2016, 2019)	Advanced recession after immediate placement is predictable only when **all** of the favourable factors are present — intact buccal wall, thick phenotype, flapless surgery, correct 3-D position, immediate provisional	Gives you the pre-requisite list in a form you can recite.
Chen and Darby (2017)	At extraction in the anterior maxilla, **fewer than half the sites had an intact facial wall**; the remainder had fenestrations or dehiscences	The honest answer to "why not place it immediately?" — you often cannot know until the tooth is out.
Kan and colleagues (2007)	Facial defect morphology predicts the soft tissue result: **V-shaped defects behaved best, U-shaped worse, and the wide "ultra-U" worst**	A shape you can assess intra-operatively and abandon on.
Tarnow and colleagues (2014)	Grafting the gap **and** placing an immediate provisional preserved facial-palatal ridge contour better than either alone	If you place immediately, do both.
Capelli and colleagues (2013)	The **implant-to-buccal-plate distance** is a usable diagnostic parameter for deciding whether and how to graft the gap	Turns "graft the jumping gap" into a measurement.
Ragucci and colleagues (2020)	Immediate placement into **molar** sockets showed high survival with limited marginal bone loss in selected cases	A technique-sensitive extension, not a default.
Zhao (2016), Lee (2018), Chen (2018), Blus (2015) — infected sockets	With debridement and antibiotic cover, immediate placement into **chronically infected** sockets showed no significant difference in failure or bone loss; reported survival around 98.8% non-infected, 100% chronically infected, and lower (about 94%) in acute infection	*"Chronic apical infection is not an absolute bar if the socket is thoroughly debrided; acute suppuration is."*

Three-dimensional position — the numbers that decide aesthetics

Paper	The number	Consequence
Grunder, Gracis and Capelli (2005)	About **2 mm of buccal bone** is needed at the shoulder for stable facial tissue	The single most quoted figure in aesthetic-zone planning.
Buser, Martin and Belser (2004, 2007)	Comfort and danger zones in three planes; the shoulder placed palatal to the point of emergence and about **3 mm apical to the future gingival margin**	Gives you a structured way to answer "where exactly would you place it?"
Tarnow, Magner and Fletcher (1992)	Contact point to bone crest **5 mm → papilla present almost always; 6 mm → about half; 7 mm → about a quarter**	The papilla answer, and the reason you look at the neighbouring bone, not the implant.
Tarnow and colleagues (2000)	Less than **3 mm between adjacent implants** increased crestal bone loss laterally, of the order of 1.3–1.4 mm from each shoulder	Why two adjacent implants are a different problem from two single ones.
Tarnow and colleagues (2003)	Only about **3.4 mm of soft tissue on average** forms over the inter-implant crest	Sets the honest expectation for the papilla between two implants.
Spray and colleagues (2000)	Facial bone loss fell sharply once the **buccal plate reached roughly 1.8–2 mm** in thickness at placement	The measurement behind "place it palatally, and graft the buccal contour".

Paper	The number	Consequence
Scutellà and colleagues (2015)	Buccolingual position and the **abutment finish-line geometry** together determine emergence and therefore cleansability	Links the surgical decision to the restorative one, which is exactly what a domain 3 question is testing.

The edentulous mandible and the overdenture

Paper	What it showed	Consequence
McGill consensus (2002–2003) and the **York statement (2009)**	A **two-implant mandibular overdenture** should be the first-choice standard of care for the edentulous mandible, rather than a conventional complete denture	Learn one sentence of this verbatim; it appears in almost every edentulous case.
Roccuzzo and colleagues (2012)	For the **mandible**, two implants are sufficient for a removable reconstruction; for the **maxilla** the evidence to recommend a number is far weaker	Lets you answer the "how many implants?" question honestly.
Schimmel and colleagues (2014)	All three loading protocols give high survival, but **early and conventional loading are better documented** than immediate	The safe recommendation, with a reason.
Sanda and colleagues (2019)	Marginal bone loss around **immediately loaded** mandibular overdenture implants was comparable with conventional loading	The counter-argument you should be able to state before rejecting it.
Rignon-Bret and colleagues (2019)	Immediate loading of **two non-splinted** mandibular implants remained satisfactory at **10 years**	Useful when the examiner pushes on immediate loading.
Stoker and colleagues (2007)	Over eight years, a **bar on two implants** was the most efficient design in terms of aftercare and cost	The maintenance argument for bar over solitary attachments.
Harder and colleagues (2011)	A **single** midline implant overdenture improved quality of life and chewing in older edentulous patients	A legitimate option for the frail or financially constrained patient.
Zitzmann and Marinello (1999)	In the **maxilla**, a fixed design suits only minimal resorption with a favourable maxillomandibular relationship; otherwise an overdenture restores lip support better	The lip-support answer for the maxillary case.
Slot and colleagues (2019)	**Four versus six** implants supporting a maxillary overdenture gave comparable five-year outcomes	More implants is not automatically better.
Allen and McMillan (2003)	Implant-retained prostheses produced significant improvement in satisfaction and oral health-related quality of life	The patient-centred outcome, which domain 2 rewards.

Grafting, ridge preservation and the materials

Paper	What it showed	Consequence
Hämmerle and colleagues (2002); **Donos and colleagues (2008)**	Implants placed in sites augmented with barrier membranes survive at rates **comparable with pristine bone — of the order of 96–100% over 2–8 years**	The reassurance figure for consent to GBR.
Wessing, Lettner and Zechner (2017)	GBR with **particulate graft and a resorbable collagen membrane** is effective for lateral ridge augmentation	The technique a general dentist with implant experience can defend using.
Milinkovic and Cordaro (2014)	**Dehiscences and fenestrations** are treated predictably by simultaneous GBR; larger horizontal and vertical defects need staged and more advanced approaches	Draws your competence line for you.
Buser (2013); **Dahlin and colleagues (2009)**	Early placement with **contour augmentation** and GBR with a xenogeneic filler remained stable at five years and beyond	Justifies over-contouring the buccal aspect with a slowly resorbing material.

Paper	What it showed	Consequence
MacBeth and colleagues (2016)	Alveolar ridge preservation **reduces but does not abolish** dimensional change — a benefit of roughly 0.7–0.8 mm of mid-buccal height over an untreated socket	Say "reduces, does not prevent". That distinction is the mark.
Jung and colleagues (2018); Lim and colleagues (2019)	**Deproteinised bovine bone mineral with collagen under a native bilayer collagen membrane** performed well for ridge preservation, including molar sites left to heal open	The material combination to name if asked what you would actually use.
Karageorgiou and Kaplan (2005)	Pore size of at least **100 μm** is needed for cell migration; **above about 300 μm** favours vascular ingrowth and direct osteogenesis	Explains why particle and scaffold architecture matter, not just chemistry.
Zhou and colleagues (2011); Zaner and Yukna (1984)	**Smaller particles** give more contact with new bone but less inter-particle space; larger particles maintain space better	The trade-off behind particle-size selection.

The maxillary sinus — mostly a referral, but know the argument

Paper	What it showed	Consequence
Esposito and colleagues (2014) — Cochrane review	**Short implants without augmentation** may be preferable to longer implants with sinus augmentation in the atrophic posterior maxilla, with fewer complications	The answer that keeps a case within your competence.
Annibali and colleagues (2012)	Short implants gave a cumulative survival of about **99%** over roughly three years	The figure that supports the paragraph above.
Thor and colleagues (2007); Fouad and colleagues (2017)	Bone formed at the sinus floor after **membrane elevation without graft material**, with simultaneous implant placement	Useful for transcrestal cases; the graftless argument.
Testori and colleagues (2019)	Risk factors for **lateral window** surgery — anatomy, septa, membrane thickness, sinus health, the medical history	Cite it to explain **why you would refer**, which is the point of the question.

Peri-implant disease

Paper	The figure	Consequence
Zitzmann and Berglundh (2008)	Mucositis around **80% of subjects and 50% of implants**; peri-implantitis around **28–56% of subjects and 12–43% of implants**	Quote as a range and say the ranges are wide because case definitions differ.
Costa and colleagues (2012)	Patients with existing mucositis and **no maintenance** had a much higher incidence of peri-implantitis at five years	The single strongest argument for the recall interval.
Fransson and colleagues (2010)	Bone loss was **non-linear and accelerating**; a substantial minority of affected implants lost 2 mm or more	Why "watch and see" is not a plan.
Katafuchi and colleagues (2018)	Peri-implantitis was significantly more common where the **emergence angle exceeded 30°** at bone-level implants (about **31% versus 15%**)	Restorative contour is a biological decision. Excellent to volunteer.
Matarasso and colleagues (2010)	Over ten years, **periodontally compromised** patients showed higher bone loss rates than periodontally healthy ones	Supports the history-of-periodontitis risk statement.
Renvert and colleagues (2007, 2008)	Non-surgical therapy is **effective for mucositis** and adjunctive antiseptics help, but non-surgical therapy alone is **insufficient for peri-implantitis**	The treat/refer boundary in one sentence.

Paper	The figure	Consequence
Heitz-Mayfield and colleagues (2008, 2012)	Named the risk indicators, and showed an **anti-infective surgical protocol** with decontamination and systemic antimicrobials to be effective at 12 months	The structure of a surgical answer if pushed.
Schwarz and colleagues (2014)	Combined therapy — access flap, **implantoplasty** of supracrestal exposed surfaces, grafting the intrabony component	Name it as a specialist procedure, and refer.
Lang and colleagues (2009)	Sites bleeding consistently lost markedly more attachment than sites consistently non-inflamed	The reason bleeding on probing is your monitoring endpoint.

Prosthetics, retention and maintenance

Paper	What it showed	Consequence
Pjetursson and colleagues (2007)	Compared survival and complications of tooth-supported FDPs, implant-supported FDPs and single crowns; complication rates for implant reconstructions are **higher than survival figures suggest**	Always separate survival from complication-free survival.
Papaspyridakos and colleagues (2012)	For full-arch fixed rehabilitation, prostheses free of any complication were about **29% at five years and 9% at ten**	The number that makes the maintenance conversation honest.
Wittneben and colleagues (2014)	No difference in survival, but **screw-retained reconstructions showed fewer technical and biological complications** and are retrievable	The defensible default in most cases.
Linkevicius and colleagues (2013)	**The deeper the cement margin, the more undetected cement** — and radiographs are not reliable for detecting it	The sentence to quote whenever cement is mentioned.
Vindasiute and colleagues (2015)	Undercuts and margin position govern how much excess can be removed at all	Design decision, not a cleaning problem.
Vigolo and colleagues (2014)	At four years, no difference in peri-implant bone or soft tissue between cemented and screw-retained single crowns **when margins were supragingival**	The fair counter-argument.
Batista and colleagues (2018, 2019)	**Splinting versus not splinting** adjacent implants made no difference to marginal bone loss; a small survival advantage to splinting in some analyses, particularly short implants	Answer with "it depends on length, bone and hygiene access".
Papaspyridakos (2014), Moreira (2015), Lee (2008)	**Splinted, open-tray** impressions were the most accurate for multiple implants, particularly in the edentulous jaw	The impression answer, with a reason.
French and colleagues (2019)	**Interproximal contact loss** occurred at about **17%** of implants and increased with time	A complication candidates routinely forget; volunteering it scores.
Misch and colleagues (2008)	The ICOI health scale for reporting outcomes	Use its four categories when asked to grade an existing implant.

Stability, micromotion and loading

Paper	The figure	Consequence
Szmukler-Moncler and colleagues (1998, 2000)	There is a **critical micromotion threshold — commonly quoted at around 50 to 150 μm** — above which fibrous encapsulation replaces bone	The mechanism behind every loading rule you give.
Kohi and colleagues (2021)	Systematic review confirming a tolerable-micromotion threshold, with variation between models	Say "the threshold exists and is of that order" rather than a single figure.

Paper	The figure	Consequence
Meredith (1998); Sennerby and Meredith (2008)	**Primary stability is mechanical**, from bone quality, implant design and technique; **secondary stability is biological**. RFA measures stiffness at the interface	Explains the **stability dip at around two to four weeks** — the most examinable curve in the subject.
Ito and colleagues (2008)	RFA values did not correlate directly with histological bone-to-implant contact	Use ISQ as a trend in one patient, not as an absolute threshold.
Hsu and colleagues (2016)	Primary stability was highest with **indirect (transcrestal) sinus elevation** among the fixation techniques compared, though differences were not significant	Minor, but useful if pressed on transcrestal technique.

Imaging and radiation

Paper	What it established	Consequence
EAO guidelines (Harris and colleagues, 2012)	Consensus on when cross-sectional imaging is justified in implant dentistry	Cite when justifying, or declining, a CBCT.
Jaju and Jaju (2015)	Move from **ALARA to ALADA** — as low as *diagnostically acceptable*	*"I would justify the CBCT for this site individually, at the smallest field of view that answers my question."*

WATCH OUT

Do not recite this section. Nobody is examined on their ability to list papers, and a candidate who produces four citations for a question about a loose denture has misread what is being tested. Each of these is a **single clause** that licenses a clinical statement you were going to make anyway. If you cannot say the clinical statement without the citation, learn the clinical statement first.